



Review Article

A comprehensive review of time series classification: Traditional, deep learning, and few-shot learning Methods

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ABSTRACT

As a core task in time-series data analysis, time series classification has extensive application value across healthcare, industry, and the military. Recently, time series classification methods based on deep learning have developed rapidly, significantly surpassing traditional approaches in their ability to represent features and recognize patterns. However, the current field still lacks a structured and practical systematic review tailored to real-world application scenarios. Therefore, we present a systematic review of recent advances in time series classification research. We propose a multi-level review framework comprising three horizontal and three vertical dimensions. The three horizontal dimensions cover traditional analytical, deep learning-based, and few-shot learning-based methods. The three vertical dimensions span data, methods, and applications. We categorize deep learning methods into local pattern mining-based, temporal dependency modeling-based, hybrid model architectures-based, and spatiotemporal fusion modeling-based, which highlights the characteristics of these methods in deep learning model structures and feature extraction. Moreover, we discuss their respective advantages and limitations. Next, we introduce commonly used public datasets and evaluation metrics for time series classification. To promote the application of methods, we outline four typical application domains and development trends of deep learning classification methods for time series. Finally, we explore future research directions that address challenges in practical applications such as generative augmentation, multimodal fusion, few-shot learning, and explainable AI. Existing literature either overemphasizes models while neglecting scenarios or overemphasizes methods while neglecting integration. This review aims to provide researchers with a structured knowledge framework to guide research on time series classification and its applications.

1. Introduction

Time series data serves as the primary format for describing the evolution of dynamic systems and has a wide range of applications in finance, healthcare, industry, transportation, and the military [1–3]. The temporal dependencies, multi-scale dynamic patterns, and cross-dimensional correlations inherent in time series data facilitate the extraction of long-term trends, seasonal fluctuations, and sudden anomalies within systems, providing crucial insights into their temporal behavior. Uncovering temporal patterns within time series provides a quantitative basis for modelling complex systems, such as monitoring and analyzing the periodic variations of long-term vital signs and their correlation with clinical diseases. On the other hand, recognizing patterns in historical data enables prediction and early warning of future states, supporting critical applications such as industrial predictive maintenance and early disease screening [4–6]. With the proliferation of

IoT technologies, multidimensional, high-frequency time-series data is experiencing explosive growth. Extracting meaningful insights from vast amounts of time-series data has become a core challenge in current interdisciplinary time-series research.

Time series classification (TSC) is crucial in analyzing time series data. It aims to categorize sequences with similar patterns into pre-defined classes, providing essential support for automated decision-making in dynamic systems. Pattern recognition is achieved by capturing local features (e.g., waveform inflection points) and global structures (e.g., periodic patterns) within time series. TSC has a wide range of applications, including medical signal classification, radar target recognition, industrial fault diagnosis, and network traffic detection [2,7]. Due to the explosive growth of temporal data and increasingly complex feature patterns, the field has evolved from traditional distance-based and statistical methods towards deep learning-driven end-to-end modelling. Time series classification has

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recently incorporated paradigms such as multimodal learning and meta-learning to overcome challenges like scarce sample labeling and cross-domain generalization. Research in TSC enhances the analytical efficiency of large amounts of real-world time-series data, profoundly impacting the accuracy and timeliness of decision-making. Against the backdrop of parallel advancements in big data and edge computing, time series classification is evolving towards lightweight and cross-modal approaches. This progress is driving technological breakthroughs in fields such as the Internet of Things (IoT), smart healthcare, and Industry 4.0. It is serving as a critical bridge connecting time-series data with practical applications.

Deep learning has made significant strides in computer vision [8] and natural language processing [9–11], gradually becoming a core driving force for progress in time series classification. From a technological development perspective, the application of deep learning to time series classification has experienced explosive growth. Initially, Convolutional Neural Networks (CNNs) [12,13] and Recurrent Neural Networks (RNNs) [14,15] were applied to this field. CNNs extract local features within time series effectively, while RNNs capture temporal dependencies within sequences. However, given the unique characteristics of time series data, researchers found that deep learning models from fields such as computer vision were not directly applicable to time series classification. Simple CNNs and RNNs gradually revealed their limitations: CNNs struggled to handle long-range dependencies effectively, and RNNs were susceptible to vanishing or exploding gradients during training. Researchers have developed improved models tailored for time series classification to address these limitations of the original deep learning models. For example, fully convolutional networks (FCNs) [16,17] enable end-to-end learning for time series data by replacing the fully connected layers in traditional CNNs with convolutional layers. This approach avoids complex feature engineering and significantly improves classification performance. Notably, InceptionTime [18] achieved leading performance on the UCR benchmark by constructing an ensemble model that integrates multiple Inception modules with distinct architectures, demonstrating the immense potential of structural innovation in time series classification. Concurrently, models based on attention mechanisms, such as Transformers and their variants [19,20], emerge as promising solutions. These models adaptively focus on critical information within time series, effectively addressing challenges related to long-range dependencies. Mamba [21], a state-space model, has unique advantages in time series processing. Leveraging its linear time complexity, Mamba efficiently processes long sequences without requiring attention mechanisms, reducing computational costs and memory requirements. Kolmogorov-Arnold Networks (KAN) [22] has robust function-fitting capabilities and interpretability, which enable it to better handle complex relationships and nonlinear patterns in time series classification. It demonstrates outstanding performance in challenging tasks such as complex time series classification [23]. Additionally, frameworks such as multimodal learning, self-supervised learning, and meta-learning have emerged as new research hotspots for time series classification tasks with limited data [3].

Deep learning-based time series classification methods are currently developing rapidly. However, existing time series reviews still have many limitations. (1) Most reviews merely list and introduce methods without providing systematic and structured integration and analysis. Consequently, the developmental trajectory and classification framework of these methods remain unclear. (2) Current reviews are not updated or refined promptly, resulting in insufficient coverage of emerging technologies. There is a lack of introductions to novel time series models, such as Transformers, Mamba, and KAN. There is currently no comprehensive review that has yet been provided on few-shot time series classification characteristics and techniques. (3) Existing reviews predominantly emphasize models while neglecting typical time series application scenario analyses. These reviews lack summaries and insights from engineering practice and real-world applications. In the face of abundant and rapidly evolving time series classification

research, researchers urgently need a comprehensive review to organize the latest technical knowledge and research directions, which will allow them to swiftly grasp the research hotspots, future development, and application directions of deep learning in the field of time series classification. Therefore, we propose an innovative three-horizontal and three-vertical framework, as shown in Fig. 1, to review time series classification. The three horizontal dimensions encompass traditional, deep learning, and few-shot learning methods. The three vertical dimensions cover data, methods, and applications. The horizontal dimension reflects the evolution of modeling methodologies from feature-engineering-driven methods toward data-driven and data-efficient learning paradigms, and the vertical dimension focuses on distinct research levels, where these elements share commonalities across different methodological paradigms. This framework addresses the shortcomings of existing reviews, which often emphasize methods over integration and models over scenarios. We conduct a thorough exploration of time series classification methods, focusing on the development of deep learning-based approaches. Additionally, we provide a systematic review of datasets and evaluation frameworks for time series classification. Furthermore, we organize and analyze four typical application scenarios of time series classification and outline their future development directions. Finally, based on the current state of research, we evaluate the challenges that deep learning-based time series classification methods face and propose future research directions.

We aim to offer researchers a fresh and systematic approach to reviewing time series classification. The review will evaluate the challenges of deep learning-based detection methods and suggest potential future research directions. Moreover, it can further advance the development of time series classification methods in downstream application domains. The contributions of this paper are as follows:

- (1) We introduce the pioneering three horizontal and three vertical multidimensional research framework. Horizontally, it spans three methodological systems: traditional methods, deep learning methods, and few-shot learning methods. Vertically, it integrates three research axes: data dimensions, methodological dimensions, and application scenarios. This framework serves as a heuristic tool to systematically reveal structural research gaps often overlooked by unidimensional reviews, thereby guiding the exploration of novel research paradigms.
- (2) At the methodological level, we comprehensively review the current state of time series classification methods based on traditional analysis approaches, deep learning, and few-shot learning. We systematically map the relationships, strengths, and limitations of the various methods while analyzing emerging frontier research directions in time series classification, which fills gaps in existing reviews regarding novel research domains.
- (3) At the application level, we systematically organize four typical application scenarios across medical signal classification, radar automatic target recognition, industrial fault diagnosis, and network traffic detection. By integrating task characteristics, we outline new paradigms for applying time series classification technologies and offer insights to advance the transformation of time series classification from academic research to critical domain applications.
- (4) Finally, we analyze and summarize detailed research challenges and future directions in time series classification, providing an academic guide and research roadmap that combines theoretical and practical value.

The remainder of this paper is outlined as follows. In Section 2, we define time series and describe the characteristics of temporal data, data augmentation methods, and the challenges of time series classification. Moreover, it furnishes a comprehensive survey of the prevailing framework and pioneering research in time series classification. In Section 3, we detail the literature retrieval strategy and review

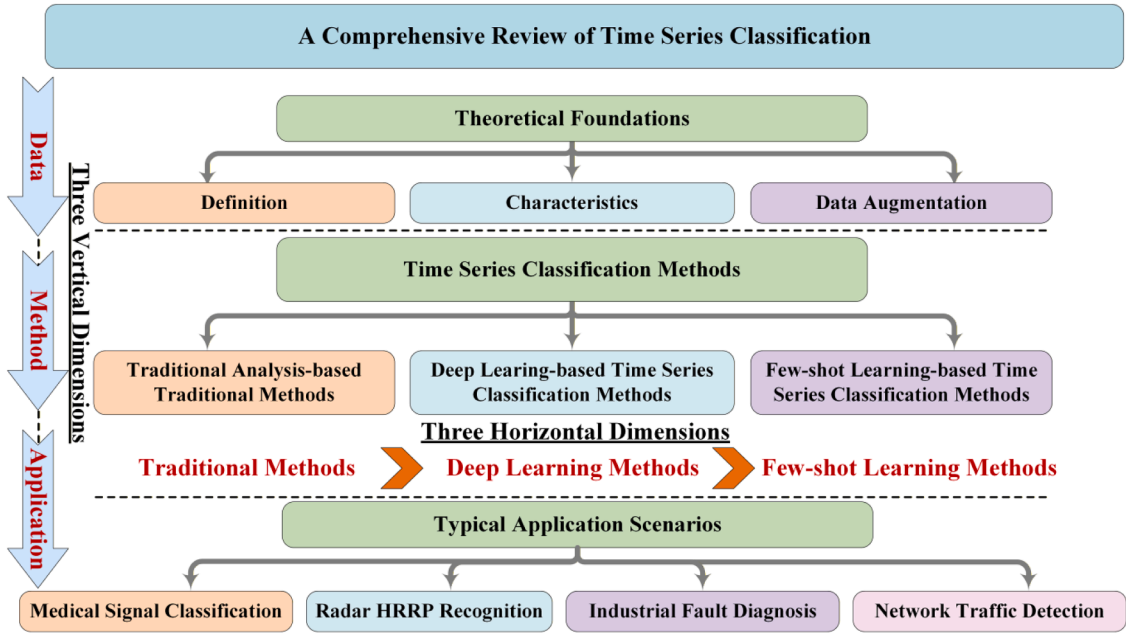


Fig. 1. The three horizontal and three vertical multidimensional research framework.

methodology. In Section 4, we provide a comprehensive review and summary of current research on traditional analysis methods, deep learning methods, and few-shot learning approaches for time series classification. It clarifies the relationships, advantages, and limitations of each category. In Section 5, we review the current state of application of time series data augmentation techniques and model architecture optimization methods. In Section 6, we introduce commonly used datasets and evaluation metrics for time series classification. In Section 7, we outline the current research status and domain-specific challenges for four typical time series classification application scenarios. In Section 8, we summarize the research challenges and future directions in time series classification, providing researchers with insights and inspiration. We summarize the entire paper in Section 9.

2. Theoretical foundations of time series classification

2.1. Definition of time series

Time series data refers to an ordered sequence of observations taken at regular time intervals. The core characteristic of time series data is the temporal dependency of data points, meaning current observations

depend on historical states and may foreshadow future trends. Time series can be categorized based on differences in variable dimensions as univariate time series (UTS) or multivariate time series (MTS). Fig. 2 shows a schematic representation of the time series.

Fig. 2(a) illustrates that a univariate time series represents a simple temporal pattern, consisting of observations of a single variable at consecutive time points. Mathematically, it is expressed as a one-dimensional sequence $\hat{\mathbf{X}} = [x_1, x_2, \dots, x_T]$, where T denotes the sequence length, x_t represents the observation at time t , and $t \in T$. Typical applications include temperature monitoring and single-channel electrocardiogram (ECG) interval sequences. The core analytical objective of the univariate time series is to capture the temporal patterns of a single variable, such as periodicity, trends, or anomalies.

Fig. 2(b) illustrates that the multivariate time series comprises temporal observations of two or more correlated variables. This mathematical structure is represented by a two-dimensional matrix or a higher-dimensional tensor $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_T]$, where T denotes the sequence length, $\mathbf{X}_t$ represents the observation at time t , and D denotes the dimension of the variable $\mathbf{X}_t \in \mathbb{R}^{D \times 1}$. Multivariate time series are ubiquitous in complex systems. One typical application is the 12-lead

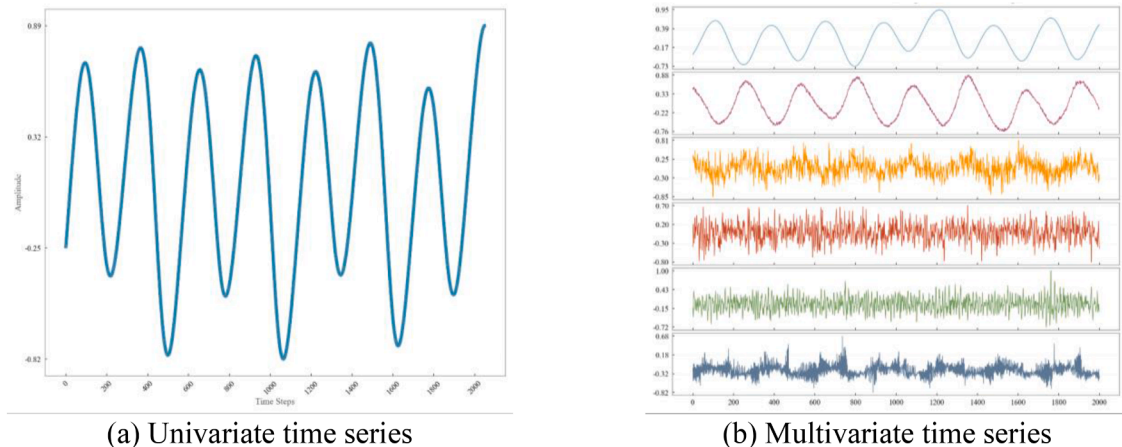


Fig. 2. Schematic Diagram of Univariate and Multivariate Time Series.

ECG, which comprises 12 spatially distributed sequences of cardiac electrical activity. Signals from each lead are collected synchronously over time, reflecting the electrophysiological characteristics of different cardiac regions. Similarly, sensor networks in industrial rotating machinery simultaneously capture multidimensional signals, such as vibration, temperature, and current. The spatio-temporal correlations among these variables reveal the operational status of the equipment. When explicit spatial topological relationships exist among the variables of a multivariable time series (e.g., node connections in sensor networks or transmission lines in power grids), the time series data can be further abstracted into a graph structure. Mathematically, this is represented as a timestamped graph signal $\tilde{G} = (\mathbf{V}, \mathbf{E}, \mathbf{X})$, where $\mathbf{V}$ denotes the node set, $\mathbf{E}$ denotes the edge set, and $\mathbf{X}$ denotes the temporal observation matrix for each node. Such data is often encountered in scenarios such as smart grids and social networks, necessitating concurrent modeling of both the temporal dynamics of nodes and the structural relationships between graphs.

Time series classification aims to map time series data to a discrete set of labels $y \in \{C_1, C_2, \dots, C_K\}$, where K represents the number of categories. Based on the amount of labeled data available, classification can be categorized as supervised (fully labeled data), semi-supervised (a small amount of labeled data and a large amount of unlabeled data), or unsupervised (completely unlabeled data). This review focuses on supervised time series classification and few-shot classification. Furthermore, the generalized definition of time series transcends numerical constraints, encompassing broader forms of dynamic data. It supports multimodal hybrid sequences, including numerical, textual, and image data, shifting from modeling temporal dependency and statistical patterns to dynamic system representation, cross-domain correlation, and processing and fusion of heterogeneous time series data. Fig. 3 illustrates the workflow for time series classification tasks.

2.2. Characteristics of time series

Due to its dynamic and complex nature, time series data presents unique characteristics and challenges in classification tasks. As time series data modalities expand, multimodal time series integrate heterogeneous data types, such as numerical, textual, image, and audio data, through synchronous or asynchronous observations across the temporal dimension. This approach introduces new features in heterogeneous data processing and fusion. Challenges stem from the complex structure of the data itself and the specific requirements of practical application scenarios. The following section systematically analyzes the

impact of these characteristics on classification tasks.

2.2.1. Temporal dependency

Temporal dependency refers to the statistical correlation between adjacent data points in a time series. The correlation arises from the physical continuity or logical causality inherent in dynamic systems. For instance, in mechanical vibration signals, velocity and acceleration at the preceding time step determine displacement at the current time step. It manifests as autocorrelation in the displacement sequence. The core characteristic of this property is that dynamic correlations decay over time, meaning dependencies between adjacent time steps are stronger than between widely separated steps, which conforms to a first-order approximation of Markov properties. Temporal dependency is an intrinsic feature of time series, manifesting as an association between current data points and historical sequences. It can be categorized as short-term local patterns or long-term global trends.

Traditional classification methods struggle to capture long-range dependencies when extracting features from time series data, primarily due to their assumption of independent and identically distributed (i.i.d.) samples, which can lead to classification errors. For example, distinguishing between normal sinus rhythm and atrial fibrillation in electrocardiogram classification relies on the morphology of individual R waves and the rhythmic regularity across multiple cardiac cycles. Models that ignore temporal dependencies may misclassify occasional premature beats as persistent abnormalities. Early-stage bearing wear in industrial equipment fault diagnosis often manifests as a subtle amplification of periodic impact in vibration signals. Accurate classification between normal operation, minor wear, and severe failure requires modeling temporal dependencies to capture the trend of progressively shortening intervals between consecutive impacts.

2.2.2. Noise sensitivity

Noise sensitivity stems from random disturbances during data acquisition, transmission, or storage. The types and distributions of noise vary significantly across different fields. Formation mechanisms include sensor electronic thermal noise, environmental electromagnetic interference, and human recording errors. Common types of noise include Gaussian and impulse noise. Noise is characterized by a broad-spectrum energy distribution that covers the entire signal frequency band. However, it exhibits regular patterns in specific scenarios, such as periodic power-frequency interference in industrial sensors. From a spectral perspective, white noise is uniformly distributed across all frequency components, while colored noise (e.g., pink noise) exhibits

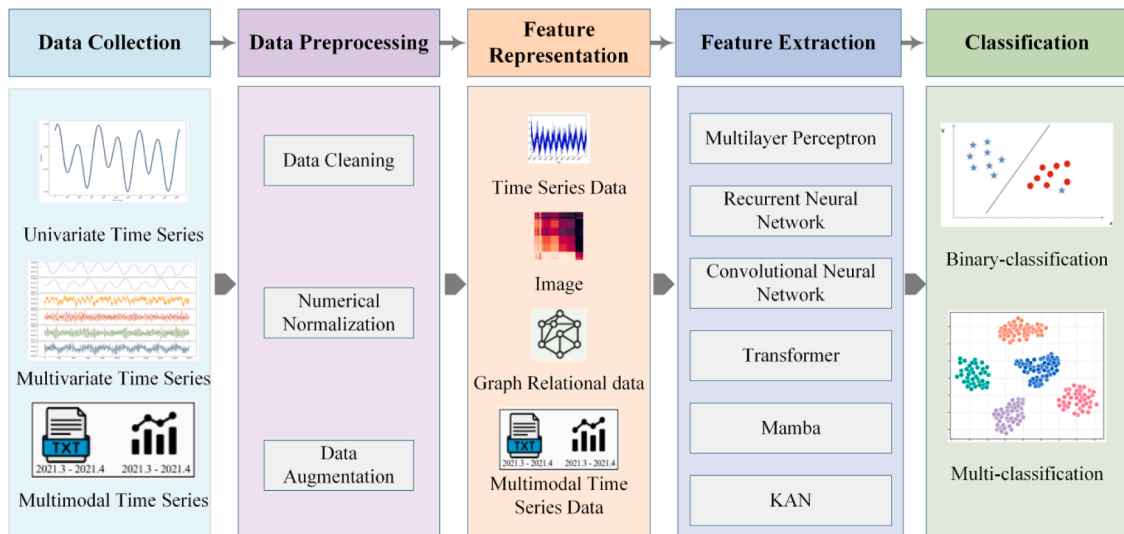


Fig. 3. The flowchart of the Time Series Classification Task.

power decay according to a power law as frequency increases.

When extracting features from time-series data, noise can obscure essential features or introduce irrelevant ones. For example, when classifying wind turbine vibration signals, the high-frequency impact signals from gearbox failures may be obscured by the low-frequency aerodynamic noise produced during turbine operation, leading to misclassification based on frequency-domain features. Wavelet denoising and modal decomposition (MD) are commonly used to mitigate the adverse effects of noise. Still, such preprocessing can compromise temporal signal details, requiring a trade-off between noise reduction and feature preservation.

2.2.3. Non-stationarity

Non-stationarity refers to the changing statistical properties of a time series, such as mean, variance, and frequency components. Alterations in internal system parameters or external environmental disturbances can cause it. For example, aging equipment in industrial manufacturing causes process parameter drift, which gradually increases the variance of sensor signals. The defining feature of non-stationarity is the variability of the data distribution over time, which starkly contrasts the constant statistical properties of stationary sequences.

In time series classification tasks, non-stationarity diminishes a model's generalization capability. Classifiers trained on historically stationary data tend to perform poorly when applied to non-stationary sequences due to shifts in feature distributions. For example, power load sequences in commercial buildings exhibit mean jumps and variance expansion during summer air conditioning operation. Models trained on winter data would misclassify normal summer peaks as anomalies. Time-varying parameter models (e.g., dynamic factor models) or modal decomposition are necessary to capture non-stationary features.

2.2.4. Pattern drift

Pattern drift refers to structural changes in key patterns within time series, such as periodicity, trend components, and anomaly patterns. These changes arise from sudden shifts in the system's state or external interventions. Pattern drift can be categorized as gradual or abrupt. Drift can be quantified mathematically by detecting differences in distributions in feature space (e.g., Kullback-Leibler divergence) or by identifying changes in key pattern statistics (e.g., variations in period length).

The core characteristic of this phenomenon is discontinuous pattern variation, which distinguishes it from gradual statistical changes associated with non-stationarity. Its impact on classification tasks is evident when models trained on static data struggle to adapt to pattern drift, resulting in a consistent decline in classification accuracy. For example, in medical time series monitoring, the progression of an ICU patient from early infection to sepsis is manifested by pattern drift in vital signs, which appears as a shift in heart rate variability frequency domain features, from high-frequency dominance (normal) to low-frequency dominance (pathological state). Traditional classification models that rely on fixed-band energy ratios fail due to this pattern transition. To address this issue, one must either decompose the time series or extract robust features that are insensitive to drift, such as similarity features based on dynamic time warping (DTW) or frequency-domain features from wavelet transforms, which capture long-term trends, rather than relying solely on raw time-series values.

2.2.5. Data heterogeneity

Data heterogeneity refers to time series that comprise multi-source, multi-type, and multi-scale observational data. This data arises from the multidimensional representation requirements of complex systems. At the physical level, heterogeneity manifests as differences in sensor types (e.g., numerical signals from vibration accelerometers versus image sequences from surveillance cameras). Semantic heterogeneity reflects dimensional differences in feature spaces (e.g., traffic volume metrics versus sentiment polarity in news texts within traffic time

series). The core characteristics of heterogeneous data are the semantic gap between modalities and asynchronous sampling.

Data heterogeneity in time series classification tasks introduces challenges in modal alignment and heterogeneous feature fusion. There are no unified standards for time calibration of asynchronously sampled data. For example, forced synchronization of ECG (1 kHz) and respiratory signals (20 Hz) in medical scenarios introduces interpolation artifacts that distort classification features. Visual features from images (e.g., edge detection results for equipment cracks) and frequency-domain features from vibration signals cannot be directly mapped. Traditional concatenation methods fail to capture cross-modal correlations, necessitating emerging multimodal learning techniques to achieve modal alignment and fusion. These techniques enhance the efficient extraction and complementary utilization of heterogeneous features.

2.3. Time series data augmentation methods

Due to the inherent characteristics of time series data and the practical requirements of time series classification tasks, commonly used feature augmentation methods for time series classification include time-frequency data augmentation, modal decomposition-based data augmentation, learning-based data augmentation, and image-based data augmentation.

2.3.1. Time-frequency data augmentation methods

1. Window Cropping

Window cropping randomly extracts consecutive sub-sequences of varying lengths w from the original time series $\hat{\mathbf{X}} = [x_1, x_2, \dots, x_T]$, thereby augmenting the data through local feature sampling. This method assumes that local patterns repeat over time and that key classification features are embedded within short time fragments. If the starting index of the cropping window is s , then the sub-sequence is $S = [x_s, x_{s+1}, \dots, x_{s+w-1}]$, where $s \sim U(1, T - w + 1)$. For a sequence of length T , theoretically, $T - w + 1$ sub-sequences can be generated. Window cropping alleviates the few-shot problem by increasing the number of samples and reducing computational complexity for long sequences. In time series classification tasks, a majority voting strategy is commonly used to aggregate the results of the sub-sequences, which enhances the model's robustness against noise and local anomalies.

2. Window Warping

Window warping uses Dynamic Time Warping principles to perform non-linear scaling on local segments of the original time series. It simulates phase shifts or velocity variations in the signal. After selecting a time range $[t_a, t_b]$, a new sequence $\mathbf{X}'$ is generated using scaling factors α to ensure that non-warped regions remain unchanged, where $x'_t = x_{\lfloor t/\alpha \rfloor}$, $t \in [t_a, t_b]$. The time warping function is defined as $w(t) = t + \delta(t)$, where $\delta(t) \sim N(0, \sigma^2)$ and $|\delta'(t)| < \varepsilon$, with first-order derivatives constrained, and ε is set to ensure sequence continuity through cubic spline interpolation. Window warping enhances the model's ability to adapt to temporal misalignments, including sampling clock offsets between sensors and operational speed fluctuations in devices. When combined with window cropping, input consistency can be maintained by adjusting the length of the warped sequence.

3. Sign Flipping

The sign-flipping technique generates a new sequence $\mathbf{X}'$ by inverting the sign of the original signal by $x'_t = -x_t$. Assuming the signal exhibits symmetry in both positive and negative amplitude directions, so the method preserves critical pattern recognition. It preserves morphological features, such as waveform periodicity and the positions of extremal points, while only altering the amplitude polarity. So it is appropriate for classification activities where shape sensitivity is crucial, but amplitude direction is less significant.

4. Noise Injection

Noise injection adds Gaussian, spike, or trend components to the original sequence. The formula defines it as $\mathbf{X}' = \mathbf{X} + \mathbf{G}$, where $\mathbf{G}$ is the noise sequence. Gaussian noise is defined as $g_t \sim \mathcal{N}(0, \sigma^2)$, and spike noise is defined as $g_t = \eta \cdot \delta(t - t_0)$, where η is the random amplitude and t_0 is the random position. The trend component is defined as a linear function as $g_t = k_t + b$, where k_t is the slope and b is the intercept. By simulating sensor noise or environmental disturbances, the model learns to recognize noise-resistant features. Impulse noise enhances the sensitivity of anomaly detection models to sudden failures, and linear trend functions simulate drift in non-stationary sequences.

5. Spectral Perturbation

Spectral perturbation transforms the time-domain sequence $\mathbf{X}$ into a frequency-domain representation using a Fourier transform as $\mathbf{F} = f(\omega_k) = A(\omega_k)e^{j\theta(\omega_k)}$. In this representation, $A(\omega_k)$ denotes the amplitude spectrum, and $\theta(\omega_k)$ denotes the phase spectrum. After applying random masking or amplitude adjustments to the spectral components, an inverse transform generates a new sequence. The perturbed spectrum is $\mathbf{F}' = f'(\omega_k) = f(\omega_k) \cdot (1 + \gamma \cdot m(\omega_k))$, where $m(\omega_k)$ is the binary mask that randomly preserves some frequency bands and γ is the perturbation intensity. This method forces the model to learn robust features across frequency bands, thereby preventing overfitting to specific frequency components. Preserving perturbations in the phase spectrum maintains the fundamental shape of the time-domain waveform.

6. Short-Time Fourier Transform Disturbance

The Short-Time Fourier Transform (STFT) perturbation method generates enhanced samples by distorting time-frequency features through frequency or time masking. Frequency masking involves zeroing consecutive frequency bands, while time masking involves zeroing consecutive time steps. This method is achieved by applying the STFT to a sequence to produce a time-frequency matrix $\mathbf{S}(t, f)$. A typical method, such as SpecAugment, performs frequency and time masking on the Mel spectrum by zeroing consecutive frequency bands and time steps, respectively. Given the original time-frequency matrix $\mathbf{S} \in \mathbb{R}^{T \times F}$, the frequency masking region $[f_1, f_1 + f]$, and the time masking region $[t_1, t_1 + t]$, the augmented matrix becomes

$$\mathbf{S}'(t, f) = \begin{cases} 0, & f \in [f_1, f_1 + f] \text{ or } t \in [t_1, t_1 + t] \\ \mathbf{S}(t, f), & \text{others} \end{cases} \quad (1)$$

This method improves the robustness of the model in the event of partial information loss by simulating real-world scenarios involving missing frequency bands or temporal interruptions, such as sensor failures. It applies to tasks like speech recognition and time-frequency analysis of vibration signals.

2.3.2. Decomposition-based data enhancement methods

The sequence data x_t is decomposed into trend τ_t , seasonal s_t , and residual components r_t using methods such as Seasonal-Trend Decomposition using Loess (STL), where $x_t = \tau_t + s_t + r_t$. The weights of each component can be adjusted or the residuals replaced to generate a new sequence $x'_t = \alpha \cdot \tau'_t + \beta \cdot s'_t + r'_t$, where τ'_t, s'_t represents the adjusted trend and seasonal components, r'_t denotes the new residuals $r'_t = \sum_{i=1}^p \phi_i r'_{t-i} + \epsilon_t$ generated via an autoregressive model, and $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$. This method separates the deterministic and random components of the sequence and then reconstructs a new sample that preserves the statistical characteristics of the original data.

Another commonly used time series enhancement method is modal decomposition. It breaks down the original sequence into several physically meaningful components, which are then manipulated and recombined independently to generate enhanced samples that align

with the temporal dynamic characteristics. It integrates decomposition theories from signal processing, such as time-frequency analysis and linear algebra decomposition, with data augmentation requirements. It makes it suitable for non-stationary and multi-component mixed time series data. For example, Empirical Mode Decomposition (EMD) decomposes a time series $\tilde{\mathbf{X}}$ into n intrinsic mode functions (IMFs) and a residual term $r(t)$, where $X(t) = \sum_{i=1}^n \text{IMF}_i(t) + r(t)$. Each IMF satisfies the condition that the number of extrema equals the number of zero crossings and has a local mean of zero. Enhancement generates new sequences through the following operations: (1) modal reorganization, which involves randomly selecting partial IMF components (e.g., retaining the high-frequency IMF corresponding to fault features) while discarding noise-dominated components, and (2) modal perturbation, which involves adding band-limited noise (e.g., $\text{IMF}'_i(t) = \text{IMF}_i(t) + \sigma \cdot \text{IMF}_i(t) \cdot \sin(2\pi f_0 t)$) to IMF components to simulate signals with varying fault severity. Let the enhanced IMF set be $\{\text{IMF}'_1, \dots, \text{IMF}'_k\}, k \leq n$, the residual term be $r'(t) = r(t) + \delta(t)$, and the trend perturbation be $\delta(t)$. Then, the enhanced sequence is $X'(t) = \sum_{i=1}^k \text{IMF}'_i(t) + r'(t)$. Modal decomposition adaptively decomposes non-stationary signals while preserving features at various scales. Selectively enhancing key modes improves the classification model's sensitivity to target features.

Additionally, variational modal decomposition enhances precise frequency component separation and prevents modal aliasing. Singular value decomposition improves the extraction of low-rank signals, enabling feature enhancement under strong noise conditions. Wavelet transform enhancement facilitates multi-scale, spatiotemporal feature manipulation with greater adaptability. Time-series decomposition-based data augmentation methods are suitable for enhancing non-stationary sequences. Compared to direct time-domain augmentation, these methods have the advantage of preserving the physical semantics of the original data through component-level operations, thereby avoiding feature distortion caused by blind perturbation.

2.3.3. Learning-based data enhancement methods

Learning-based time series data augmentation methods use neural network models to learn the characteristics of the distribution and patterns. This process generates augmented samples with authentic temporal dependencies. Compared to traditional signal processing methods, such as window clipping and spectral perturbation, these methods automatically capture high-dimensional, nonlinear temporal patterns. It makes them suitable for data augmentation in complex dynamic systems. End-to-end training or generative modeling eliminates the need for manual feature engineering, shifting from rule-driven to data-driven augmentation paradigms.

1. Generative Adversarial Network Enhancement

Generative adversarial networks (GANs) generate temporal data through adversarial training between a generator G and a discriminator D . The generator receives random noise z or conditional information c , such as class labels, and produces simulated sequences $\tilde{\mathbf{X}} = G(z|c)$. The discriminator distinguishes between the real sequence $\mathbf{X}$ and the generated sequence $\tilde{\mathbf{X}}$. The loss function is defined as

$$\min_G \max_D \mathbb{E}_{\mathbf{X} \sim p_{\text{data}}} [\log D(\mathbf{X})] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))]. \quad (2)$$

A series of improved models has been developed to address the characteristics of time-series data. For example, the Recurrent GAN (RGAN) model [24] uses RNN structures for its generator and discriminator to capture long-range temporal dependencies. Time-GAN [25] introduces an embedding network E and supervised loss. It ensures temporal consistency through stepwise conditional distribution learning. The generator loss is defined as $L_G = L_{\text{adv}} + \lambda_1 L_{\text{step}} + \lambda_2 L_{\text{rec}}$, where L_{adv} represents the supervised loss, and L_{step} constrains

the temporal conditional probability, L_{rec} ensures embedding space reconstruction accuracy. The TimeGAN generator produces sequences $h_t = \text{LSTM}(h_{t-1}, x_{t-1})$ via recursive mapping, where $\tilde{x}_t = \text{MLP}(h_t)$. The embedding network E maps sequences to the latent space $z_t = E(x_t, h_t)$ and aligns the embedding distribution z_t with real sequences via adversarial training. GANs can capture complex temporal dynamics and generate samples that exhibit real data's distributional characteristics and dependencies. They support the conditional generation of y samples, making them suitable for scenarios with class imbalance.

2. Variational Autoencoder Enhancement

Variational autoencoders (VAEs) map sequences $\mathbf{X}$ to latent space $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{X})$ via the encoder E and reconstruct $\hat{\mathbf{X}} = D_\theta(\mathbf{z})$ through $\mathbf{z}$ by the decoder D , optimizing for a variational lower bound $\mathcal{L} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{X})}[\log p_\theta(\mathbf{X}|\mathbf{z})] - \text{KL}(q_\phi(\mathbf{z}|\mathbf{X}) \parallel p(\mathbf{z}))$. During data augmentation, new sequences $\mathbf{X}' = D_\theta(\mathbf{z}')$ are generated by sampling from the latent space $\mathbf{z}' \sim \mathcal{N}(m, s^2)$, where m, s represent the mean and variance of the encoder output, respectively. The encoder outputs the distribution parameters of the latent variables $\mu, \log \sigma^2 = E(\mathbf{X})$, and $\mathbf{z} = \mu + \sigma \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. The decoder reconstructs sequences through a recurrent layer that satisfies $\hat{x}_t = \text{GRU}(h_{t-1}, z_t)$, $h_t = \text{GRU}(h_{t-1}, \hat{x}_{t-1})$, where $\text{GRU}(\cdot)$ denotes a gated recurrent unit. VAEs can learn compact representations of sequences, and interpolation in latent space can generate semantically coherent transition samples. Furthermore, the KL divergence constraint prevents generator overfitting and ensures sample diversity.

3. Auto-Regressive Network Enhancement

Auto-Regressive models generate time series based on causality $p(x_t|x_{t-1}, \dots, x_1)$. These models assume that the value at the current time step x_t depends solely on historical values $[x_{t-1}, \dots, x_{t-k}]$. These models use deep learning techniques, such as long short-term memory (LSTM) and transformer, to model conditional probabilities. Examples of typical models include WaveNet [26], which uses expanded causal convolutions to capture long-range dependencies while ensuring real-time generation. The Transformer model [19] generates sequences through the self-attention mechanism that capture global dependencies. During Transformer generation, the output at each time step depends on all previously generated tokens as

$$\tilde{x}_t = \text{MLP}\left(\sum_{i=1}^{t-1} \alpha_{t,i} \cdot \text{Embedding}(\tilde{x}_i)\right)$$

$$\alpha_{t,i} = \frac{\exp\left(\text{Embedding}(\tilde{x}_t) \cdot \text{Embedding}(\tilde{x}_i)\right)}{\sum_{j=1}^{t-1} \exp\left(\text{Embedding}(\tilde{x}_t) \cdot \text{Embedding}(\tilde{x}_j)\right)}, \quad (3)$$

where $\text{MLP}(\cdot)$ represents a feedforward neural network, $\text{Embedding}(\cdot)$ denotes the Transformer encoder, and Transformer can efficiently model long-range dependencies. Furthermore, it facilitates parallel generation, as evidenced by the autoregressive sampling employed in Transformers, thereby enhancing the efficiency of large-scale data augmentation.

2.3.4. Image-based data enhancement methods

Time series imaging is a technique that converts one-dimensional temporal data into two-dimensional images using specific mapping rules. The core concept leverages the spatial features of images, such as pixel values, texture, and color distribution, to preserve the dynamic characteristics, statistical properties, and underlying patterns of temporal data. It can transform temporal analysis problems into image processing or computer vision challenges. Furthermore, by employing well-established image enhancement techniques from computer vision, like rotation, cropping, and noise injection, to create varied samples,

indirect augmentation becomes possible. Image-based Data Enhancement increases data diversity by utilizing the extensive enhancement strategies available in the image domain.

1. Gram Angular Field Enhancement

The Gram Angle Field (GAF) technique maps one-dimensional time series into two-dimensional images. Using trigonometric identities, it encodes the numerical information of the time series into pixel relationships within the image, thereby preserving the sequence's temporal dynamics and periodic characteristics. The core concept involves transforming the time series into a matrix with angular correlations, which enables visualization of long-term dependencies within the sequence. GAF maps one-dimensional time series $\mathbf{X} = [x_1, x_2, \dots, x_T]$ onto two-dimensional matrices using trigonometric identities $\cos^2\theta + \sin^2\theta = 1$ or sum-difference formulas $\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$ to construct Summation fields (GASFs) or difference fields (GADF). The specific steps are: (1) Normalize the time series as $[-1, 1]$ and calculate the angular mapping $\theta_t = \arccos(x_t)$. (2) Construct the $T \times T$ matrix, where the elements are $m_{ij} = \cos(\theta_i \pm \theta_j)$, taking "+" for GASF and "-" for GADF.

GAF preserves the periodicity and trend characteristics of the time series. Diagonal textures in GAF images correspond to sequence periods. It transforms the temporal dependencies of one-dimensional time series into spatial relationships in two-dimensional images. Additionally, image enhancement operations correspond to phase shifts in the time series, and scaling corresponds to changes in period length.

2. Recurrence Plot Enhancement

The Recursive Phase (RP) method uses state space reconstruction techniques to visualize the dynamic characteristics of time series as a two-dimensional binary image, representing the recursive behavior of phase points within the state space. RP is used to analyze nonlinear features, such as periodicity and chaos, in time series. Determining the proximity of phase points within the state space allows RP to construct a two-dimensional image, making it suitable for characterizing the dynamic properties of time series. The state space of a time series $\mathbf{X}$ is reconstructed as $\mathbf{X}' = [x_t, x_{t+\tau}, \dots, x_{t+(m-1)\tau}]$, where τ represents the time delay and m denotes the embedding dimension. RP is defined as

$$RP(i, j) = \begin{cases} 1, & \|Y_i - Y_j\| < \epsilon \\ 0, & \text{others} \end{cases}, \quad (4)$$

where ϵ represents the distance threshold. This method visualizes recursive patterns in time series. Periodic behavior appears as diagonal stripes, while chaotic states appear as disordered spots. This facilitates the discovery of hidden dynamic rules within time series.

3. Markov Transition Field Enhancement

Markov Transition Fields (MTF) generate new sequences that conform to the statistical patterns of original data by constructing probabilistic models of sequential state transitions. This approach utilizes the memoryless and steady-state distribution properties of Markov chains. MTFs treat time series as Markov processes in discrete state spaces, wherein state transitions at each time step depend solely on the preceding state. MTF is suitable for time-series data with stable statistical properties. Simulating state transition processes enhances data diversity while preserving the sequence's long-term dependencies and dynamic patterns.

Divide the continuous time series $\mathbf{X} = [x_1, x_2, \dots, x_T]$ into discrete states $\mathbf{S} = \{s_1, s_2, \dots, s_N\}$. Common methods include equidistant partitioning, such as dividing the numerical range $[\min(\mathbf{X}), \max(\mathbf{X})]$ into equal N segments, and adaptive partitioning based on clustering, such as K-means clustering of sequence feature vectors. Next, construct a transition matrix by counting the frequency of state transitions between

adjacent time steps to generate a transition probability matrix $P \in \mathbb{R}^{N \times N}$. In this matrix, $P_{ij} = \frac{\text{count}(s_i \rightarrow s_j)}{\sum_{k=1}^N \text{count}(s_i \rightarrow s_k)}$ represents the probability of transitioning from state s_i to state s_j . Validation $\pi P = \pi$ ensures that the model converges, π is the steady-state distribution vector, meaning that the long-term state probabilities remain unchanged over time.

The Markov transition function preserves the statistical properties of time series data by generating state transition probabilities for the sequence. The MTF's steady-state distribution aligns with the original data, ensuring that visualized samples adhere to the dynamic patterns of real data while remaining suitable for long-term dependency modeling. The MTF effectively handles time series with periodic or trend-based characteristics by capturing short-term dependencies through Markov chains. Additionally, the transition matrix intuitively reflects the logic of state transitions, enabling the integration of domain knowledge to adjust and enhance strategies.

2.4. Current state of the review on time series classification

A review of time series classification is crucial for integrating and applying emerging technologies. However, current literature on time series classification is scarce, revealing significant gaps in rapidly evolving research and application scenarios. While existing reviews comprehensively critique time series classification methods, most focus on model enumeration rather than systematic classification and evaluation. Introductions to emerging, deep-learning-based methods are scarce, and research on cutting-edge topics, such as multimodal sequence fusion and few-shot learning, is largely unexplored. Additionally, existing reviews lack diversity in their analysis of application scenarios for time series classification, with few integrated case studies and in-depth analyses across domains such as healthcare and industry.

Abanda et al. [31] constructed a classification framework for distance-based TSC methods early on. This framework encompasses three major categories: k-NN, distance feature transformation, and distance kernel construction. The researchers analyzed each method's technical pathways and application scenarios to provide a systematic framework for domain research. However, their review lacked study and commentary on deep learning methods and emerging paradigms, resulting in an incomplete perspective on time series classification. Gupta et al. [6] systematically summarized early time series classification approaches and categorized them into four groups: prefix-based, shape-based, model-based, and other. They aimed to provide a systematic framework and future research directions for time series classification. However, their discussion of deep learning and emerging paradigms was minimal. Haghighi et al. [27] systematically constructed a mathematical framework for functional time series (FTS) research spanning univariate to multivariate approaches. This framework integrates functional analysis, multivariate statistics, and time series analysis methods, demonstrating interdisciplinary applications through case studies in healthcare, industry, and environmental contexts. Nevertheless, the paper lacks a discussion of deep learning methods and fails to provide a systematic summary of the current state of time series classification research.

With the advancement of deep learning, Fawaz et al. [1] systematically reviewed its application in TSC. Through empirical studies comparing nine deep learning architectures, the researchers developed a unified classification framework and validated its effectiveness through experimentation. However, the paper focuses on introducing the method and does not cover new time series models or emerging technologies, making it insufficiently timely to meet researchers' needs. Ruiz et al. [28] constructed a benchmark comparison framework for MTSC algorithms, systematically evaluating the practical performance of 16 algorithms to guide algorithm selection in real-world applications. However, the paper's focus on specific scenarios and time series classification methods is limited. It does not include new models, such as

Transformers, nor does it provide sufficient coverage of emerging paradigms. Founmani et al. [2] reviewed deep learning approaches for time series classification and external regression, categorizing methods from a neural network architecture perspective. These include self-supervised learning and data augmentation techniques and applications such as human activity recognition and satellite Earth observation. However, the paper excludes emerging topics such as multimodal and few-shot learning. Additionally, its application scenarios lack representativeness and fail to provide a comprehensive evaluation perspective for time series classification.

Additionally, some reviews focus solely on individual methods or application domains within time series classification. For example, Zhang et al. [3] concentrate on applying self-supervised learning (SSL) in time series analysis, categorizing methods into generative, contrastive, and adversarial approaches. However, their paper primarily addresses SSL and lacks a comprehensive evaluation of time series classification. It also insufficiently explores downstream application scenarios. Jin et al. [7] systematically reviewed the application of graph neural networks (GNNs) across four major time series analysis tasks: prediction, classification, anomaly detection, and interpolation. The authors constructed a dual-dimensional classification framework based on task orientation and methodology. This framework reveals the technical evolution and future directions of GNNs for time series analysis. However, this paper exclusively focuses on GNN applications in time series analysis and lacks a discussion of time series classification. Theissler et al. [29] reviewed explainable AI (XAI) applications in TSC and proposed a classification framework based on time points, subsequences, and instances. They discussed attribution methods, attention mechanisms, prototype comparison, and other explanation techniques, covering medical and industrial applications. However, the paper emphasizes explanation methods while inadequately classifying and reviewing deep learning models, and it lacks coverage of emerging technologies. Wang et al. [4] systematically reviewed non-deep learning techniques in TSC for biomedical applications. They analyzed common algorithm types and performance metrics to provide methodological guidance. However, the study is heavily biased toward biomedical contexts, limiting its applicability. Furthermore, the discussed methods are outdated and offer limited insight into advancing TSC classification. Li et al. [30] systematically reviewed the application of few-shot learning in biomedical time series and categorized methods into data-driven, model-driven, metric-driven, optimization-driven, and hybrid-driven approaches. The paper analyzes the application of these methods across various modalities, including electroencephalograms (EEG) and ECG, and discusses the challenges in knowledge transfer across subjects and datasets. The core innovation is establishing a classification framework for few-shot learning in biomedical time series, which provides references for clinical diagnosis and personalized medicine. However, its applicability is limited to biomedical domains, thereby restricting its generalizability.

In summary, the current review of time series classification has yet to establish a scientific framework for classifying this field. It has failed to provide a comprehensive, systematic evaluation. References [31,6], and [27] focus primarily on traditional time series classification methods. References [1,28], and [2] review deep learning-based time series classification approaches but lack the latest temporal models and pay limited attention to emerging paradigms. Reference [3] focuses on self-supervised learning models and lacks a comprehensive evaluation perspective for time series classification. References [7] and [29] focus on review studies of graph neural network models and model interpretability. References [4] and [30] primarily focus on time series classification methods in the biomedical field. These references lack a comprehensive evaluation perspective and exhibit limitations in generalizability. Unlike existing reviews that typically adopt a linear, algorithm-centric perspective, we propose a three horizontal and three vertical structure to bridge the disconnect between theoretical models and application constraints. By intersecting the horizontal dimension of

methodological evolution with the vertical dimension of data characteristics, this framework serves as a heuristic tool to expose structural research gaps. The results of the content organization of each review article are summarized in Table 1.

Where, $\sqrt{}$ indicates inclusion, $\square$ indicates partial inclusion, $\times$ indicates exclusion.

The statistical results in Table 1 suggest that current reviews of time series classification tend to offer limited discussions of classification methods and lack comprehensive evaluation perspectives. Additionally, existing reviews often lack integrated case studies and in-depth analyses across various domains, including healthcare and industry. Overall, the existing literature exhibits shortcomings in technical comprehensiveness, scientific rigor of evaluation, application depth, and theoretical foresight. There is an urgent need for more systematic, practice-oriented review literature that bridges the gap between academic research and engineering applications. Therefore, this paper proposes a multi-level review framework of three horizontal and three vertical dimensions. The three horizontal dimensions encompass traditional analysis-based methods, deep learning-based methods, and few-shot learning-based methods. The three vertical dimensions traverse data, methods, and applications. The framework emphasizes emerging deep learning model approaches and few-shot learning paradigms. It aims to provide researchers with a rapid overview of the latest advancements in time series classification and offer a more insightful and comprehensive review perspective. It is worth noting that while this review specifically targets Time Series Classification, the underlying logic of the proposed framework, intersecting the evolutionary timeline of representations with the fundamental constraints of data and applications, possesses intrinsic extensibility. The challenges identified (e.g., data scarcity, non-stationarity) and the methodological paradigms discussed (e.g., temporal dependency modeling, few-shot adaptation) are universal foundations applicable to broader temporal tasks, including time series forecasting and anomaly detection.

3. Review strategy

This review focuses on time series classification using deep learning as its core research subject. The goal is to systematically integrate the development of traditional methods and deep learning technologies to create a multidimensional knowledge framework that encompasses data characteristics, research methods, and scenario applications. To ensure the scientific rigor and timeliness of the review, we develop a rigorous search strategy that defines the scope of retrieval across four dimensions: domain boundaries, data types, methodological scope, and temporal scope.

- (1) Domain boundaries: The primary focus is on time series classification tasks within supervised learning scenarios, while also considering traditional time series classification methods and

those tailored for few-shot scenarios. We exclude peripheral areas such as unsupervised clustering and reinforcement learning-driven sequence decision-making.

- (2) Data types: It includes univariate time series, multivariate time series, and graph-structured time series data. These cover typical application scenarios such as medical signal classification, high-resolution radar range image recognition, industrial fault diagnosis, and network traffic detection.
- (3) Methodological Scope: Centered on deep learning models, it incorporates traditional approaches and emerging few-shot learning methods.
- (4) Timeframe: The focus is on literature published between January 2020 and June 2025, with foundational classical time series classification methods used to trace historical evolution. It ensures a balance between cutting-edge and fundamental research.

Six major academic databases, such as IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, Wiley, and Google Scholar, are selected as retrieval tools. Combined with Web of Science's citation tracking functionality, a composite retrieval strategy of "core database precision search + citation network expansion" is formed. To enhance retrieval accuracy, Boolean logic operators were used to combine keywords such as "time series classification", "time series analysis", "deep learning", "feature representation", and "data augmentation". The language was set to English, and non-academic literature, such as conference abstracts and short reviews, was excluded. Approximately 280 relevant papers were initially identified, and after further screening, about 130 were selected as the primary subjects of the review. This chapter systematically outlines the review methodology. Through a defined search strategy and a rigorous literature screening process, the chapter ensures the relevance of the research subjects and the reliability of the literature. This approach avoids omitting key studies while eliminating duplicative, low-level work, laying a solid foundation for the in-depth reviews in subsequent chapters.

4. Time series classification methods

As time series data has become more complex and deep learning technologies have advanced rapidly, time series classification methods have developed quickly. A multi-level review framework has been proposed to provide a clarified overview. The framework is structured as three horizontal and three vertical dimensions. First, we categorize time series classification methods based on the temporal progression of research subjects, including traditional analysis-based methods, deep learning-based methods, and few-shot learning-based methods. Next, we perform a secondary classification based on extracted temporal feature patterns. Figure 4 illustrates the detailed classification framework.

Table 1
The Overview of Existing Time Series Classification Reviews.

Reference	Years	Traditional Methods	Machine Learning Methods	Deep Learning Methods	Few-shot Learning Methods	TSC Applications
Abanda et al. [31]	2019	$\square$	$\times$	$\times$	$\times$	$\times$
Fawaz et al. [1]	2019	$\times$	$\times$	$\square$	$\times$	$\times$
Gupta et al. [6]	2020	$\sqrt{}$	$\sqrt{}$	$\square$	$\times$	$\times$
Ruiz et al. [28]	2021	$\sqrt{}$	$\sqrt{}$	$\square$	$\times$	$\times$
Theissler et al. [29]	2022	$\sqrt{}$	$\sqrt{}$	$\square$	$\times$	$\times$
Wang et al. [4]	2022	$\sqrt{}$	$\sqrt{}$	$\times$	$\times$	$\square$
Zhang et al. [3]	2024	$\times$	$\times$	$\sqrt{}$	$\times$	$\times$
Jin et al. [7]	2024	$\times$	$\times$	$\square$	$\times$	$\times$
Hagbin et al. [27]	2024	$\sqrt{}$	$\times$	$\times$	$\times$	$\sqrt{}$
Foumani et al. [2]	2024	$\times$	$\times$	$\square$	$\times$	$\sqrt{}$
Li et al. [30]	2025	$\times$	$\square$	$\square$	$\sqrt{}$	$\square$
Ours	2025	$\sqrt{}$	$\sqrt{}$	$\sqrt{}$	$\sqrt{}$	$\sqrt{}$

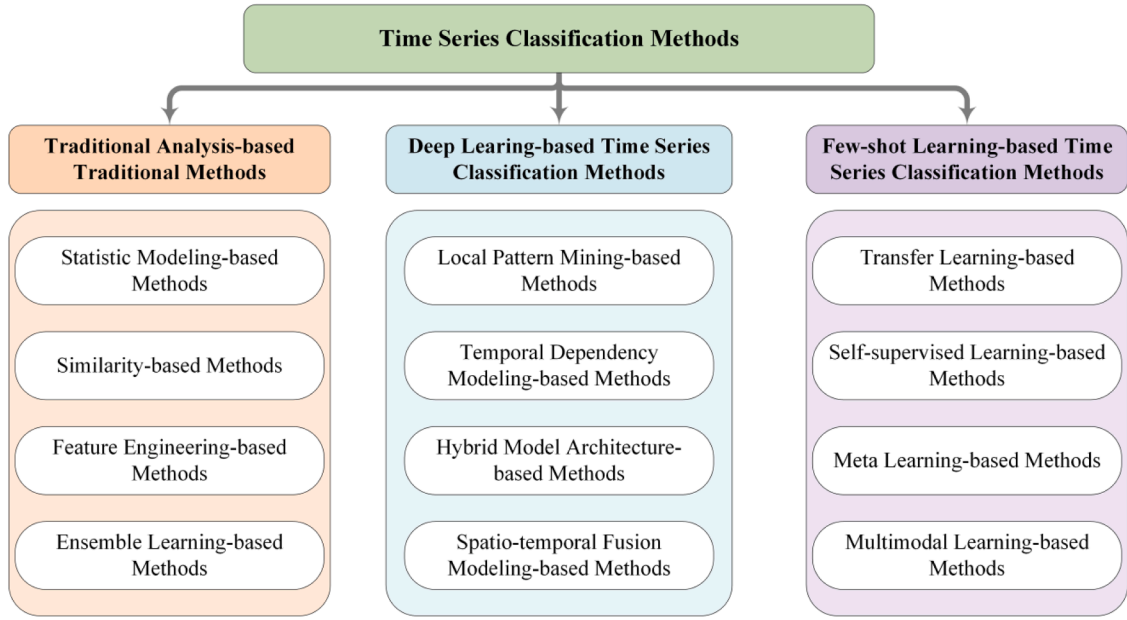


Fig. 4. Classification framework for time series classification methods.

4.1. Traditional analysis-based time series classification methods

Traditional time series classification methods focus on statistical and manually designed features, relying on domain knowledge to develop classification logic. While these methods still hold value in specific scenarios during the deep learning era, they face challenges such as cumbersome feature engineering and limited generalization capabilities. Therefore, traditional analysis-based time series classification methods can be categorized as statistical modeling-based methods, similarity-based methods, feature engineering-based methods, and ensemble learning-based methods.

4.1.1. Statistical modeling-based methods

Statistical modeling-based methods utilize statistical theory and models to capture patterns, structures, or distributional characteristics in time series data, thereby enabling classification. Typically, classifiers are constructed through parameter estimation or nonparametric inference based on data's statistical properties (such as mean, variance, autocorrelation, etc.) and assumptions about probability distributions.

Gupta et al. [32] proposed a fault-tolerant early-stage time series classification method based on FECM. This method uses ARIMA models to identify fault sequences and Gaussian processes to estimate the minimum required length. It also uses k-means clustering to calculate feature correlations. Then, a partial order set is used to select key features for classification. Johnpaul et al. [33] proposed a trend-based global feature representation primitive. This method extracts global statistical characteristics (e.g., slope, curvature) from time series trends and integrates fuzzy logic to construct classification features. However, its ability to capture trends in short-period time series data is limited. Eftimov et al. [34] proposed a landscape analysis-based benchmark selection process for time series datasets. They used TSFresh to extract statistical and frequency-domain features, mapped high-dimensional features onto a two-dimensional grid via self-organizing maps, and generated dataset clusters through hierarchical clustering. Finally, they selected representative datasets based on coverage matrices and thresholds to address redundancy and insufficient representativeness in traditional benchmarks. Taranenko et al. [35] captured dynamic dependencies in non-stationary signals by analyzing the time-frequency correlations of time series using wavelet autocohérence and autocohérence. Introducing wavelet autocohérence into time series

correlation analysis provides a new tool for analyzing non-stationary time series and enhances the accuracy of dynamic system modeling. While wavelet autocohérence performs better on non-stationary data, it involves higher computational complexity.

Statistical modeling approaches have rigorous theoretical foundations and strong parameter interpretability. They enable the quantification of prediction uncertainty through confidence intervals. However, their limitations include stringent assumptions about data distribution (e.g., ARIMA requires stationarity), difficulty in modeling highly nonlinear or long-range dependencies, and reliance on manual feature engineering for high-dimensional sequence classification, with adaptability constrained by data complexity.

4.1.2. Similarity-based methods

Similarity-based methods use morphological similarity between sequences as the core discriminative criterion. They quantify the similarity between samples by defining distance metrics (e.g., Euclidean distance, Manhattan distance, Dynamic Time Warping, etc.), and then combine these with nearest neighbor rules (e.g., KNN) or clustering strategies to achieve category classification.

Ye et al. [36] proposed defining time-series shapelets as sub-sequences that are maximally representative of categories. They screen for optimal sub-sequences using information gain and construct a decision tree classifier for classification. All possible subsequences are extracted using a sliding window. Then, it uses decision trees to split nodes based on sub-sequence distance thresholds to achieve classification. This approach provides novel primitives for time series data mining, advances the application of local feature analysis in classification tasks, and lays the groundwork for subsequent Shapelet-related research. Mueen et al. [37] extended traditional shapelets to logical shapelets, combining multiple sub-sequences via logical operators (AND, OR, NOT) to construct more complex classification rules. First, basic Shapelets are extracted. Then, composite rules are generated through logical combinations. Finally, decision trees or support vector machines (SVMs) map the logical expressions to classification decisions. These logical expressions enhance the representational power of Shapelets by enabling the capture of multi-feature interaction patterns. They also address the expressive limitations of traditional Shapelets in complex classification scenarios. However, optimizing logical rules on large-scale datasets is inefficient and struggles with high-dimensional

feature interactions.

Liang et al. [38] proposed a time series classification method based on class-specific shapelet learning (CSSL). Class-specific variable-length shapelets are automatically discovered by mining frequent sub-sequences via the Sequitur algorithm, clustering to select highly discriminative shapelets, and optimizing weights through logistic regression. The class-specific shapelet design reduces redundant features, and automatic length determination based on frequent sub-sequences eliminates the need for manual parameter tuning. However, frequent sub-sequence mining is sensitive to noise, and the computational complexity increases for high-dimensional time series. Li et al. [39] proposed a BSPCOVER-based time series classification framework. It generates candidate shapelets via Symbolic Aggregate Approximation (SAX), trims redundant patterns using Bloom filtering, screens similar patterns through non-metric distance, and ultimately models shapelet selection as a weighted bitmap coverage problem solved by heuristic algorithms. Schäfer et al. [40] proposed a time series classification method based on symbolic Fourier approximation (SFA). This approach converts time series into a frequency-domain representation, reduces dimensions through discretization, and constructs indices to enable fast similarity searches. By combining the frequency-domain compression capability of Fourier transforms with the noise resistance of symbolization, SFA enables the efficient storage and retrieval of high-dimensional time series data. However, the symbolization process may lose critical phase information, affecting exact matching. Lhache et al. [41] proposed a time series classification method based on local extremes dynamic time warping (LE-DTW). This method reduces the dimension by extracting local extreme points (minima, maxima, and start/end points) from the time series. It then separates these points into two sub-sequences by type. Next, dynamic time warping is applied to compute the similarity for each sub-sequence. Finally, it calculates the global distance via weighted summation. Shape features are preserved through parameter-free extreme point extraction and separation, and classification efficiency for long sequences is enhanced by integrating extreme point representation with DTW. Zhang et al. [42] proposed a belief Rényi divergence (BRD)-based time series classification method that combines fuzzy belief structures with Rényi information metrics to establish a novel distance measure. The proposed BRD is suitable for noisy and incomplete data scenarios because it can quantify uncertainty and variability in time series. BRD is more robust than traditional divergence metrics but more computationally complex for high-dimensional time series. Wu et al. [43] proposed the COPP-Miner time series classification algorithm, which mines top-k contrast order preservation patterns (COPPs) for classification. COPP-Miner compresses time series through extreme point extraction, generates candidate patterns via forward/backward mining, and employs group pattern fusion strategies with support rate calculations for efficient pruning. Experimental results demonstrate that COPP-Miner achieves superior classification accuracy and computational efficiency compared to conventional pattern mining methods. However, improvements are needed for multi-class classification and processing high-dimensional time series. Tan et al. [44] enhanced the Proximal Forest algorithm by incorporating Amended Dynamic Time Warping (ADTW), cost function tuning, and HYDRA dictionary transformation. The approach facilitates the measurement of similarity between entities using ADTW, cDTW, LCSS, and Minkowski distance, while enhancing computational efficiency by employing strategies that terminate the process early. However, its feature representation capability remains limited for high-dimensional data.

Similarity-based methods do not require complex feature engineering, enabling direct classification based on raw sequence morphology. These methods are adaptable to nonlinear and non-stationary sequences and feature intuitive principles and straightforward implementation. However, its limitations are significant, as distance calculations exhibit high computational complexity (particularly DTW, which shows quadratic complexity for long sequences), are sensitive to noise and

outliers, and the validity of distance metrics depends on the data characteristics. Additionally, classification performance can be easily compromised in high-dimensional multivariate sequences or ambiguous category boundaries.

4.1.3. Feature engineering-based methods

Feature engineering-based methods extract discriminative features, such as time-domain statistics, frequency-domain spectral features, and morphological features, from time series through manual or semi-automated processes. These methods can transform high-dimensional time series data into low-dimensional feature vectors used in traditional machine learning models, such as support vector machines, random forests, and gradient-boosted trees, to perform classification. These methods leverage domain knowledge or statistical methods to uncover latent patterns within sequences, thereby transforming time-series classification problems into conventional classification tasks within feature spaces.

Karan et al. [45] proposed a temporal classification method based on topological data analysis. It converts univariate time series into point clouds via temporal delay embedding to extract topological features and uses the support vector machine classification. However, it incurs high computational costs with high-dimensional data and has weak physical interpretability of topological features, requiring parameter tuning. León-López et al. [46] proposed an anomaly detection and classification framework based on hidden Markov models (HMMs). It trains HMMs using normal data, detects anomalous parcels via likelihood assessment, locates anomaly cycles through temporal segmentation, and classifies anomaly types using SVM. However, HMM parameter training is time-consuming, and classification accuracy for complex anomaly patterns is limited. Yahyaoui et al. [47] introduced MinR, a minimal time series dimensionality reduction technique. MinR standardizes time series, segments them into equal-length fragments, extracts the mean and standard deviation as features, and designs a new distance metric, MinRDist, which mathematical proofs demonstrate outperforms Partial Aggregate Approximation. The method's effectiveness was validated using a 1NN classifier. However, it cannot handle multivariate time series, relies on a fixed segmentation strategy, and has difficulty capturing complex patterns. Feremans et al. [48] proposed a pattern embedding-based classification framework named PETSC. PETSC extracts symbolized subsequences from time series via sliding windows, mines variable-length sequential patterns, and constructs a pattern occurrence frequency matrix. Ultimately, it classifies using a linear model (Elastic Net). Although multi-resolution ensembles enhance generalization capabilities, they remain sensitive to noise. Therefore, additional dimensionality reduction is necessary for high-dimensional time series. Baldan et al. [49] systematically analyzed the application of complexity metrics (e.g., Lyapunov exponents and sample entropy) in time series classification. They compared the performance of different complexity features across benchmark datasets. The study revealed that certain complexity indicators (e.g., permutation entropy) outperform others in nonlinear time series classification. Bradde et al. [50] proposed a multi-class classifier based on sparse centroids. The binary space is partitioned using a decision tree structure. Each node employs a sparse centroid classifier. This classifier selects the most discriminative feature subset. It enables fast time series classification with linear complexity. This method maps high-dimensional time series data to a low-dimensional sparse space, making it suitable for scenarios with resource constraints, albeit with slightly reduced accuracy. Erazo-Costa et al. [51] introduced a time series classification method based on fuzzy probability representation learning. It extends probability-weighted fuzzy time series (PWFTS) from traditional prediction tasks to classification scenarios. They constructed fuzzy time patterns and calculated weights using empirical probabilities, combining PWFTS features with classifiers such as random forests and support vector machines. Limitations include the inability to handle heteroscedastic time series and the necessity of manually tuning optimal window strategy parameters.

Fulcher et al. [52] employed a greedy algorithm to forward-select optimal feature subsets from numerous time series features and utilize Linear Discriminant Analysis (LDA) for classification. Predefined features (e.g., mean, autocorrelation, approximate entropy) are computed for each time series. The features that most significantly enhance classification accuracy are sequentially added via a greedy algorithm, culminating in the construction of an LDA-based classifier. However, feature computation requires significant computational resources, making it impractical for real-time applications. Additionally, it inadequately covers emerging temporal patterns (e.g., nonstationary signals), necessitating the continuous expansion of the feature library. Li et al. [53] proposed a classification method based on spectral envelopes and optimal scaling. Gaussian process modeling captures oscillatory patterns in time series. The spectral envelope characterizes the frequency-domain energy distribution. Furthermore, optimal scaling quantifies the temporal transition features between categories. These features are ultimately integrated via a Bayesian classifier. Integrating frequency-domain analysis with classification enables the spectral envelope and scaling parameters to provide an interpretable rationale for classification, thereby addressing the "black box" issue in time series classification. However, it is ineffective for non-periodic sequences and becomes computationally complex exponentially with the number of categories. Baldan et al. [54] proposed an interpretable multi-time-series classification framework. The framework extracts key temporal patterns through feature selection and dimensionality reduction and combines them with traditional models, such as decision trees, for classification. It enhances model interpretability. Feature-importance-based temporal pattern extraction addresses the interpretability issue of black-box models, and domain knowledge is leveraged to construct interpretable classification rules.

The advantages of feature engineering-based machine learning methods lie in their clear physical significance of features, strong model interpretability, high training efficiency on small-to-medium datasets, and stable generalization performance. The limitations include excessive reliance on domain expertise and feature design capabilities, difficulty automatically capturing complex, non-linear, and temporal patterns, high feature engineering costs when handling high-dimensional and heterogeneous time-series data, and weak adaptability to manually designed features to unseen pattern drift.

4.1.4. Ensemble learning-based methods

Ensemble learning-based methods enhance classification performance by constructing multiple heterogeneous or homogeneous base classifiers (such as decision trees trained on different feature subsets or variants of time-series models), then integrating their outputs through voting, weighted fusion, or stacking strategies. The core principle is leveraging the diversity of base models to counteract individual model biases and variance.

Shifaz et al. [55] proposed a TS-CHIEF-based random forest algorithm that combines elastic distance splitting of proximity forests with the hybrid splitting criteria of dictionary and interval features to construct scalable decision forests. A greedy algorithm maximizes information gain, and the results from multiple trees are integrated for classification. TS-CHIEF balances accuracy and efficiency through hybrid feature splitting. However, hyperparameter tuning for feature splitting strategies is complex and requires an automated search to enhance usability. Bai et al. [56] proposed a temporal classification framework that combines multi-feature dictionary representation and ensemble learning. The framework extracts time-domain, frequency-domain, and time-frequency domain features to construct a joint dictionary. It generates feature representations via sparse coding and integrates classification through random forests. The multi-feature space dictionary representation captures multidimensional patterns in time-series data, and the ensemble learning strategy balances classification strengths across different feature spaces. However, dictionary construction incurs high computational costs, and the feature dimension

disaster problem remains unresolved. He et al. [57] proposed the Feature Selection-based Multi-view Clustering (FS-MVC) algorithm for multivariate time series classification. It comprises multidimensional feature extraction, multi-view clustering, and multi-view ensemble classification. It integrates feature selection with multi-view learning and enhances stability through an ensemble-group framework. Rotti et al. [58] proposed a short-term classification framework for solar energetic particle events based on a multivariate time series classifier. They constructed an ensemble model comprising Time Series Forest (TSF), Supervised Time Series Forest, and Bag of SFA Symbols (BOSS) by integrating three proton channels from GOES satellites and long-wave X-ray flux data. However, the model is limited in its robustness to imbalanced classes, requiring manual tuning of the window parameters. Ren et al. [59] combined the time series ensemble learning algorithm HIVE-COTE2.0 (HC2) with the SHAP interpretation method to create an interpretable seismic phase classification framework. HC2 enhances accuracy by utilizing multi-classifier ensembles (STC, DrCIF, etc.), and SHAP determines the importance of features to optimize combinations of features.

Ensemble learning-based methods exhibit strong generalization capabilities, effectively handling noise, nonlinearity, and pattern diversity in time-series data while mitigating overfitting risks associated with single models. They demonstrate robustness in multi-class and complex time-series scenarios. However, these approaches feature complex model structures, incur high training and inference costs, rely on empirical design for base model diversity, and offer limited interpretability of ensemble results, making it difficult to trace key classification criteria.

Additionally, we summarized the results of traditional analytical methods using standard datasets, as shown in Table 2.

4.2. Deep learning-based time series classification methods

Deep learning overcomes the bottleneck of manual feature design in traditional methods by learning features end-to-end. It demonstrates exceptional performance, particularly in high-dimensional, long-sequence, and multivariate scenarios. Given the feature patterns extracted from temporal data and the modeling objectives, the methods can be categorized into four types: local pattern mining-based, temporal dependency modeling-based, hybrid model architecture-based, and spatio-temporal fusion modeling-based. Among them, local pattern mining-based methods treat time series as static objects to extract time-invariant local morphological features, distinctively focusing on identifying short-range discriminative patterns rather than modeling global sequential ordering. Temporal dependency modeling-based methods view time series as sequential streams to capture global evolutionary dynamics and long-range dependencies, relying on recurrent processing or attention mechanisms to model the contextual relationships between distant time steps. Hybrid model architecture-based methods integrate heterogeneous modules to simultaneously capture multi-scale temporal representations, aiming to overcome the limitations of single-paradigm models within the univariate temporal dimension. Spatio-temporal fusion modeling-based methods are designed for multivariate time series or graph-structured data to explicitly model inter-variable correlations (spatial) alongside temporal dynamics, distinguishing themselves from hybrid models by their unique focus on the interaction between sensors or channels.

4.2.1. Local pattern mining-based methods

Local pattern mining-based methods use convolutional transforms (e.g., CNN, TCN) or the wavelet transform to extract features from locally contiguous subsequences within time series. They emphasize extracting short-time-domain morphological patterns (e.g., peaks, abrupt changes, local trends). Extracting local features at varying scales through sliding convolutional kernels and enhancing robustness via pooling operations are well-suited for scenarios where local patterns

Table 2

The summary of important traditional analysis-based time series classification methods.

Author	Year	Dataset	Method	Accuracy (%)	Advantages	Disadvantages
Johnpauet al. [33]	2021	128 datasets of UCR	Trend-Global-TS-Rep	83	Introducing global trend features. The generated representation primitives are dimensionally controllable, achieving improved accuracy while maintaining computational complexity.	The global feature extraction logic requires further refinement to enhance representational specificity. Classification performance is weak in complex noise scenarios.
Ye et al. [36]	2009	5 domain datasets: Synthetic Lightning EMP, Projectile Points, Historical Documents, Gun/NoGun, Wheat Spectrography. Trace dataset of UCR	ree-Shapelet-TS	About 80	Shapelets provide interpretability over traditional methods. They offer fast classification speeds with low computational complexity.	Not suitable for multivariate temporal scenarios. Shapelet search efficiency requires optimization.
Mueenet al. [37]	2011	24 datasets of UCR	Logical-Shapelets	About 90	The pruning strategy is 27 times faster than the Shapelets method. It exhibits strong generalization capabilities and can automatically adapt to the characteristics of data across different domains.	Logical expressions are prone to overfitting.
Lahreche et al. [41]	2021	118 datasets of UCR	LE-DTW	About 82	Achieve fast and accurate classification of long-term time series. Parameter-free, eliminating the need for manual parameter tuning. Computational speed is several orders of magnitude faster than DTW.	Insufficient capability to characterize high-noise, long-time-series data. Enhance processing speed for large-scale, long-term data series.
Lianget al. [38]	2021	25 datasets of UCR	CSSL	88.53	Proposed a class-specific shape learning paradigm to enhance computational efficiency in shapelet updates and improve the interpretability of classification decisions. Automatically determined the number, length, and initial values of shapelets while reducing iteration counts. Implemented parallel acceleration using OpenMP.	Hyperparameter tuning is complex. Not suitable for multivariate time-series scenarios.
Abanda et al. [60]	2022	10 datasets of UCR	Perturb-Interp-TS	Average loss $\Delta eLoss$ 0.3-0.7	Four temporal perturbation methods with semantic interpretability. A two-stage explanation method balancing global robustness and local interpretability.	Not suitable for multivariate time-series scenarios.
Liet al. [39]	2022	45 datasets of UCR	BSPCOVER	86	Bloom filters enable rapid pruning, boosting model training efficiency. Overall computational complexity is significantly reduced, thereby markedly improving processing efficiency for long-term series data. Supports integration with any sub-learning method, maintaining stable performance across high-noise, long-time-series, and small-sample datasets.	Not suitable for multi-variable scenarios. Adaptive tuning strategies for hyperparameters are complex.
Zhanget al. [42]	2024	3 PhysioNet datasets	BRÉDOD	Length 300:89.56 Length 140:80.56	Resolves the issue of ambiguous temporal sequences that traditional methods struggle to handle, based on D-S evidence theory. Effectively reflects temporal data characteristics by measuring the internal divergence of the sequence twice.	Time-series boundaries depend on domain knowledge and require manual configuration. Requires offline processing of complete time-series sequences, lacks incremental computation mechanisms, and cannot be directly applied to real-time streaming time-series scenarios. Not suitable for multi-variable time-series scenarios.
Wuet al. [43]	2024	10 datasets of UCR	COPP-Miner	89.5	Solve balanced binary classification problems. The pruning strategy is efficient and avoids parameter sensitivity issues.	Not applicable to multi-class classification problems. Requires manual parameter tuning, with significant variations in optimal parameters across different datasets.
Tan et al. [44]	2025	112 datasets of UCR2018	PF2.0	89	PF2.0 offers the highest precision and fastest speed. Introduces ADTW, enhancing temporal alignment flexibility through adjustable non-diagonal penalty terms. Integrates three representations—raw time series, first-order derivatives, and HYDRA dictionary transformation—covering multidimensional features including trends, local fluctuations, and repetitive patterns.	There is room for improvement in fine-tuning distance metric sets for individual datasets.

(continued on next page)

Table 2 (continued)

Author	Year	Dataset	Method	Accuracy (%)	Advantages	Disadvantages
Fulcheret al. [52]	2014	20 datasets of UCR2011	HC-Feature-TSC	80.9	Covers multi-dimensional temporal attributes, eliminating the subjectivity of manual feature selection. Significantly reduces dimensionality, lowering computational and storage costs while adapting to large-scale datasets. Features possess interpretability, revealing the mechanisms of inter-class differentiation. Suitable for long-term time series and variable-length data.	The initial feature database contains many redundant features, resulting in time-consuming feature computation. It relies on linear discriminants, exhibiting weaker classification capabilities for strongly nonlinear time series compared to nonlinear methods.
Feremans et al. [48]	2022	22 datasets of UCR 8 datasets of UEA	PETSC	76.8	Extended pattern mining technology extracts inter-class differences. Multiple embedding representations characterize features, combined with mutual information for feature selection, yielding interpretable results. Handles variable-length time series with robustness against high noise or missing values.	Further research is needed on various temporal classification scenarios, such as resource-constrained settings. Pattern mining efficiency is low, requiring further optimization of pruning strategies.
Erazo-Costa et al. [51]	2024	24 datasets of UCR	PWFTS	80.35	PWFTS exhibits low computational time and is suitable for real-time TSC applications. It demonstrates high stability during the characterization construction process.	Limited to handling univariate time series data, with constraints in processing heteroscedastic and multivariate scenarios. It relies on manual parameter tuning without proposing methods to determine window length and overlap degree.
Yahyaoui et al. [47]	2025	109 datasets of UCR	MinR	72.3	Extract the minimal number of salient features from time series data. A novel distance metric that serves as a lower bound for Euclidean distance offers a tighter bound than the piecewise aggregation approximation (PAA). Features possess interpretability by corresponding to explicit physical meanings.	The number of segments depends on manual parameter tuning. Further optimization is possible for MinR's feature extraction, distance calculation, and classification algorithm execution times. It only processes single-variable time series and cannot capture correlations between multiple variables.
Bai et al. [56]	2021	82 datasets of UCR	TBOPE	69.9	Combining mean-based and trend-based features to capture key characteristics and enhance classification accuracy. Introducing confidence-weighted voting to mitigate single-parameter bias and improve overall accuracy.	Unable to capture correlations among multiple variables. Some hyperparameters require manual tuning.
He et al. [57]	2024	26 datasets of UEA	FS-MVC	73.05	Integrating multiple sequence transformation methods to capture correlations and trends among variables. Combining dimension-dependent and dimension-independent PPPV convolutional features to enhance accuracy for highly nonlinear data. An ensemble group feature selection framework is introduced to strengthen feature stability through feature vector aggregation. Supporting multi-sensor signal fusion with interpretability.	The large number of features requires significant memory, limiting its operation on small computing devices. Feature selection requires offline processing of complete time series, lacks incremental learning mechanisms, and cannot update models in real-time.

Note: The accuracy values are reported based on the standardized fixed train/test splits of the UCR/UEA archives to ensure data consistency.

exhibit strong discriminative capability.

Campo et al. [61] proposed an adaptive MLP named A-MLP, which dynamically adjusts the number of hidden layer nodes and learning rate while avoiding overfitting through Bayesian optimization and early stopping strategies. However, it exhibits insufficient scalability for multivariate time series. Liu et al. [62] proposed a Tree-Wave Neural Network (T-WaveNet), combining the multi-resolution analysis of wavelet transforms with a tree-structured network to extract sensor data features hierarchically. However, model parameters exhibit exponential growth with tree depth, necessitating regularization (e.g., weight decay) to prevent overfitting. Wang et al. [63] proposed a deep learning model based on CNNs and fully connected layers, enabling end-to-end training on raw time series without complex preprocessing or feature engineering. The input sequences undergo feature extraction via 1D

convolutional layers and max-pooling layers, then are mapped to categories through fully connected layers, optimized using cross-entropy loss. The advancement of deep learning has driven more research toward model innovation rather than feature engineering, yet explicit modeling of long-range dependencies remains lacking. Tang et al. [64] proposed a multi-scale CNN model named Omni-scale CNN. By dynamically combining convolutional layers with varying kernel sizes (1, 3, 5, 7), it automatically captures multi-scale time series features without manual kernel parameter tuning. However, the multi-kernel combination increases training time, and classification accuracy for complex patterns requires optimization. Zhang et al. [65] introduced a dilated convolutional fully-connected network named DM-FCN as a strong baseline model for time series classification. The method expands the receptive field using multi-scale dilated convolutional kernels (3, 8,

8, 8) with varying dilation rates (1, 5, 10) and time-axis average pooling to ensure output consistency. However, it lacks an adaptive mechanism for dynamically adjusting the dilation rate. InceptionTime [18] addressed the Receptive Field Dilemma by deploying parallel Inception modules with varying kernel lengths (e.g., 10, 20, 40) to simultaneously capture short-term fluctuations and long-term patterns. While this multi-scale mechanism establishes it as a robust, high-accuracy baseline for general-purpose TSC, its increased network width results in higher computational costs compared to standard ResNets. Cui et al. [66] introduced a multiscale convolutional neural network named MCNN. It extracts features at different temporal scales through a multi-branch architecture (downsampling branch, frequency domain transformation branch), achieving end-to-end classification by integrating convolutional and pooling layers. The multi-branch network automatically learns optimal scale features, eliminating the need for manual feature engineering. Tabassum et al. [67] introduced a simple and fast segmented word embedding neural time series classifier, SAFE. It converts time series into symbolic representations using Simple Symbolic Aggregate Approximation, extracts fixed-length “words” via sliding windows, and combines natural language processing word embedding models with neural networks for classification. However, its adaptability to high-dimensional non-stationary time series requires improvement. Tahan et al. [68] proposed end-to-end models (FCN-DISC, LSTM-FCN-DISC, ALSTM-FCN-DISC) that embed time-discretization into deep learning architectures. These models optimize both discretization quality and classification accuracy via dual loss functions. By integrating the discretization process with deep learning training, they address the issue of traditional methods treating discretization and training as independent steps. Dempster et al. [69] introduced a Rocket-based time series classification method. It transforms time series using stochastic convolution kernels (with random lengths, dilation, and padding) and combines them with a linear classifier (logistic regression) for rapid classification. The diversity of stochastic kernels captures multiple temporal patterns, while PPV features quantify these patterns to achieve linear computational complexity. However, its ability to represent strongly nonlinear temporal patterns remains limited. Tan et al. [70] proposed a MultiRocket-based time series classification model. It extends the Rocket algorithm by incorporating multiple pooling operations (e.g., max, average, and product pooling) and transformations (e.g., differencing and absolute value) to enhance feature representation. Combined with stochastic convolution kernels, this approach generates efficient feature vectors. Dempster et al. [71] developed MiniRocket by refining Rocket to reduce model complexity. It employs a fixed kernel set and deterministic dilation strategy, retaining only PPV features while optimizing convolution through pre-computations and bitwise operations. MiniRocket balances efficiency and accuracy, making it suitable for real-time classification tasks, although its kernel design relies on empirical parameters. Xu et al. [72] proposed the Generalized Time-Warping Invariant Dictionary Learning (GTWIDL) method. It employs Adaptive Linear Space Approximation (ALSA) for dimensionality reduction and dynamically determines discretization cut-off points using the golden ratio to achieve symbolic representation of time series. It fits mean distances using Gaussian distributions and employs the Rocket classifier to identify homogeneous healthy sub-cohorts (HBCs). However, dictionary update efficiency requires optimization in non-stationary signal scenarios.

To reduce model complexity and optimize efficiency, He et al. [73] proposed a relational convolutional neural network (Rel-CNN)-based time series classification framework that enhances classification performance by explicitly learning relational features within time series. The framework consists of stacked Local Pattern Global Relationship (LPGR) blocks that transform global relational features into local representations, reducing parameter complexity through hierarchical convolutions. Xiao et al. [74] proposed a TSCaps-based time series classification model that integrates multi-head convolutional neural networks with capsule networks (CapsNet). This model extracts features

through multi-scale convolutions and preserves spatial relationships via capsule routing mechanisms, eliminating the need for reconstruction modules. This approach incorporates capsule mechanisms into temporal classification, preventing information loss from CNN pooling. The multi-head architecture, combined with multi-scale features, enhances representational capacity. However, it suffers from large parameter scales. Chen et al. [75] introduced a multi-task representation learning method named MTRL. MTRL shares parameters between a deep wavelet decomposition network and a residual network to improve classification performance by minimizing classification triplet loss. While the joint optimization strategy effectively integrates task correlations, the method suffers from a large parameter size and time-consuming training. Xiao et al. [76] introduced a Dynamic Sparse Network (DSN) that learns sparse connections in convolutional kernels through dynamic sparse training. DSN divides time series windows into multiple groups, each exploring effective receptive fields within constrained regions. Gong et al. [77] introduced KDCTime, a knowledge distillation and calibration framework. KDCTime enhances InceptionTime through label smoothing, knowledge distillation, and calibration strategies. Incorporating soft label calibration into time series classification addresses prediction bias in teacher models and improves the generalization capability of student models. These approaches offer lightweight model optimization solutions for time series classification, balancing accuracy and inference efficiency.

Local pattern mining-based methods are sensitive to subtle local patterns, exhibit high computational efficiency, are easily parallelizable, and possess some noise suppression capabilities. However, they struggle to model long-range temporal dependencies, have weak global feature integration capabilities, and perform poorly in classification tasks that rely on long-term trends.

Additionally, we summarize the time series classification methods based on local pattern mining using standard datasets, with the results shown in Table 3.

4.2.2. Temporal dependency modeling-based methods

The temporal dependency modeling-based methods center on recurrent neural networks and their variants (LSTM, GRU) alongside Transformers. These methods utilize memory or self-attention mechanisms to capture sequences' short-term and long-term temporal dependencies. They focus on depicting the dynamic evolution of variables over time. These methods are ideal for scenarios that require extracting temporal sequential relationships.

Bianchi et al. [78] applied Reservoir Computing (RC) to multi-variable time series classification. By decoupling and coupling multi-variable time series features, they enhanced the modeling capability for complex systems and employed a dynamic Reservoir structure to adapt the network topology. Wang et al. [79] proposed a time series classification method based on GRAE-ESN, combining Echo State Networks (ESN) with Global Reversible Autoencoders. Input weights are iteratively updated by pulling back decoder layer outputs, replacing the stochastic reservoir with a Simple Cyclic Reservoir (SCR). Reversible autoencoders optimize input weights to enhance feature correlations, while SCR's deterministic generation of reservoir weights reduces randomness. However, its data compression efficiency remains insufficient for high-dimensional data, requiring manual adjustment of the reservoir scale. Silva et al. [80] proposed the Multi-Level Horizontal Visibility Graph (MHVG) to capture lag dependencies in multivariate time series through cross-level visibility. Time series are mapped onto a multi-level graph where intra-layer edges are based on horizontal visibility and inter-layer edges on cross-level visibility, enabling analysis of temporal dependencies via graph topology features. Yu et al. [81] compared the time series classification performance of RNN variants, including LSTM, GRU, and BiLSTM. They systematically analyzed how different gating mechanisms perform in the temporal feature extraction process. Results indicate that gating mechanisms have a significant impact on temporal feature extraction. Bidirectional RNNs are better

Table 3

The summary of important time series classification methods based on local pattern mining.

Author	Year	Dataset	Method	Accuracy (%)	Advantages	Disadvantages
Cui et al. [66]	2016	44 datasets of UCR	MCNN	86.5	Integrate feature extraction with classification tasks, utilizing backpropagation to learn model parameters adaptively. Extract features across different frequency and temporal scales, enhancing accuracy by fusing multi-scale and multi-frequency features.	It can only handle univariate time series and not model relationships between multiple variables.
Dempster et al. [69]	2020	128 datasets of UCR	Rocket	78.8	It covers multi-scale/frequency features without requiring kernel parameter learning through extensive random kernels. It employs positive proportionality (PPV) to enhance discriminative power, performing better in high-noise environments.	Not applied to multivariate temporal classification. Random kernels lack clear meaning and offer weak interpretability. Increasing kernel count improves accuracy, but excessive kernels increase memory consumption.
Wanget al. [63]	2017	44 datasets of UCR	MLP/FCN/ResNet	FCN:86.7 ResNet:85.2 MLP:78.3	The benchmark training model requires no manual feature engineering. Input raw time-series data for end-to-end training.	Unable to capture dynamic relationships among multiple variables. Lacks adaptive hyperparameter tuning. ResNet is prone to overfitting when the sample size is small.
Xiao et al. [74]	2021	128 datasets of UCR	TSCaps	87.3	Simultaneously encodes feature attributes and spatial relationships to prevent information loss from CNN pooling. Possesses multi-scale feature fusion capabilities.	Only single-variable time-series data was processed. No research was conducted on multivariable time-series data. Hyperparameters significantly impact performance, requiring cross-validation tuning, which increases costs.
Chen et al. [75]	2021	36 datasets of UCR	MTRL	88.2	Jointly optimize classification and retrieval tasks by applying supervised representation learning to classification and unsupervised representation learning to retrieval. Distance-weighted sampling strategy efficiently identifies discriminative samples. Demonstrates strong generalization, outperforming single-task methods across 36 diverse domain datasets.	DTW computation has limitations and is not suitable for extremely long-scale time series. It only processes univariate time series and has not been tested on multivariate data.
Dempster et al. [71]	2021	109 datasets of UCR	MiniRocket	-	High computational speed, up to 75 times faster than the original Rocket method. Achieves an optimal balance between accuracy and computational cost. Capable of handling multivariate applications. Adaptive hyperparameters.	Its performance potential may not have been fully realized without deep integration with nonlinear classifiers. Innovation in foundational theory remains relatively limited.
Campoet al. [61]	2021	61 datasets of UCR	A-MLP	88.5	Low computational complexity, minimal resource requirements, and training achievable with a CPU alone. Simple model architecture, easy to implement, and highly accessible.	Low accuracy on high-dimensional time series. Not suitable for multivariate data.
Xiao et al. [76]	2022	85 datasets of UCR 3 datasets of UCI	DSN	UCR: 89.6 UCI: 98.37	Automatically learns eNRF through dynamic sparse training without manual hyperparameter tuning. Achieves 2x reduction in memory and computational costs while maintaining higher accuracy performance. Delivers high accuracy on both univariate and multivariate datasets, demonstrating superior robustness to noise and temporal shifts compared to CNN-based methods.	It is not deployed on actual edge devices; it remains at the theoretical and dataset experimentation stage. Unable to determine whether existing performance advantages can be maintained in resource-constrained edge scenarios.
Gong et al. [77]	2022	113 datasets of UCR	KDCTime	91	Soft labels via label smoothing will be introduced to mitigate overfitting. Utilizing knowledge distillation from a teacher model to automatically generate soft labels, eliminating manual parameter tuning. Employing calibrated knowledge distillation to correct erroneous soft labels predicted by the teacher model.	Initial training of the teacher model is time-consuming. Soft labels struggle to distinguish fine-grained categories and perform poorly on complex datasets. Corrections only address erroneous soft labels, failing to calibrate samples with inherently incorrect predictions, thus limiting calibration scope.
Tabassumet al. [67]	2022	30 datasets of UCR	SAFE	88.7	Compact model parameters with minimal memory consumption. Preserves all temporal information through tokenization. Hyperparameters can be tuned as needed to adapt to diverse domain data.	Applicable only to univariate time-series data. Applicable only to univariate temporal classification. Requires manual hyperparameter tuning across different datasets, with time-consuming tuning processes. Fixed-length words fail to capture long-term semantic patterns.
Tan et al. [70]	2022	109 datasets of UCR	MultiRocket	86.92	Multiple pooling operators and transformation operations have been added to the MiniRocket architecture to enhance the diversity of generated	Adaptability and feature efficiency are limited, as it can only process univariate data and requires expanding multi-channel

(continued on next page)

Table 3 (continued)

Author	Year	Dataset	Method	Accuracy (%)	Advantages	Disadvantages
					features. Extended from the Rocket framework, it is easy to implement and requires no complex hyperparameter tuning.	convolutional kernels. Sensitive to small samples, with accuracy dropping significantly when training samples are fewer than 50, as first-order differencing further reduces data volume. Bias sampling from convolutional outputs easily leads to loss of high-frequency features. Pooling operations are weak at capturing high-frequency details in long time series.
Tanget al. [64]	2022	85 datasets of UCR	Omni-scale CNN	84.77	Easy to implement, with only three convolutional layers and no complex modules. Convolution kernel size hyperparameters require no fine-tuning. Full-scale design reduces redundant kernels through receptive field coverage, resulting in a compact parameter size.	The study is limited to univariate time-series data and does not address multivariate time-series data. When the training sample size is less than 50, the full-scale kernel is prone to overfitting, resulting in lower accuracy than Rocket.
Tahanet al. [68]	2023	40 datasets of UCR	FCN-DISC, LSTM-FCN-DISC, ALSTM-FCN-DISC	85.97	End-to-end model architecture with temporal discretization to enhance computational efficiency. Extraction of fundamental time-series features while reducing noise. Adaptation to multiple network architectures, including FCN, LSTM-FCN, and ALSTM-FCN.	Processes only single-variable data. Multi-channel discretization is not supported. Lacks direct supervision and constraints on the discretization process, potentially leading to biased results.
He et al. [73]	2023	85 datasets of UCR 8 datasets of UEA	Rel-CNN	UCR: 86.92 UEA: 80.89	By mining relational features, learn local patterns and global characteristics. Combine with multi-layer deep learning structures to enhance accuracy. Visualize relational feature matrices to identify critical temporal segments, achieving interpretability.	Accuracy is relatively low on high-dimensional multivariate data. Relational features require computation based on full-volume data, making real-time adaptation to streaming time series impossible.
Xuet al. [72]	2025	4 datasets of UCR	GTWIDL	97.2	Effectively captures latent features in time-series data affected by temporal distortion, supporting multi-dimensional datasets with varying sequence lengths and demonstrating strong adaptability. Dictionary interpretability enables clear capture of intra-class patterns through learned atomic features, while sparse coefficients visualize sample distributions.	Critical discriminative information in the data may be overlooked without incorporating sparse coding coefficients. Dictionaries require training on the entire dataset, necessitating re-optimization whenever new samples are added.
Zhanget al. [65]	2025	128 datasets of UCR 30 datasets of UEA	DM-FCN	UCR: 82.36 UEA: 75.84	By expanding the receptive field through multi-scale convolutional kernels, the model adapts to the complex structures of multivariate data. The model architecture can be customized according to specific time series application requirements, accommodating the differentiated needs of various scenarios.	Customizing network architectures for different tasks incurs high adaptation costs. Optimal architecture selection is inefficient, requiring full training-testing cycles to match optimal variant structures to specific tasks.

Note: The accuracy values are reported based on the standardized fixed train/test splits of the UCR/UEA archives to ensure data consistency.

suit for capturing long-range dependencies but incur higher computational costs. Hyperparameter optimization relies on random search and lacks theoretical guidance. Xiao et al. [82] proposed an RTFN-based time series classification model comprising a Time Feature Network (TFN) and an LSTM Attention Network (LSTMaN). TFN employs a residual convolutional structure to extract local features, integrating multi-head CNNs and self-attention mechanisms. LSTMaN captures global dependencies through a two-layer LSTM-attention hybrid structure, combined with BiLSTM to enhance long-term feature extraction. Bakirtzis et al. [83] proposed a deep learning-based multivariate time series classification framework, constructing the Convolutional Avoided Space Pyramid Pooling-Attention LSTM (CAP-ALSTM) model. Time series from sensor data are extracted via sliding windows, with self-attention mechanisms focusing on key features. The ASPP module captures multi-scale contextual relationships, and classification is ultimately performed through two LSTM layers. Huang et al. [84] proposed a strAPI-based time series classification model, achieving efficient classification through Attention Power Iteration (API). It utilizes Time Convolutional Networks (TCN) for feature extraction, dynamically updates compact representations via Incremental Weighted Principal Component Analysis (PCA), and assigns sample weights through an attention mechanism. Zhang et al. [85] proposed an ACS-ATCN-based time series classification framework. It employed an adaptive top-k

differential evolution algorithm (ATDE) to optimize the class cost weights and hyperparameters of the attention-based temporal convolutional network (ATCN), addressing imbalanced time series classification. By jointly training class cost optimization and network hyperparameters, the minimum recall function via LSE smoothing is approximated to enhance recognition of the minority class. Mbouopda et al. [86] introduced a scalable Subsequence Transformation (SST) model that enhances time series classification efficiency through subsequence segmentation, attention-weighted fusion, and global feature integration. Long sequences are decomposed into overlapping sub-sequences, leveraging the self-attention mechanism of Transformers to capture cross-segment dependencies. The lightweight combination of subsequence transformation and attention addresses computational bottlenecks in long-sequence classification. Tang et al. [87] introduced the hierarchical Transformer network named Hformer. It segments sequences via temporal slicing layers, progressively aggregating multi-scale features across stages by combining Time Absolute Position Encoding (TAPE) and Efficient Relative Position Encoding (ERPE). The TAPE-ERPE fusion enhances temporal sequence sensitivity while reducing self-attention computational complexity through slicing layers. Hformer performs strongly in multivariate classification but incurs high memory consumption in long sequence scenarios.

Temporal dependency modeling-based methods effectively capture

long-range temporal dependencies and excel at detecting overall trends in sequences. However, their training processes are susceptible to vanishing/exploding gradients and lack sensitivity to local key patterns. Transformer-based models exhibit high computational complexity, yielding limited benefits for short-sequence tasks.

Additionally, we summarize the time series classification methods based on temporal dependency modeling using standard datasets, as shown in Table 4.

4.2.3. Hybrid model architecture-based methods

Hybrid model architecture-based methods integrate local feature extraction with temporal dependency modeling modules. Through multi-feature fusion mechanisms, these methods simultaneously capture local details and global temporal associations. The goal is to combine the strengths of different models to improve classification generalization. These methods are widely used in complex temporal scenarios.

Wu et al. [89] proposed a spectral convolutional network named SCNet to extract multiscale temporal features. It converts 1D time series into frequency-domain features via enhanced Discrete Fourier Transform (DFT), prioritizes key frequencies through spectral energy-based filtering, and captures short- and long-term trends using multiscale convolutional modules. The proposed method offers a new paradigm for frequency-domain optimization in sequence classification but lacks robustness for non-stationary signals. Lyu et al. [90] introduced a time series classification approach based on the Multiscale Echo Self-attention Memory Network (MESAMN). It encodes high-dimensional features using a multi-echo state network (ESN), captures long-term dependencies through a multi-head self-attention mechanism, and integrates features via a multi-scale convolutional network. Combining ESN's short-term memory with self-attention's long-term modeling capability addresses long-term dependencies in multivariate time series data. Aslan et al. [91] proposed a hybrid model based on sliding window-LSTM. It extracts mean, standard deviation, median, and multiscale entropy features from time series via sliding windows, feeds them into an LSTM for deep feature extraction, and then uses classifiers like k-NN or SVM for recognition. This approach outperforms standalone statistical features or traditional machine learning

methods but relies on specific time series preprocessing for the dataset. Elen et al. [92] proposed a time series classification method integrating Long Short-Term Memory (LSTM) with Random Forest. First, LSTMs extract temporal dependency features from the time series. Then, Random Forests handle nonlinear relationships. Finally, classification results are integrated through weighted voting. The hybrid model performs excellently in non-stationary, multi-scale feature scenarios, outperforming standalone LSTMs and Random Forests. However, parameter tuning requires domain knowledge. Liu et al. [93] proposed a multi-scale temporal dynamic learning method that captured sequence dynamic patterns by extracting features at different temporal granularities hierarchically. The method combines convolutional and recurrent networks to model long-term and short-term dependencies across different time scales, achieving detailed representations or incorporating attention mechanisms to enhance key time points, outperforming single-scale models. Ma et al. [94] proposed a differential-guided representation learning network named DGRL-Net, which simultaneously processes raw and differential sequences via differential gated LSTM modules, followed by multi-scale CNNs to extract cross-dimensional features. Differential information significantly enhances the model's sensitivity to abrupt changes, while multi-scale convolutions improve feature generalization. Xiao et al. [95] proposed a dual-network architecture: Perceptual Capsule Network (PCapN) and Deep Transformer Capsule Mutual Distillation (DTCM). PCapN comprises a Local Feature Network (LFN) and a Global Relationship Network (GRN) to extract local features and global patterns. However, its computational complexity remains high, and its adaptability to high-dimensional data requires optimization. Mu et al. [96] proposed a multi-scale periodic time series network named MPTSNet. It decomposes time series into multi-scale periodic components via Fast Fourier Transform (FFT), then performs local feature extraction and global dependency capture sequentially using PeriodicBlock modules. Local feature extraction employs an Inception-based architecture to process convolutional kernels at different scales, while global capture models cross-periodic dependencies through a multi-head attention mechanism. Multi-scale features are ultimately aggregated using amplitude-weighted aggregation.

Table 4

The Summary of time series classification methods based on temporal dependency modeling.

Author	Year	Dataset	Method	Accuracy (%)	Advantages	Disadvantages
Wanget al. [79]	2021	10 datasets of UCR 5 datasets of UEA	ESN-GRAE	UCR: 86 UEA: 83	GRAE's global reversible structure enables ESN redundancy removal and dimensionality reduction. By eliminating the need to train ESN weights, GRAE reduces computational complexity. While ESN captures local dynamic dependencies, GRAE optimizes global feature distribution.	Enhance GRAE's cross-variable fusion capabilities to improve multi-variable time-series classification performance.
Xiao et al. [82]	2021	121 datasets of UCR 30 datasets of UEA	RTFN	82.3	The TFN and LSTMAn dual networks work synergistically to balance local feature extraction with global relationship discovery. Achieves high efficiency and low redundancy through optimized architecture. Delivers stable performance on long-term, high-noise data.	The ability to capture nonlinear temporal patterns is limited.
Mbouopda et al. [86]	2024	72 datasets of UCR and UEA	SAST	84	Enhance interpretability by visualizing key discriminative sub-sequences. Combine with multiple classifiers to improve generalization.	Distance calculation efficiency is low. Module complexity is high.
Tanget al. [87]	2025	30 datasets of UEA	Hformer	77.5	Multi-scale feature capture. Enhanced temporal dependency modeling through efficient position encoding via TAPE+ERPE. Strong generalization capability with high feature space discriminability.	Insufficient capability to uncover dynamic correlation patterns among multiple variables. Failure to adapt the slice size and number of phases based on data complexity.
Liet al. [88]	2025	33 datasets of UCR	Diff-ESN	79	High efficiency, achieving 6 times faster processing than LSTM. Strong robustness, Diff-ESN is insensitive to random connections.	The number of neurons and the topological structure are fixed, with unbalanced efficiency performance.

Note: The accuracy values are reported based on the standardized fixed train/test splits of the UCR/UEA archives to ensure data consistency.

Seydi et al. [97] proposed a Dual Attention Convolutional Neural Network (Dual Attention CNN) to achieve dynamically adaptive temporal feature attention weights. This network processes time series data by combining spectral and spatial attention modules with multi-scale residual blocks. Chen et al. [98] introduced a Dual Attention Network (DA-Net) comprising Squeezing-Exciting Window Attention (SEWA) and Sparse Self-Attention within Windows (SSAW). SEWA weights key local features through window-interval dependencies, while SSAW filters activation scores using Kullback-Leibler divergence to broaden the window range and capture global dependencies. The dual attention mechanism effectively balances long-range dependencies with short-range interactions. (SSAW). SEWA weights key local features based on inter-window dependencies, while SSAW filters activation scores using Kullback-Leibler divergence to expand window coverage and capture global dependencies. This dual-attention mechanism effectively balances long-range dependencies and local details, though parameter tuning remains challenging in complex scenarios. Liu et al. [99] introduced a multi-attention mechanism model integrating temporal, graph, and channel attention. Temporal attention captures

long-term sequence dependencies, graph attention models inter-dimensional relationships, and channel attention enhances key feature weights. Squeeze-Activate networks optimize cross-channel interactions. Zhao et al. [100] introduced a Flexible Multi-Head Linear Attention (FMLA) model combining deformable convolutional blocks with collaborative linear attention, incorporating random masking and online knowledge distillation. Deformable convolutions dynamically adjust receptive fields, while linear attention reduces computational complexity. Training employs self-distilled stable masks. FMLA achieves efficient inference with high accuracy but struggles with modeling long-range dependencies and requires manual mask ratio tuning.

To improve the computational efficiency of Transformers, Mamba introduces a data-dependent tango scan mechanism, which dynamically compresses historical context into a hidden state while filtering noise. This innovation solves the quadratic complexity bottleneck of Transformers, achieving linear time complexity with global context modeling, making it uniquely suited for ultra-long sequences, despite currently lacking the mature hardware optimization ecosystem of attention-based models. Ahamed et al. [101] proposed the TSCMamba method for time

Table 5

The summary of time series classification methods based on hybrid model architectures.

Author	Year	Dataset	Method	Accuracy(%)	Advantages	Disadvantages
Ma et al. [94]	2022	18 datasets of MTS	DGRL-Net	66.7	LSTMs capture dynamic evolutionary features, addressing the limitations of traditional MTS classification methods. Dual-module collaboration balances local and global features, delivering strong generalization capabilities. Achieves optimal performance-efficiency balance with low training/testing time and minimal computational overhead, outperforming SOTA across 12 MTS datasets.	Without utilizing temporal difference information, focus on specific variable dimensions to extract highly discriminative features.
Chen et al. [98]	2022	30 datasets of UEA	DA-Net	72.4	The SEWA layer captures local key sequence fragments, while the SSAW layer captures global long-range dependencies with linear complexity. Computational speed is up to 622 times faster than traditional methods. The DA-Net approach balances accuracy, efficiency, and interpretability.	The current DA-Net window size is fixed and cannot be resized automatically.
Zhao et al. [100]	2023	85 datasets of UCR	FMLA	89.37	Its linear complexity makes it significantly more efficient than traditional Transformer-based methods. Enhancing local perception through deformable convolutional blocks, online knowledge distillation, and masking mechanisms to reduce noise interference and model redundancy.	The dynamic adaptation capability of the masking mechanism is insufficient.
Xiao et al. [95]	2024	UEA	DTCM	74.7	Outstanding classification accuracy with exceptional performance across diverse datasets. Significant lightweighting effects.	The failure to explore instance-level hierarchical associations between local feature networks and global relationship networks may result in information loss.
Mu et al. [96]	2025	10+20 datasets of UEA	MPTSNet	10 datasets: 75.4 25 datasets: 74	Extract multi-scale periodicity via FFT. Periodic module captures local and global features. Enhance classification interpretability through attention visualization.	The strategy for selecting the time scale is limited.
Ahamed et al. [101]	2025	10+20 datasets of UEA	TSCMamba	10 datasets: 76.04 25 datasets: 69.51	Multi-view feature fusion for comprehensive capture of critical temporal information. Efficient bidirectional sequence modeling. Enhanced computational efficiency for handling long sequences and resource-constrained scenarios.	Unexplored self-supervised learning techniques enhance feature representation capabilities, and model generalization is validated across complex datasets in more domains.
Wuet al. [89]	2025	UEA	SCNet	74.3	Converts 1D time-series data into a 2D frequency-domain representation through enhanced DFT. Combines spectral energy prioritization with multi-scale convolutional modules to balance frequency-domain feature capture and multi-scale dynamic analysis, offering superior parameters and computational complexity compared to Transformer-based methods.	Insufficient robustness of noise data. Inadequate adaptability of spectral feature extraction to multi-scale convolutional structures.

Note: The accuracy values are reported based on the standardized fixed train/test splits of the UCR/UEA archives to ensure data consistency.

series classification to address the quadratic complexity of self-attention mechanisms. It integrates the Mamba state space model with multi-view learning, combining continuous wavelet transform (CWT) spectral features, Rocket temporal features, and MLP global features. The tango scan mechanism enhances inversion invariance. CWT extracts time-frequency features, generating spectral representations via Conv2D and FFN. Rocket and MLP extract local and global temporal features, respectively, which are then fused through switch gating. The Mamba module employs tango scanning to capture long-range dependencies, combined with depthwise pooling classification. TSCMamba balances efficiency and accuracy in time series classification, though parameter tuning for multi-view fusion becomes complex with high-dimensional data.

Hybrid architecture-based methods can capture multi-scale features, typically yielding superior classification performance compared to single models. However, they suffer from structural complexity, challenging hyperparameter tuning, and high training costs. Additionally, improper feature fusion strategies may lead to information redundancy or dilution of key features.

Additionally, we summarized the results of time series classification methods based on hybrid model architectures using standard datasets, as shown in Table 5.

4.2.4. Spatio-temporal fusion modeling-based methods

Spatio-temporal fusion modeling-based methods address multivariate time series by incorporating spatial relationships among variables through architectures such as graph neural networks or spatio-temporal convolutions, while simultaneously modeling temporal dependencies. These approaches are suitable for classification tasks involving spatio-temporal coupling.

To enhance spatio-temporal information mining, Yao et al. [102] proposed a TSec-based time series classification framework integrating correlation analysis, time series alignment, shapelet deduplication, and Bi-GRU/GCN-GRU models, supporting both univariate and multivariate classification. Zhao et al. [103] proposed an LSTM-MFCN-based time series classification model integrating a multi-scale fully convolutional network (MFCN) with LSTM. It constructs multi-scale receptive fields through dilated convolutions to capture spatio-temporal features. Zhang et al. [104] introduced the Attention Dynamic Self-aggregation Network (ADSN), which comprises a channel attention module and dynamic self-aggregation blocks. The channel attention module adaptively weights spectral bands, while the dynamic self-aggregation block captures spatio-temporal features through multi-scale dynamic convolutions and enhanced multi-head attention. Graph learning methods leverage their relational representation advantages and are also applied to spatiotemporal modeling. Wang et al. [105] proposed a graph learning approach for handling irregularly sampled multivariate time series. They constructed spatio-temporal graphs linking time points and features, utilizing Graph Convolutional Networks (GCNs) to capture temporal dependencies and feature associations under non-uniform sampling. However, the complexity of graph construction increases exponentially with the number of variables. Cheng et al. [106] introduced a tensor graph convolutional network named Tensor-GCN. It represents time series as tensors, defines Tensor Similarity Values (TSV) to measure distance and shape similarity, constructs tensor graphs, and extracts spatio-temporal and polarization features via GCN. Liu et al. [107] introduced Scale-teaching, a framework for classifying noisy-labeled time series. It achieves robust classification through a bottom-up cross-scale fusion mechanism and multi-scale embedded graph learning. The method first samples time series at different scales to generate multi-scale views and trains multiple neural networks. It utilizes cross-scale complementary information to select clean labels via a minor loss criterion. It constructs graph structures based on label propagation theory to correct noisy labels for unscreened samples. Results demonstrate that multiscale information effectively enhances noise resistance, though input scales are fixed and computational complexity

increases with scale. Liu et al. [108] proposed TodyNet, a temporal dynamic graph neural network. It processes multivariate time series through dynamic graph construction, transformations, and temporal graph pooling to capture spatio-temporal dependencies. Time series slices generate dynamic graphs, where DyGIN aggregates neighborhood information. A temporal convolutional pooling layer enables hierarchical feature compression, supporting the extraction of multi-scale spatio-temporal features. Younis et al. [109] proposed the MTS2Graph-based time series classification framework, which converts multivariate time series into temporal evolutionary graphs for classification using graph neural networks. Nodes are constructed using sliding windows, edges defined by temporal relevance and feature similarity, and dynamic relationships captured through graph attention mechanisms. Interpretable classification results are ultimately generated via graph pooling. However, graph construction for high-dimensional time series incurs high computational costs. Han et al. [110] introduced DynIMTS for time series representation, integrating graph neural networks with instance attention mechanisms to dynamically learn graph structures for handling irregular multivariate time series (IMTS), supporting real-time streaming data processing. Spatio-temporal learning utilizes a GRU-based graph convolutional network, demonstrating strong performance on irregular time series tasks, particularly in scenarios with high rates of missing data. Du et al. [111] introduced CaLoNet, a causal and local correlation network. It constructs causal graph structures between dimensions via transfer entropy, combines sparse self-attention (SSA) for local feature extraction, and ultimately integrates these into a graph neural network for classification. Wang et al. [112] proposed SAGoG, a similarity-aware graph neural network for multi-variable time series classification. This method constructs instance graphs using Dynamic Time Warping to capture sample similarity, while a Channel Graph Structure Evolution module (combining GCN and GRU) learns channel-dependent dynamic changes. It provides a general framework for multi-time-series classification that integrates dynamic graph structures, but suffers from high computational complexity and requires improved adaptability to high-dimensional data.

To efficiently model long-range dependencies in time series, Foumani et al. [113] proposed optimizing the Transformer's positional representations for multivariate time series by incorporating temporal absolute position encoding (tAPE) and efficient relative position encoding (eRPE). They constructed a ConvTran model that combines convolutional input encoding with tAPE/eRPE. This model extracts spatio-temporal features via 3D CNNs while capturing long-range dependencies through Transformers. The position encoding is designed for sequence length and dimensionality, addressing the anisotropy issue in traditional encodings. It enhances the Transformer's generalization capability in time series classification, though attention computation efficiency in long-sequence scenarios requires optimization. Zuo et al. [114] proposed a Shape-Level Variable-Position Transformer model (SVP-T) for multivariate time series classification. This approach extracts time-series sub-sequences as "shape" inputs through clustering, designs Variable Position encoding layers (VP-layers) to fuse variable and positional information, and enhances attention weights for overlapping shapes via VP self-attention mechanisms. It upgrades traditional timestamp-level inputs to shape-level while modeling variable and temporal dependencies. Furthermore, image-based approaches have been employed to analyze spatial relationships in time series. Anand et al. [115] proposed a DeLTa-based time series classification method that converts 1D time series into 2D images. It utilizes a pre-trained ResNet to extract layout-aligned features (DeLTa-LA) and layout-independent features (DeLTa-LI). LA preserves spatial layout information within the image, while LI eliminates layout dependencies through global max pooling. Combining both optimally enhances classification performance.

Spatio-temporal fusion modeling-based methods can capture spatio-temporal interaction features, thereby overcoming modeling limitations based solely on the temporal dimension. However, they heavily rely on

prior knowledge of spatial structure, may introduce redundant computations in purely temporal data scenarios, and exhibit high model complexity, making them prone to overfitting on small-scale datasets.

Additionally, we summarized the time series classification methods based on spatio-temporal fusion modeling using standard datasets, as shown in Table 6.

4.3. Few-shot learning-based time series classification methods

The real world presents numerous scenarios with scarce data. Few-shot learning aims to address the scarcity of labeled data. Techniques such as transfer learning, self-supervised learning, meta-learning, and

multimodal learning are employed to enhance the adaptability and generalization capabilities of deep learning in few-shot scenarios.

4.3.1. Transfer learning-based methods

The few-shot time series classification method based on transfer learning reuses knowledge by transferring pre-trained temporal feature extraction capabilities from large-scale labeled data (the source domain) to the few-shot target domain. Typical strategies include fine-tuning pre-trained models, updating only the classification head while retaining the parameters or fixed feature extraction layers trained on general datasets. The primary objective is to minimize the discrepancy in distributions between the source and target domains.

Table 6

The summary of Time Series Classification Methods Based on Spatio-Temporal Fusion Modeling.

Author	Year	Dataset	Method	Accuracy (%)	Advantages	Disadvantages
Anand et al. [115]	2021	85 datasets of UCR	DeLTa	85	Converting 1D time series into 2D images, leveraging the visual feature extraction capabilities of pre-trained ResNet50. DeLTa-LA preserves layout information, adapting to scenarios sensitive to temporal alignment. 2D image conversion exhibits strong noise resistance. Simultaneously captures variable and temporal dependencies, addressing the limitation of traditional timestamp-level Transformers that only model temporal dependencies. The VP-layer better adapts to multi-variable temporal data by preserving key information through variable-position encoding. Self-attention complexity scales with the length of the input shape, thereby reducing computational demands while maintaining strong generalization capabilities. Integrating spectral, local, and global features, leveraging graph structures to capture dynamic dependencies among variables. K-shape clustering reduces MHAP redundancy, while graph embedding enhances the discriminative power of features.	Fails to select DeLTa-LA/LI, requiring manual selection automatically. Insufficient capability for representing nonlinear temporal data. The 2D image conversion process introduces delays in temporal classification scenarios. Shape generation based on k-means clustering relies on manual parameters. Enhancing shape representation without integrating frequency-domain features.
Zuo et al. [114]	2023	30 datasets of UEA	SVP-T	73.3	Design models tailored to data characteristics to address the limited adaptability of traditional unified models. Utilize Pearson matrices and maximum complete subgraphs to partition multivariate data into correlated/unrelated subsets, improving computational efficiency and accuracy compared to full-variable modeling. Leverage temporal alignment to reduce intra-class noise and generate multi-scale shapes via sliding windows.	Not optimized for streaming temporal data. Graph construction and embedding processes are computationally complex, making them unsuitable for real-time scenarios.
Younis et al. [109]	2024	8 datasets of UEA and UCR	MTS2Graph	86	Dynamic graph modeling addresses temporal dynamics and demonstrates strong generalization across multi-domain datasets. Temporal graph pooling mitigates information loss. Module-level joint training eliminates complex preprocessing, outperforming traditional and Transformer-based methods in efficiency and parameter scale.	Variable association modeling is static and unsuitable for scenarios involving multiple variables that change over time. Shape sub-generation and model inference workflows require optimization and do not support incremental classification of streaming time-series data.
Yao et al. [102]	2024	112 datasets of UCR 26 datasets of UEA	TSec	UCR: 88 UEA: 76	Quantify causal relationships between variables using Transfer Entropy (TE) to construct explicit causal graphs. Combine CBAM and SSA to capture local temporal dependencies, achieving 3-5 times higher computational efficiency than fully connected self-attention. Fuse causal graph structures with local correlation features through GIN.	The fixed window approach fails to adapt to time-series data of varying complexity. Insufficient feature extraction performance for high-dimensional datasets.
Li et al. [108]	2024	26 datasets of UEA	TodyNet	72.6	Modeling dual-layer graph structures to capture global-local correlations. Utilizing GCN+GRU to track channel correlations over time, adapting to dynamic temporal characteristics. Hierarchical hard clustering pooling of channel graphs preserves core correlation information while reducing node count.	The dynamic GNNs studied do not capture the evolution of causal relationships over time. They lack integration with frequency-domain features to enhance local correlation extraction and are insufficiently adapted to highly nonlinear data.
Duet et al. [111]	2025	21 datasets of UEA	CaLoNet	75.5		
Wang et al. [112]	2025	22 datasets of UEA	SAGoG	74		No research has been conducted on the real-time capture of association for non-concurrent temporal data. Graph contrastive learning pre-training has not been introduced, resulting in overfitting issues on small-sample datasets.

Note: The accuracy values are reported based on the standardized fixed train/test splits of the UCR/UEA archives to ensure data consistency.

To leverage high-performance pre-trained computer vision models, Oh et al. [116] transformed multidimensional time series into multi-channel analog images, employing a pre-trained ResNet-50 Model combined with transfer learning for classification. The generation of grayscale channels was based on feature dimensions to preserve temporal correlations in time series. The initialization of network parameters was conducted using pre-trained weights from the ImageNet dataset. This approach provides a novel image-based representation for multi-dimensional time series, while transfer learning effectively mitigates overfitting in scenarios with limited data. Antonijevic et al. [117] proposed a transfer learning framework for crop classification based on satellite image time series. This method utilizes pre-trained models to extract spatio-temporal features from Sentinel satellite data. These features are then adapted to target agricultural scenarios via domain adaptation techniques. Finally, the crop phenological feature weights are enhanced through attention mechanisms. Sun et al. [118] proposed a classification method based on Multi-Scale Dilated Convolutional Neural Networks (MSDCNN) and Meta-Learning of Transfer Metrics (MLS) for time series classification in intelligent transportation systems. MSDCNN reduces parameters through dilated convolutions and multi-scale feature fusion. At the same time, MLS optimizes the transfer learning process using a scale function, thereby enabling the model to learn from tasks with varying numbers of categories. The integration of metric learning with transfer learning has been demonstrated to enhance the capabilities of traffic flow detection in low-sample scenarios. Cheng et al. [119] propose a CrossTimeNet-based time series representation method that utilizes pre-trained language models (PLMs) for cross-domain self-supervised pre-training. The proposed methodology discretizes time series into tokens via vector quantization, employs BERT as the backbone network, and learns universal representations through masked token prediction tasks while supporting cross-domain transfer. This approach utilizes TCN as an encoder/decoder for time series segmentation and reconstruction, maps continuous values to codebooks via vector quantization, optimizes models during pretraining through masked token prediction, and adapts for downstream tasks via linear evaluation or full fine-tuning. The integration of PLMs into the time series domain facilitates the transfer of knowledge across domains. A time series discretization module has been developed to bridge modal differences, while cross-domain self-supervised pretraining has been shown to enhance model generalization. However, the pretraining stage is associated with high computational costs, and its theoretical foundation requires further refinement. Luo et al. [120] proposed a time series classification model based on VTSAPF. The model integrates sensor values, trend states, and type embeddings, employing a bidirectional GRU to extract temporal features and incorporating masked pre-training tasks. The process of abstracting numerical trends into discrete states serves to mitigate noise interference. The multi-task pre-training design is an effective solution to the problem of insufficient labeled data. Nevertheless, the trend state definition is simplistic and does not account for gradual transition patterns.

Transfer learning-based few-shot time series classification methods effectively leverage existing labeled resources, significantly reducing the required sample size in the target domain, and demonstrating outstanding performance in cross-scenario time series classification. However, these methods are contingent upon the presence of similarity between the source and target domains. In the event of excessive domain differences, negative transfer can occur, and the temporal feature representations of pre-trained models may become misaligned with the demands of the target task. There is a lack of large-scale pre-training datasets in the time series domain. Consequently, the majority of approaches rely on time series visualization techniques and pre-trained computer vision models.

4.3.2. Self-supervised learning-based methods

Self-supervised learning methods have been shown to learn general feature representations from large amounts of unlabeled sequential data

by designing sequence-specific pretext tasks (such as sequence prediction, sequential jigsaw puzzles, and noise restoration). These representations are subsequently employed to train classifiers on a limited number of labeled samples. The core approach utilizes the inherent structure of sequential data to construct supervisory signals, thereby enabling feature learning in unlabeled scenarios.

García et al. [121] proposed a self-supervised method for time series classification. To this end, they applied linear transformations to generate self-labeled training sets, employing four linear scaling operations on normal traffic time series to produce pseudo labels. They then employed a Random Interval Spectral Ensemble (RISE) classifier to learn to distinguish transformation types, ultimately performing recognition through decision rules. Bendou et al. [122] introduced a few-shot classification framework based on ensemble learning and self-supervised learning. This approach utilizes ResNet12 as the primary network, integrating a "Y-shaped" architecture with classification loss and self-supervised rotation prediction loss, while enhancing feature robustness through Manifold Mixup. It has been demonstrated that the system aggregates feature vectors from multiple backbone networks, completing classification via nearest-neighbor mean classifier (inductive) or soft K-means (transductive). The present study proposes a novel approach to few-shot classification by systematically validating the complementarity of its components. This approach entails replacing complex meta-learning with simple components, namely data augmentation and ensemble learning, and enhancing feature generalization through self-supervised rotation tasks. However, ensemble training is associated with high computational costs and is not optimized for extreme scenarios, such as one-shot classification. Eldele et al. [123] proposed a time series classification framework that utilizes the TS-TCC and CA-TCC methodologies. This framework facilitates self-supervised and semi-supervised learning by integrating Time Contrast (TC) and Context Contrast (CC) modules. The TS-TCC model is designed to generate diverse views of data by augmenting it with strong and weak data augmentation techniques. These augmentations are implemented using Transformers, which predict future temporal features across different views. The CC module, on the other hand, aims to maximize the intra-sample contextual similarity. CA-TCC employs a class-aware contrastive loss method that utilizes pseudo labels in semi-supervised settings. Zhang et al. [124] proposed a time series classification framework based on LogoRA, which enhances robustness by aligning local and global representations. The method utilizes contrastive learning to enforce consistency between local features (sub-sequences) and global features (entire sequences), while leveraging self-supervised pretraining to optimize the feature space. Nevertheless, calibrating these parameters continues to present significant challenges in complex scenarios. Han et al. [125] proposed a content-aware balanced spectral encoding method. The preservation of frequency-domain features through spectral decomposition during masked modeling, in conjunction with applying balanced encoding strategies to masked tokens, enhances the time series representation capabilities. This approach establishes a novel framework for self-supervised time series learning, employing frequency-domain prior-guided masking strategies to preserve periodic features and enhance representation. However, pre-training entails substantial computational expenses when dealing with large-scale datasets and demonstrates restricted adaptability to non-stationary time series. In their seminal work, Tang et al. [126] introduced a novel framework termed Language Pre-training Guided Masked Representation Learning (LPMRL). This innovative approach employs CLIP text encoders to generate semantic embeddings for labels, marking a significant advancement in the field. This approach facilitates the semantic region masking of time series patches and employs dual-information contrastive learning (comparing local patches with global instances) to enhance representational discriminative power. The proposed methodology involves replacing random masking with language-guided semantic masking to focus on critical regions, designing a hard negative sample generation mechanism, and

integrating slice-level and instance-level contrastive learning. The ACF-COS metric was developed to quantify temporal sequence importance and guide model application scenarios. It establishes a new semantic-driven paradigm for self-supervised representation learning in time series. However, the efficacy of this approach is contingent upon the use of pre-trained language models, which necessitates substantial computational resources and a large amount of unlabeled text data. In their seminal work, Qian et al. [127] proposed a novel learning method called DGMSCL, an acronym for Dynamic Graph Mixed-Supervision Contrastive Learning. It has been demonstrated that this method reconstructs time series as dynamic graphs, extracts features via hierarchical attention graph neural networks (HAGNN), and combines weighted augmentation-supervised contrastive learning (WAIGC) with minority class-aware contrastive learning (MCAC) losses. The DGMSCL approach effectively addresses class imbalance. However, it requires validation for generalization in complex dynamic scenarios. Additionally, its graph construction incurs a high computational overhead. Fawaz et al. [128] introduced a deep unsupervised domain adaptation (UDA) benchmark framework for time series classification. This approach integrates nine UDA algorithms (e.g., VRADA, CoDATS, InceptionDANN) with backbone networks, such as Inception, to construct models through causal relationship matrices and local correlation extraction. The proposed approach involves the introduction of a generalized temporal distortion operator, which aims to achieve dynamic time alignment through linear combinations of continuous basis functions. The optimization of distortion paths, dictionaries, and sparse coefficients is achieved by implementing block coordinate descent. The contribution of this study lies in establishing a standardized evaluation framework. However, it does not fully address real-time adaptation requirements in dynamic scenarios. In their seminal work, Kim et al. [129] proposed a novel approach, termed the Patch Order-aware Pretext Task (PPT) method, for self-supervised pretraining of time series. This approach captures temporal and channel-dimensional order dependencies through channel-level patch permutations and contrastive learning. The ACF-COS metric, which was recently introduced, quantifies the order importance of time series. This metric guides the application of PPT in specific scenarios and enhances the representation capabilities of time series in the context of few-shot learning. However, this method entails a higher computational overhead compared to traditional masking techniques, and its performance is unstable when dealing with non-stationary signals. In addressing the issue of domain drift in time series classification, Yang et al. [130] propose a hierarchical domain adaptation method based on virtual labels. The construction of a hierarchical feature alignment structure in conjunction with a virtual label generation mechanism enables multi-level feature matching between source and target domains. The proposed methodology employs adversarial training and self-supervised learning to enhance cross-domain consistency, a process that has been demonstrated to be effective in the context of cross-domain time series data.

In scenarios where the availability of labeled data is limited, Liu et al. [131] proposed an iTimes-based time series classification method. This method enhances semi-supervised classification through generative contrastive learning frameworks by exploiting irregular sampling patterns. This approach involves constructing a temporal sampling pattern generator and employing contrastive learning to align feature representations across disparate sampling patterns, thereby reducing reliance on labeled data. The extraction of temporal patterns inherent in irregular sampling as supervisory signals has been demonstrated to improve the efficiency of semi-supervised learning. The findings indicate that iTimes achieves 85 % classification accuracy with only 10 % labeled data, a performance comparable to that of fully supervised models. In their seminal work, Liu et al. [132] put forward a novel approach to semi-supervised time series classification. This pioneering method ingeniously integrates diffusion models with Shapelets, a sophisticated machine learning technique, to generate pseudo labels. Diffusion models are trained to learn latent semantic spaces and to generate

high-confidence pseudo labels for training. In addition, language models enhance the semantic representations of Shapelets. Integrating diffusion models within the framework of semi-supervised learning reduces the need for manual annotation, thereby providing an effective solution for scenarios involving limited data instances. However, the generalization of these findings in few-shot settings necessitates additional validation. Jin et al. [133] proposed a semi-supervised multi-instance discriminative contrastive learning framework, SmdCLR, to address false negatives and class imbalance in time series classification. The model's approach to supervised contrastive learning involves utilizing multi-instance discriminative tasks (MID) to enhance inter-instance relationships by implementing graph convolutional projection heads.

Additionally, the model incorporates semi-supervised semantic consistency regularization (SC-Reg), aiming to optimize representations of the minority class. Theoretical insights have been demonstrated to reveal the intrinsic mechanisms that link false negatives and class imbalance in contrastive learning. By dynamically allocating weights to strengthen minority class features, it establishes a universal representation learning framework for imbalanced time series. Self-supervised learning methods are autonomous, relying on unlabeled data to learn robust temporal features, thus reducing reliance on costly annotations. The limitations of this approach stem from the need to ensure that the pretext task design aligns with temporal characteristics. Failure to do so can result in the acquisition of redundant features. In scenarios involving samples of minimal size, the fine-tuning stage of the classifier is susceptible to overfitting.

4.3.3. Meta-learning-based methods

The meta-learning-based few-shot classification paradigm prioritizes the acquisition of mechanisms that facilitate the learning process. The model acquires a general strategy for rapidly adapting to new tasks through a meta-training phase across diverse temporal tasks. Representative approaches include gradient-based meta-learning (e.g., Model-Agnostic Model Learning (MAML)) and metric-based meta-learning (e.g., temporal prototype networks), with the core objective being to enhance the model's learning efficiency and generalization performance on novel temporal categories.

In addressing the challenge of classifying few-shot time series, Narwariya et al. [134] employed meta-learning to facilitate sparse time series classification. The Model-Agnostic Meta-Learning framework was adopted to enhance classification performance in sparse scenarios through cross-task knowledge transfer. The meta-learning optimizer adapts to sparse temporal features, addressing the sample scarcity issue in traditional methods and providing solutions for emerging fields in temporal analysis. Zhao et al. [135] proposed a temporal traffic classification framework based on MetaRockETC. This framework involves the modeling of packet length sequences as multivariate time series. Multi-scale features were extracted via stochastic convolution kernel transformations, and task-adaptive loss functions were designed using meta-learning. The optimization of feature extractor parameters is achieved by implementing scale functions, which facilitate the integration of multiple views via hard-voting classification. However, this approach requires substantial meta-training data and exhibits considerable performance variability in the extreme few-shot scenarios.

Metric-based meta-learning methods offer relatively higher training efficiency and better adaptability for edge devices. Park et al. [136] proposed a meta-feature fusion framework for time series classification, integrating multi-scale features (e.g., time-domain statistics, frequency-domain transforms) through meta-learning and optimizing sample representations via prototype networks. The meta-feature extractor captures cross-task shared patterns, while attention mechanisms are engineered to weight and fuse multi-modal features. This work aims to enhance the efficacy of classification generalization in scenarios involving few shots. In their seminal work, Tang et al. [137] advanced an interpretable classification framework, underpinned by meta-learning and attention mechanisms, for scenarios characterized by

limited sample sizes. This framework utilizes few-shot transfer learning to discern temporal patterns, marking a significant advancement in the field. This method utilizes a Prototype Network as the meta-learner, leveraging attention mechanisms to prioritize key time points and compute prototype distances between support and query sets. The system incorporates an interpretable module to highlight decision-relevant subsequences. By dynamically focusing on key features through attention and rapidly adapting to new categories via a meta-learning optimizer, it addresses the issue of overfitting in few-shot settings. This advancement has implications for few-shot learning applications in temporal classification, though it relies on the generalization capabilities of pre-trained models. In their seminal work, Wu et al. [138] put forward the Multi-Channel Transformer (MCTNet) as a solution for multivariate temporal classification, leveraging the innovative slice and channel aggregation technique. The model extracts global features via channel attention, captures local patterns through slice attention, and enhances few-shot generalization by class prototype learning. Tensor decomposition has been demonstrated to reduce computational complexity. Xie et al. [139] proposed a few-shot learning method based on DeepBDC, utilizing Brownian Distance Covariance (BDC) to assess disparities in joint distributions. The term BDC refers to a specific configuration of deep neural networks that incorporates a pooling layer. The model achieves few-shot classification by integrating a meta-learning framework (Meta DeepBDC) and a transfer learning framework (STL DeepBDC), incorporating deformable convolutional blocks to enhance local perception. While DeepBDC has been demonstrated to exhibit exceptional performance in the context of few-shot classification, the necessity of computational optimization of the BDC matrix persists to handle high-dimensional features.

Metric-based meta-learning is particularly well-suited for real-time inference, edge deployment, and classification tasks, offering significant advantages in terms of training and inference efficiency. Gradient-based meta-learning has been demonstrated to apply to scenarios characterized by complex requirements, cross-domain generalization, and high-performance demands. However, it should be noted that this approach imposes higher hardware requirements. Meta-learning-based few-shot temporal classification demonstrates a rapid convergence with minimal samples and exhibits strong adaptability to incremental category scenarios. Nevertheless, meta-training necessitates the implementation of extensive heterogeneous temporal tasks, imposes substantial computational demands, and manifests considerable performance deterioration in contexts characterized by markedly divergent temporal patterns. The efficacy of meta-training's generalization capability is profoundly influenced by the degree of distributional consistency between meta-training and target tasks.

4.3.4. Multimodal learning-based methods

Multimodal learning-based methods have been shown to enhance feature discriminability for few-shot classification by integrating temporal data with other modalities (e.g., textual descriptions, image features, and sensor-correlated data), thereby leveraging complementary insights across modalities. The core mechanism addresses the limitations of single-modal data through modal alignment and fusion techniques.

He et al. [140] proposed the WEFormer model, which integrates wavelet decomposition with a time series foundation model for classifying few-shot physiological signals. The model employs a decomposition process that involves segmenting signals into multiband subbands using the maximum overlap discrete wavelet transform (MODWT). It then extracts universal features through a frozen-parameter Time Series Foundation Models (TSFM) approach, and subsequently utilizes a dynamic weighting mechanism to determine band importance through SE-Net. Xu et al. [141] proposed ADGCN, which integrates an autoencoder (AE) and a deep graph convolutional network (GCNII), for the classification of traffic patterns with limited data. The AE is a reconstruction method that can replace zero-padding in time series, thereby

mitigating information loss resulting from length normalization. GCNII employs a graph structure construction method that utilizes KNN and heat kernels to facilitate the capture of relationships among traffic data. Indrasiri et al. [142] proposed an approach for encoding time series as images, followed by deep learning ensemble methods, suitable for small datasets. The implementation of these models may encompass time-frequency transformations, such as spectrograms, and multi-model ensemble voting mechanisms. The innovation lies in leveraging pre-trained models from the image domain to enhance few-shot generalization, thereby outperforming single models on small datasets. However, it should be noted that image encoding may result in the loss of some temporal information. In a seminal paper, Zhang et al. [143] introduced the Sequence-Spectrogram Fusion Network, a pioneering approach that achieved multimodal feature fusion by integrating sequence models and spectrogram models. The spectrogram model generates spectrograms via Short-Time Fourier Transform and performs random erasure data augmentation, while the sequence model processes time series using 1D convolutions. The process of hierarchical feature fusion is facilitated by a Hybrid Layer-Wise Feature Aggregator and a Fusion Connection Module. However, the model contains a substantial number of parameters. Hu et al. [144] proposed a cross-modal fusion framework named CMINet, comprising a Cross-modal Transfer Module (CMTM) and a Cross-attention Fusion Module (CAFM). CMTM corrects modal features through channel and spatial transfers, while CAFM exchanges semantic context using a cross-attention mechanism. CMTM employs a strategy of global pooling and MLP to generate weight masks, while CAFM utilizes a method of multimodal feature fusion through self-attention. The hierarchical feature fusion strategy balances intra-modal dependencies and cross-modal interactions, thus facilitating cross-attention for the enhancement of semantic alignment.

With the rapid advancements in large language models, Jang et al. [145] proposed the Language-Scaffolded Time Series Transformer. This approach reframes time series classification as a contextual question-answering task, integrating time series embeddings with LLM semantic knowledge through language-scaffolded prompts. The method utilizes 1D convolutions to generate time series embeddings, addresses irregular sequences through real-time position encoding, and integrates these embeddings into the LLM's token sequence. Subsequently, the model undergoes end-to-end training to optimize the classification head. Integrating the LLM's semantic reasoning capabilities into time series classification, in conjunction with the dynamic management of varying sequence lengths, culminates in the realization of language-time bimodal fusion. However, it has been observed that this approach incurs high inference costs for large models and encounters challenges in semantic modeling for ultra-short sequences. Cheng et al. [146] proposed an InstructTime-based time series representation method, reframing time series classification as a multimodal language generation task. Time series data are converted into tokens via vector quantization, further enhanced by incorporating domain-specific instruction text as input. The implementation of GPT-2 facilitates the generation of label text, thereby actualizing a paradigm wherein learning and generation are inextricably intertwined. The team developed a time-series discretization module and constructed prompt templates incorporating task instructions, contextual features, and time-series tokens. The model underwent a series of refinements through autoregressive pre-training and domain-specific fine-tuning processes. The large model can transform classification tasks into language generation, unify multimodal inputs, and design instruction templates to integrate domain knowledge, thereby enhancing interpretability. The multimodal alignment module bridges the representation gap between time series and text. However, this method relies on the generative capabilities of pre-trained language models and may not capture complex time series patterns.

The efficacy of multimodal learning methods stems from their capacity to harness multi-source information, thereby facilitating more comprehensive characterization of temporal patterns and enhancing classification robustness. This approach is particularly well-suited for

temporal scenarios involving auxiliary information. However, challenges emerge from the inherent heterogeneity between modalities, such as the continuity of time series versus the discreteness of text, which complicates alignment. Furthermore, the acquisition of multi-modal data incurs substantial financial costs. Additionally, modal imbalance has the potential to amplify noise from the dominant modality.

Additionally, we summarize the results of time series classification methods based on few-shot learning using standard datasets, as shown in Table 7.

4.4. Analysis of computational complexity and applicability

While accuracy remains the primary metric in academic benchmarks, real-world applicability is constrained by model complexity. Since most original studies lack unified reporting of these metrics, we provide a qualitative complexity analysis of common time series models based on the algorithmic mechanisms of both established and emerging paradigms. Let T denotes sequence length, C denotes input channels, K denotes kernel size, D denotes hidden/feature dimension, D_{dil} denotes dilation coefficient, and S denotes number of spline grid points in KAN.

1. Established Deep Baselines: CNNs and RNNs

Convolutional Neural Networks (ResNet, InceptionTime, FCN): The complexity is low to moderate, approximately $O(T \times C \times K \times D)$. 1D Convolutions allow for massive parallelization on GPUs. InceptionTime and ResNet serve as the gold standard for deployment, offering an excellent trade-off between accuracy and latency. Furthermore, lightweight convolutions, such as depthwise separable convolutions, have emerged in recent years, making these models highly compatible with edge-AI accelerators (e.g., TensorRT, NPU). CNNs are ideal for signal-based tasks (e.g., fault diagnosis, ECG) due to their robust extraction of local, translation-invariant patterns from noisy environments.

Recurrent Neural Networks (LSTM, GRU): The complexity is moderate, approximately $O(T \times D^2)$, but strictly sequential. Despite the linear complexity with respect to T , the lack of parallelization results in significantly longer wall-clock time for training and inference. Consequently, they are increasingly being superseded by CNNs or Transformers in high-throughput applications.

2. The High-Accuracy/High-Cost Regime: Transformers and Ensemble-learning based Models

Transformers (TST, Hformer): The complexity is high, approximately $O(T^2 \times D)$. The quadratic complexity of self-attention leads to prohibitive memory usage for long-sequence training. While they often achieve State-of-the-Art accuracy, deployment on resource-constrained devices typically requires aggressive quantization or pruning. It is worth noting that linear attention mechanisms have developed rapidly in recent years, offering a better trade-off between accuracy and complexity. Transformers excel in complex multivariate scenarios (e.g., macro-behavior analysis) where capturing long-range dependencies and channel interactions is critical for accuracy.

Ensemble Models (HIVE-COTE v2, TS-CHIEF): The complexity is very high, equal to the sum of the complexities of all base classifiers. By aggregating hundreds of classifiers (shapelet, dictionary, spectral), this approach reaches the theoretical ceiling of accuracy. However, the computational cost is often too prohibitive for real-time industrial scenarios. Ensembles are optimal for high-stakes, small-data applications (e.g., precision medicine) where maximizing accuracy takes precedence over computational efficiency.

3. Efficient Architectures: Rocket Series Models

Randomized Kernels (Rocket, Hydra): The complexity is low. By utilizing random convolutional kernels and linear classifiers, Rocket achieves remarkable classification accuracy with a fraction of the training cost of deep learning methods. Its unique advantage lies in

enabling ultra-low latency deployment on CPU-only hardware. Rocket series models are the best solution for resource-constrained edge deployment (e.g., IoT sensors) by delivering SOTA accuracy with ultra-low latency on CPU-only hardware.

4. The New Frontier: KANs, Mamba and Time Series Foundation Models

KAN: The complexity is high, approximately $O(T \times D \times S)$. Unlike MLPs that use fixed activation functions on nodes, KANs use learnable activation functions on edges. While theoretically parameter-efficient and interpretable, current implementations lack the hardware optimization of MLPs (e.g., optimized CUDA kernels), resulting in slower training and inference speeds. They are currently better suited for scientific discovery than real-time classification. KANs are suited for scientific discovery tasks where model interpretability and the revelation of underlying physical laws are as important as classification performance.

Mamba (State Space Models): The complexity is low, approximately $O(T \times D)$. Mamba represents a breakthrough, combining the global receptive field of Transformers with the linear complexity of RNNs. Its hardware-aware tango scan mechanism enables extremely fast inference, making it a promising candidate for long-sequence processing on edge devices. Mamba is uniquely positioned for processing ultra-long sequences (e.g., high-frequency seismic data) by combining global receptive fields with linear computational complexity.

Time Series Foundation Models (MOMENT, InstructTime): The complexity is extremely high. Adapted from Large Language Model, these models possess zero-shot learning capabilities. However, their massive size (7B+ parameters) creates significant memory bandwidth bottlenecks. Deployment is strictly limited to cloud-based GPU clusters, making them unsuitable for latency-sensitive edge applications without knowledge distillation. Time Series Foundation Models serve as the primary solution for cold-start or cross-domain scenarios, leveraging zero-shot capabilities when domain-specific training data is scarce.

As illustrated in our analysis, the field faces a bifurcation. TSFMs and Transformers pursue generalization and accuracy at the cost of computational resources, while linear attention, Mamba, and CNNs offer viable pathways for efficient deployment. We strongly advocate that future research explicitly reports floating-point operations and throughput (samples/sec) to guide model selection in practical engineering contexts.

This section systematically reviews the three major methodological frameworks for time series classification. Traditional analysis-based methods centered on manual feature design remain valuable in small datasets and short sequence scenarios. Deep learning has emerged as the prevailing approach, facilitated by automatic feature learning, which has been demonstrated to exhibit remarkable efficacy, particularly in long sequences and multivariate scenarios. A pioneering solution offered by few-shot learning addresses the issue of scarce labeling. The observed discrepancies in performance among these methodologies are primarily attributable to their divergent capabilities in modeling the characteristics of time-series data, including dependency, noise, and dimensionality. Traditional methods rely on domain knowledge to compensate for algorithmic limitations, while deep learning overcomes computational bottlenecks through architectural innovation. Conversely, few-shot learning has been demonstrated to broaden the scope of applications by enhancing data utilization efficiency. Each approach has strengths and weaknesses, and researchers can select the most suitable time-series classification method based on specific application scenarios. Despite experimental heterogeneity, the comparative analysis herein remains robust due to the community's strict adherence to standardized UCR/UEA fixed splits, which effectively eliminates data partitioning bias. To further enhance rigorous evaluation, we advocate that future research prioritizes unified benchmarking frameworks and statistical significance testing. This shift is essential to decouple algorithmic efficacy from implementation variances, fostering a truly objective and reproducible

Table 7

The summary of time series classification methods based on few-shot learning.

Author	Year	Dataset	Method	Accuracy (%)	Advantages	Disadvantages
Oh et al. [116]	2022	5 datasets of UEA	Multivariate Time Series Classification Method Based on Multi-Channel Image Transformation	F1 score: 53.8	Employing multi-channel approaches preserves inter-variable correlations. Utilizing transfer learning mitigates overfitting in few-shot scenarios. Implementing image transformations preserves temporal global features.	Validated only on 5 UEA datasets. Hyperparameters require manual tuning.
Sun et al. [118]	2024	71 datasets of UCR	MSDCNN	69.2	Enhance feature extraction capabilities through multi-scale features. The MLS strategy demonstrates excellent pre-training convergence rates.	The pre-training process is time-consuming. The model requires training on the entire dataset and cannot adapt to new tasks in real time.
Cheng et al. [119]	2025	HAR, EEG, ECG	CrossTimeNet	74.2	The same model supports both classification and prediction tasks with low fine-tuning costs. The tokenization module eliminates domain-specific data differences, enabling strong cross-domain transfer capabilities. Masked tokens eliminate the need for manual annotation, leveraging unlabeled data to enhance performance in few-shot scenarios.	Reliance on large models involves high computational complexity, requiring substantial computational resources and time. Discretization of continuous-time series may result in the loss of subtle temporal dynamics or precision.
Luo et al. [120]	2025	HAR, ISRUC-S3	VTSAPF	89.79	Enhanced ability to extract key trend features, with reduced sensitivity to noise. Effective pre-training strategies provide initial data, reducing reliance on large amounts of labeled data. Flexible architecture capable of modeling complex relationships among multiple variables.	Unable to capture changes in stability and other factors. Highly reliant on pre-assumed assumptions about training data, resulting in weak domain adaptability. Pre-training strategies exhibit high sensitivity to hyperparameters, requiring precise tuning and increasing computational resource costs.
Eldele et al. [123]	2023	HAR, Sleep-EDF, Epilepsy FD, 7 datasets of UCR	Self-supervised TS-TCC	1 % label: 76.1, 5 % label: 80.6	Combining weak and strong enhancement for time-series features effectively avoids information degradation caused by image enhancement. The dual contrast learning mechanism enhances feature robustness.	The framework is designed solely for univariate time series and lacks research on multivariate time series. It exhibits computational complexity when processing extremely long sequences.
Liu et al. [132]	2024	106 datasets of UCR	DiffShape	10 % label: 85.1	The diffusion model generates sub-shapes with over 70 % similarity to the original data. These sub-shapes correspond to specific temporal patterns, enabling visual interpretation of classification criteria. By incorporating semantic-level supervision, superior differentiation between distinct categories is achieved.	It is susceptible to hyperparameters. Limited to univariate time series, unable to model variable interactions in multi-sensor data.
Zhang et al. [124]	2024	4 datasets of UCR	LogoRA	86.6	The diffusion model generates sub-shapes with over 70 % similarity to the original data. These sub-shapes correspond to specific temporal patterns, enabling visual interpretation of classification criteria. By incorporating semantic-level supervision, superior differentiation between distinct categories is achieved. Integrating global and local features, it fully explores multi-scale information within time series. Designing an effective time-shift-invariant loss function enhances feature robustness.	High computational complexity, with diffusion model sampling being time-consuming and unsuitable for real-time scenarios. The shape sub-length requires manual tuning and is sensitive to parameters. Increased model complexity and parameter count lead to high computational overhead.
Tang et al. [126]	2025	106 datasets of UCR	LPML	85.1	Employing multi-strategy alignment methods significantly improves cross-domain feature alignment performance. Incorporating linguistic semantics into time series representation learning transcends traditional purely numerical representation learning methods. Targeted sampling of spatio-temporal patches enables the acquisition of more discriminative features. The dual-information contrastive learning mechanism captures both local details and global patterns within time series.	Optimization opportunities remain for the alignment loss function, with adaptive alignment methods offering greater efficiency. Dependence on “label-to-text” conversion quality and the potential for high computational complexity. LPML has yet to fully leverage repetitive or similar patterns that may exist across time intervals within the sequence. The core “language-guided” approach requires domain-specific expertise.
Han et al. [125]	2025	10 datasets of UCR	CBD	86.7	The dual-channel decoding architecture learns complementary information from both the time and frequency domains. Through strategies such as trend simplification, the method effectively reduces noise interference.	Although robustness is improved, the ability to capture complex patterns is sacrificed. The method is sensitive to device configurations in training data. It is sensitive to hyperparameters and

(continued on next page)

Table 7 (continued)

Author	Year	Dataset	Method	Accuracy (%)	Advantages	Disadvantages
Fawaz et al. [128]	2025	12 cross-domain datasets (bearing, ECG, har, etc.)	UDA-TSC Benchmark Framework	InceptionRain: 77, InceptionDANN: 75; Raincoat: 76	It efficiently utilizes large amounts of unlabeled data. Provides the first comprehensive benchmark evaluation in the temporal UDA domain, filling the gap in system performance comparisons. Constructs high-quality datasets and baselines tailored to domain-specific offsets. Employs a unified time budget hyperparameter tuning strategy to ensure fairness and reliability in algorithm comparisons.	lacks adaptive adjustment mechanisms. Computational efficiency for long sequences is low. No novel UDA algorithm was proposed, with the focus of the work being on evaluation rather than methodological innovation. Research conclusions may be constrained by the specific datasets and evaluation frameworks selected, and their generalizability requires further validation.
Kim et al. [129]	2025	24 datasets of UEA	PPT	Self-supervised: 82.71, Supervised: 88.2	First modeling of patch order to capture temporal and channel dependencies. Highly universal, plug-and-play without network restructuring. Single-GPU batch replacement takes only 0.018 seconds, achieving training efficiency close to traditional masking methods.	Patch size sensitivity requires manual adjustment of patch dimensions, with improper parameters causing accuracy fluctuations. Channel-level replacement fails to model cross-variable collaboration. The collaborative mechanism between consistency and contrastive learning lacks mathematical proof.
Jin et al. [133]	2025	HAR, Sleep-EDF, and four other real-world datasets	SmdCLR	HAR: 93.20, TUSZ: 69.00	Effectively addresses the false negative issue in contrastive learning by enhancing representation quality through multi-instance discrimination and graph-based constraints. Significantly mitigates class imbalance in data by boosting minority classes using semantic consistency loss on sparse labeled data.	Framework performance partially relies on a small amount of labeled data and cannot fully leverage its advantages in unsupervised scenarios. The graph-based projection head and pseudo-label constraints increase computational complexity and training difficulty. The quality of pseudo-labels may compromise the method's effectiveness, which poses potential risks of error accumulation and propagation.
Tang et al. [137]	2020	18 datasets of UCR	DPSN	6shot: 82.67 8shot: 85.41	Provides dual-granularity interpretability. Interpretable results provide fresh perspectives for data analysis and facilitate a deeper understanding of model decision-making.	The model structure is relatively complex, and the dual prototype mechanism introduces additional parameters. Model performance depends on the discovery and optimization of Shapelets, making it sensitive to the quality of the training data.
Wu et al. [138]	2023	26 datasets of UEA	MCTNet	78	Integrating CNN and Transformer architectures to provide a novel solution for MTSC. Designing efficient slicing and channel tokenization inputs to reduce Transformer parameters effectively. Combining channel and slice attention mechanisms to capture global and local multi-scale features effectively.	The fusion method between CNN and Transformer still relies on manual design, failing to automatically learn the optimal combination and strength of sensing bias. The model's parameter count still has room for optimization, and computational efficiency requires further improvement.
Cheng et al. [146]	2025	5 datasets of UCR, ECG and HAR	InstructTime	UCR: 78.7, ECG and HAR: 94.17	Reframe temporal classification as a multimodal generative task. Time series discretization enables cross-domain parameter transfer, enhancing model generalization capabilities.	Only smaller language models, such as GPT-2, are utilized, without leveraging the latest large language models. Synergistic collaboration with traditional deep learning models is overlooked, failing to integrate complementary strengths.

Note: The accuracy values are reported based on the standardized fixed train/test splits of the UCR/UEA archives to ensure data consistency.

evaluation ecosystem.

5. Data augmentation and model architecture optimization techniques

Deep learning-based time series classification methods encounter challenges in real-world deployment, primarily at the data and model design levels. The augmentation of data and the optimization of model architecture are foundational to the practical deployment and implementation of time series classification models. Data augmentation has been demonstrated to expand the training distribution and enhance the

model's adaptability to data variations, including noise, drift, and sparse samples. Consequently, this approach mitigates the risk of performance degradation caused by data distribution shifts in real-world scenarios. This approach is particularly beneficial in the context of high-cost data acquisition domains, such as industrial sensors and medical monitoring, as it leads to significant enhancements in deployment stability and reliability. The optimization of model architecture is a multifaceted process that addresses fundamental practical challenges, such as computational constraints and real-time requirements, through the implementation of lightweight design, enhanced inference efficiency, and improved adaptability to temporal features. Maintaining

classification accuracy while reducing deployment costs and engineering complexity accelerates the transition of time series classification technology from laboratory models to industrial applications.

5.1. Data augmentation techniques

Data augmentation techniques generate new samples by applying constrained transformations to the original time series, expanding the training set size, and enhancing data diversity. Their core principle involves introducing reasonable perturbations while preserving key temporal patterns, mitigating data scarcity or class imbalance issues.

Yuan et al. [147] conducted an empirical investigation into the impact of 11 data augmentation techniques (e.g., noise injection, temporal distortion, interpolation resampling) on the classification of pixel-level satellite image time series (SITS). The researchers implemented these 11 augmentation methods across four deep learning architectures (CNN, RNN, Transformer, and hybrid model) to evaluate the model's generalizability. The analysis revealed that interpolation, resampling, and temporal distortion emerged as the most effective methods for augmenting SITS. However, it is crucial to note that these methods impose computational costs that necessitate careful balancing to ensure optimal performance and cost-effectiveness in practical applications. Zhu et al. [148] proposed an Over-sampling with Hierarchical Transformation (OHIT) method to address imbalanced time series classification. The OHIT methodology involves capturing patterns of the minority class using Density Ratio Shared Nearest Neighbor Clustering and subsequently generating multivariate Gaussian synthetic samples by implementing a process known as covariance shrinkage. The OHIT model exhibits robust performance in imbalanced classification tasks. However, implementing clustering algorithms on high-dimensional data sets is associated with substantial computational costs. Time series visualization constitutes an additional prevalent data augmentation technique. Wang et al. [149] proposed a Gram matrix-based T-CNN method that preprocesses time series with wavelet threshold denoising and utilizes Gram matrices to losslessly convert time series into time-domain images, thereby preserving temporal attributes. To address these limitations, they proposed the adoption of a Toeplitz convolution kernel matrix to supplant conventional convolutions and integrating a triplet network to optimize the loss function. This approach was designed to enhance the similarity among similar samples and to accentuate the differences between dissimilar ones. However, this approach is unsuited for high-dimensional non-stationary time series and necessitates manual parameter tuning. Velasco-Gallego et al. [150] proposed a fault classification framework based on time series imaging. The proposed methodology involves converting time series into transition matrix images using first-order Markov chains. Subsequently, image classification is performed using ResNet50V2 and CNN. Markov chains have been employed to model state transition characteristics, while time series imaging has been demonstrated to capture fault dynamic patterns effectively. This renders the model highly sensitive to gradual faults. Chen et al. [151] proposed a novel time-series classification framework based on the CycleTime metric. The proposed methodology involves converting time series into two-dimensional images using Gram Angular Difference Fields, implementing CycleGAN for data augmentation, and incorporating a fully convolutional network for classification. The cross-domain fusion achieved through two-dimensional transformation and generative adversarial network augmentation preserves temporal dynamic features. CycleTime has been demonstrated to enhance the precision of classification, yet its computational demands for high-dimensional data remain substantial.

Generative models have demonstrated efficacy in addressing class imbalance and few-shot problems. Zhu et al. [152] developed a time series generation method based on a bidirectional long short-term memory convolutional neural network generator for generating electrocardiograms. The generator employs a two-layer bidirectional long short-term memory model to capture bidirectional temporal

dependencies. At the same time, the discriminator employs a convolutional neural network to differentiate between real and synthetic sequences. Adversarial training combines the bidirectional temporal modeling capability of BiLSTM with the local feature discrimination of CNN, preserving the PQRS waveform characteristics of ECG. To address data scarcity, Yang et al. [153] introduced TS-GAN, a framework employing LSTM-based generator and discriminator networks for sensor health data augmentation. The architecture distinguishes itself by integrating a Sequence Squeeze-and-Excitation module within the discriminator to refine spatial dependency extraction. Furthermore, to ensure training stability, the authors adopted the Gradient Penalty strategy from Wasserstein GAN during the optimization phase. Yoon et al. [25] proposed a TimeGAN-based method for generating time series. By integrating supervised loss with adversarial training, the system optimizes the Embedding Network, Generator, and Discriminator to produce sequences consistent with temporal dynamics. Integrating a reversible embedding space and a conditional generation mechanism enables the concurrent optimization of temporal dynamic capture and distribution matching. It addresses the challenge posed by traditional GANs in preserving temporal correlations in time-series data.

Data augmentation methodologies effectively enhance model generalization, particularly in scenarios involving limited data samples. However, their limitations include a high dependency on domain knowledge for augmentation strategies, potential introduction of spurious features or disruption of temporal dependencies through inappropriate transformations, and restricted applicability to highly time-sensitive data.

5.2. Model architecture optimization techniques

In machine learning, model architecture optimization techniques employ automated algorithms to explore optimal neural network structures. The techniques include methods such as neural architecture search and hyperparameter optimization. These approaches enhance the model's ability to capture and generalize temporal features by adapting network structures, training mechanisms, or introducing time-series-adaptive components (e.g., attention mechanisms, dynamic regularization, time-sensitive activation functions). The primary objective is to enhance the model's precision in modeling temporal patterns through the automated design of its architecture. This approach entails implementing adaptive learning rate scheduling to address temporal non-stationarity and designing lightweight networks to optimize the balance between performance and efficiency.

Early on, Falanti et al. [154] proposed the third-generation Pareto Optimal Neural Architecture Search algorithm, which automatically designs time series classification networks through multi-objective optimization (accuracy, parameter efficiency, inference speed). The proposed method dynamically integrates reinforcement learning and weight inheritance to generate lightweight architectures within the search space, balancing model capacity and computational cost. However, the diversity of the search space still requires expansion. Markovic et al. [155] proposed an enhanced particle swarm optimization (PSO) algorithm, termed PSOGO, to optimize long short-term memory (LSTM) networks for classifying gait time series in Parkinson's disease. The system's limitations stem from its dependence on particular sensor layouts, which restrict its applicability to specific data collection scenarios. In their seminal work, Chen et al. [156] introduced POCKET, a pioneering algorithm that utilizes feature selection to prune random convolution kernels efficiently. The model under consideration employs a group elastic net problem through a two-stage optimization process to emulate the redundant kernel pruning inherent to the Rocket framework. Stage 1 utilizes dynamic penalties to achieve group sparse classification, while Stage 2 employs regularization to reconstruct the classifier. The proposed methodology utilizes norm constraints to eliminate irrelevant kernels. The process of kernel pruning is thus transformed into a feature selection procedure for the classification

layer and group sparse learning with adaptive thresholds. However, it is important to note that hyperparameters remain sensitive to pruning rates. Wang et al. [157] proposed the lightweight spatio-temporal decoupled Transformer framework, which integrates fuzzy position encoding, TS-separable linear self-attention, and convolutional feed-forward networks. The automated model design is accomplished using a covariance matrix adaptive evolution strategy and global adaptive pruning. This approach has been demonstrated to outperform baseline models in lightness and accuracy, thus offering an efficient solution for real-time time series classification. However, the information loss issue in linear attention requires optimization through multi-branch feature fusion. King et al. [158] proposed a hardware-aware neural architecture search framework named MicroNAS. The proposed framework delineates a search space comprising time-reduction and sensor fusion units, facilitating low-power classification through the integration of delay lookup tables and dynamic convolutions. The findings indicate the efficacy of MicroNAS in resource limitations, though the scope of search space coverage necessitates further augmentation.

When optimizing model architecture, enhancing the interpretability of deep learning constitutes a key area of focus. Beyond standard model interpretation methods based on SHAP, Younis et al. [109] generated interpretable classification results through graph pooling to enhance the explainability of deep learning models. In their seminal work, Tang et al. [137] pioneered the concept of interpretable modules, which have since become a cornerstone of research in the field. These modules have been instrumental in elucidating decision-relevant subsequences, a critical step in analyzing few-shot scenarios. Attention mechanisms were incorporated to focus on key features dynamically. Cheng et al. [146] transformed classification tasks into language generation, unified multimodal inputs, designed instruction templates to integrate domain knowledge, and enhanced interpretability. A multimodal alignment module was developed to address the representation gap between time series and text. The optimization of models has been shown to enhance their adaptability to complex temporal patterns, obviating the necessity for data transformations. This quality renders them well-suited for high-dimensional or strongly nonlinear scenarios. However, these models are not without their limitations. These limitations include high design complexity, requiring domain knowledge and parameter tuning expertise, significant training costs, and susceptibility to overfitting in few-shot settings due to mismatched model complexity and data volume.

This section systematically reviews data augmentation and model optimization techniques in time series classification. Data augmentation is a process that aims to enhance the diversity of data through the utilization of methods such as time-domain, frequency-domain, and adversarial generation approaches. This process mitigates issues related to small sample sizes and noisy data. The optimization of models has been demonstrated to enhance the capabilities of engineering deployment by focusing on lightweight structures, real-time performance, and interpretability. Subsequent research endeavors should delve deeper into the realm of adaptive augmentation strategies and automated optimization frameworks. These frameworks hold immense potential in facilitating the efficient deployment and trustworthy application of TSC models in complex, real-world scenarios.

6. Datasets and evaluation metrics

6.1. Datasets

Public datasets are essential to time series classification research. The diversity and standardization of these datasets directly impact the reproducibility of model performance and the fairness of cross-method comparisons. Based on data dimensions, domain characteristics, and task types, mainstream datasets can be categorized into three main types: univariate, multivariate, and domain-specific. These datasets cover typical healthcare, industry, and military application scenarios.

This paper reviews commonly used time series classification datasets, focusing on four application domains: medical signal classification (particularly electrocardiograms and electroencephalograms), radar automatic target recognition, industrial fault diagnosis, and network traffic detection. For ease of reference and comparison, dataset information is summarized in Table 8.

The following trends can be identified for future research through detailed introductions and comparisons of various datasets:

1. The development of large-scale, real-world datasets. Advances in data collection technologies and increased awareness of data sharing will lead to the development and sharing of larger-scale, real-world time-series datasets.
2. Integration of multimodal datasets. Multimodal datasets that combine time series data with other modalities, such as images or text, will emerge as a research hotspot to improve classification tasks' performance and generalization capabilities.
3. Research on dynamic and non-stationary datasets. Datasets emphasizing dynamic changes and nonstationary characteristics will be developed to better reflect real-world time series properties.
4. Privacy Protection and Data Security. As data privacy and security concerns grow, developing and sharing datasets that prioritize these issues will become a significant research direction.

6.2. Evaluation metrics

A scientifically rigorous evaluation metric system is essential for time series classification research. It requires balanced consideration of classification performance, computational efficiency, and scenario-specific requirements. Table 9 summarizes the assessment metrics for time series classification based on evaluation dimensions, covering the calculation formulas, characteristics, and typical application scenarios of commonly used metrics.

This chapter systematically reviews the core datasets and evaluation frameworks used in time series classification. Public datasets encompass a range of standard univariate and multivariate datasets and domain-specific datasets. These datasets serve as benchmarks for various research objectives. Evaluation metrics establish a multidimensional framework that addresses both performance and scenario specificity. This framework guides the comparison of scientific models and the implementation of engineering solutions. Researchers can select the appropriate datasets and evaluation metrics based on the specific requirements of their task, thus avoiding misjudgments of performance caused by relying on a single metric.

7. Typical application scenarios

The value of time series classification ultimately lies in its application to real-world problems. This section focuses on four scenarios: medical signal classification, radar high-resolution range profile recognition, industrial fault diagnosis, and network traffic detection. The section provides an in-depth analysis of the temporal data characteristics, core classification tasks, mainstream methods, and domain-specific challenges in each field. This analysis builds a bridge between TSC theoretical research and practical application, highlighting the transformation paths of TSC technology in different fields.

7.1. Medical signal classification

Medical signal classification is a significant field in biomedical engineering and data science research. It aims to extract valuable patterns and features from time-series physiological signals for clinical applications, such as disease diagnosis, health status assessment, and monitoring treatment efficacy. Advancements in sensor technology and medical devices have led to the generation of vast amounts of time-series data in the healthcare field, including ECGs, EEGs, electromyograms

Table 8

The summary of time series classification datasets.

Dataset Categories	Dataset Name	Sample size	Time Series Length	Number of variables	Application domains	Characteristics
Univariate Time Series	UCR Time Series Classification Archive	128	Wide range of variation	1	Benchmarking of Time Series Classification Algorithms	Univariate time series data with multiple characteristics
Multivariate Time Series	UEA Multivariate Time Series Classification Archive	30	Wide range of variation	Wide range of variation	Benchmarking of Multivariate Time Series Classification Algorithms	Multivariate time series data spanning multiple domains
Univariate/ Multivariate Time Series	MONSTER	29	Wide range of variation	Wide range of variation	Benchmark Testing of Time Series Classification Algorithms on Large Datasets	Supplement to the UCR and UEA time series classification repository, with a large collective dataset.
Univariate/ Multivariate Time Series	NASA Prognostics Data Repository	21	Wide range of variation	Wide range of variation	Industrial Equipment Failure	Contains a large amount of time-series data spanning from initial normal states to fault states.
Medical Signal	MIT-BIH Arrhythmia Database	48	30 minutes	Wide range of variation	Classification of Arrhythmias	Detailed annotations by specialized cardiologists
Medical Signal	PTB-XLECG	21837	10 seconds	12	ECG Analysis and Diagnosis	Multi-label classification encompassing 71 ECG statements
Medical Signal	CHB-MIT Scalp EEG Database	182	1-4 hours	Wide range of variation	Seizure Detection	Detailed annotation including epileptic seizures
Medical Signal	DEAP Sentiment Analysis Dataset	32	1 minute	64	Emotion Recognition	Multimodal dataset containing EEG and physiological signals
Medical Signal	VTaC Tachycardia Alarm Dataset	Over 5000	Wide range of variation	Wide range of variation	False Heart Rate Abnormality Alarms	Two independent human expert annotators marked
Medical Signal	HiRID Critical Care Dataset	34,000	2 minutes	681	Intensive Care	High data storage resolution with comprehensive sampling metrics
Medical Signal	MIMIC-III	26	Wide range of variation	Wide range of variation	Epidemiology and Clinical Rule Decision Optimization	Includes highly detailed data such as vital signs, laboratory test results, and medication administration.
RATR	MSTAR-HRRP	51730	128	1	Military Target Recognition	Converted from SAR images
Fault Diagnosis	CWRU Bearing Dataset	48	Wide range of variation	1	Bearing Fault Diagnosis	Includes different fault types and severity levels
Fault Diagnosis	HIT Shaft-bearing Dataset	2412	20480	6	Bearing Fault Diagnosis	Includes different fault types and severity levels
Fault Diagnosis	PU Bearing Dataset	80	256823	1	Bearing Fault Diagnosis	Includes different fault types and severity levels
Fault Diagnosis	XJTU-SY Bearing Dataset	9216	1024	1	Bearing Fault Diagnosis	Includes different fault types and severity levels
Network Traffic	CICIDS2017	2.5 million	79	1	Intrusion Detection	Includes various types of modern cyberattacks
Network Traffic	UNSW-NB15	2.5 million	49	1	Intrusion Detection	Includes nine new types of attacks
Network Traffic	TON-IoT	64342	45	1	Intrusion Detection	Internet traffic accounts for over 70 %
Network Traffic	BoT-IoT	73	29	1	Intrusion Detection	Some data simulates internet behavior.
Network Traffic	NSL-KDD	150000	41	1	Intrusion Detection	Improved Dataset for KDD99
Equipment Sensor Logging	HAR	30	2.56 seconds	561	Smart Wearables	Records of Activities of Daily Living (ADL)

(EMGs), and respiratory signals. These signals contain rich physiological and pathological information. However, challenges such as low signal-to-noise ratios, sparse data, and cross-device variations pose significant hurdles to accurate classification[4]. In recent years, rapid advancements in deep learning have driven substantial progress in time-series classification methods for medical signals. While traditional methods rely on manually designed features and statistical models, current approaches increasingly use end-to-end deep learning architectures that can automatically extract features from raw signals. Furthermore, emerging techniques such as few-shot learning have enhanced model generalization capabilities in the face of data scarcity. This paper reviews the application of time series classification methods to various medical signals, including ECG, EEG, EMG, and respiratory signals, and explores future directions for medical signal classification tasks.

7.1.1. Related works on the medical signal classification

1. ECG Signal Classification

The electrocardiogram is a time-series recording of the heart's electrical activity, widely used to diagnose cardiac diseases. ECG signals are characterized by distinct periodicity and comprise characteristic waveforms, such as the P wave, QRS complex, and T wave. As one of the most extensively studied applications in medical signal

time series classification, ECG signal classification is primarily employed for arrhythmia detection, myocardial infarction diagnosis, and assessment of cardiac function. Researchers have proposed various methods for ECG classification. Traditional methods rely on manually designed features and statistical models. Examples include R-wave detection techniques, morphological feature extraction, and template matching.

In recent years, deep learning has played a pivotal role in classifying medical signals. Huang et al. [159] proposed a deep learning-based method for temporal representation to address the underutilization of temporal features in ECG classification. This method transforms the mean RR interval into temporal representation coefficients, applies a scale transformation to ECG signals, and uses a CNN-LSTM network with an attention mechanism to extract spatiotemporal features. The temporal scale transformation enhances interclass differences while maintaining lower model complexity than traditional feature fusion methods. Boda et al. [160] focused on patient-specific ECG heartbeat classification by proposing an LSTM-based recurrent neural network. This research aimed to address the limited adaptability of traditional methods to individual variations. The network achieved personalized classification by capturing ECG temporal dynamic features through LSTMs and integrating patient-specific training strategies. To address category

Table 9

The summary of evaluation metrics for time series classification.

Name of the metrics	Formula for Calculation	Characteristics	Application Scenarios
Accuracy	$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$	Measures the proportion of overall correct classifications. Intuitive and straightforward, but insensitive to imbalanced data.	Foundational evaluation with balanced data categories and preliminary model performance comparison.
Precision	$\text{Precision} = \frac{TP}{TP + FP}$	Measures the accuracy of positive predictions, which focuses on the cost of false positives (the fewer false positives, the higher the precision).	Scenarios requiring reduced misclassification should be minimized to prevent negative examples from being incorrectly classified as positive examples.
Recall	$\text{Recall} = \frac{TP}{TP + FN}$	Measures the proportion of positive examples correctly identified—focusing on the cost of missed detections (the fewer false negatives, the higher the recall rate).	Scenarios where all positive examples must be captured to avoid omitting any positive examples.
Specificity	$\text{Specificity} = \frac{TN}{TN + FP}$	Measures the proportion of negative examples correctly identified—complementary to recall (focuses on the accuracy of negative example classification).	Classifying negative examples in critical scenarios to evaluate the model's ability to distinguish between positive and negative examples.
F1 Score	$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	The harmonic mean of precision and recall balances the trade-off between the two, making it suitable for imbalanced data.	In scenarios with imbalanced categories (such as anomaly traffic detection), it is necessary to comprehensively evaluate the performance of both positive and negative examples.
F-beta Score	$F\beta = (1 + \beta^2) \times \frac{\text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}}$	Adjusting the weights by β , we can prioritize recall when $\beta > 1$ and precision when $\beta < 1$.	Adjust positive and negative example priorities based on the scenario, $\beta = 2$ (high recall), $\beta = 0.5$ (high precision).
Confusion Matrix	$\begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix}$	Visualize the distribution of results across each classification category, including all fundamental data points: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).	Intensive analysis of model error types (such as multi-class confusion) and foundational evaluation of multi-class classification problems.
AUC-ROC	Plot the curve based on TPR (True Positive Rate) and FPR (False Positive Rate), then calculate the area under the curve.	Comprehensively evaluate classification performance across different thresholds, focusing on TPR and FPR to ensure insensitivity to class imbalance.	In scenarios with high category imbalance, it is necessary to compare the overall performance of different models.
AUC-PR	Plot the curve based on precision and recall, then calculate the area.	Focus more on the classification performance of positive examples, which is more intuitive than AUC-ROC when dealing with class imbalance.	Extreme category imbalance scenarios, evaluating the model's ability to capture positive examples.
Macro-average	For multi-category classification, first calculate the metrics for each category, then take the arithmetic mean.	Treat each category equally, as this approach is suitable for addressing the issue of imbalanced sample sizes across categories.	Avoid letting the larger category dominate the evaluation when multiple categories exist and their importance is equal.
Micro-average	Combine the TP, FP, TN, and FN values across all categories for multi-class classification before calculating the metrics.	Focus on overall classification accuracy, which is significantly influenced by major categories.	Multi-class classification tasks with balanced sample sizes when overall accuracy is prioritized.
Cohen's Kappa	$\kappa = \frac{p_0 - p_e}{1 - p_e}$, where p_0 is the observed accuracy, p_e and is the expected accuracy.	To account for the impact of random classification, measure the consistency of evaluators (excluding incidental factors).	Multi-annotator consistency test: Comparison of model results with human annotations.
Log Loss	$L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c})$, where $y_{i,c}$ is the true label, $p_{i,c}$ is the predictive label.	Measures the consistency between predicted probabilities and actual labels, imposing severe penalties on extreme probabilities.	Scenarios requiring probability confidence output and quantification of model uncertainty.

imbalance in medical signal samples, Khan et al. [161] employed a one-dimensional deep residual neural network to overcome traditional ECG classification's complex signal processing challenges, thereby enabling efficient and automated classification. This study directly inputs raw heartbeats into ResNet and utilizes the Synthetic Minority Over-sampling Technique to mitigate category imbalance. Xia et al. [162] proposed a Transformer-based generative adversarial network that uses transformers to capture ECG sequence dependencies and combines them with convolutional discriminators to generate high-quality heartbeat signals. Roy et al. [163] introduced ECG-NET based on LSTM autoencoders. They addressed class imbalance and insufficient labeled data by training solely on normal ECG signals and identifying anomalies through reconstruction loss thresholds. The model uses a three-layer LSTM encoder-decoder architecture and integrates Kapur's automatic thresholding method to define anomaly boundaries. El-Ghaish et al. [164] designed a classification framework based on ECGTransForm. This framework captures bidirectional temporal dependencies in ECG signals via a bidirectional Transformer. It also enhances spatial features through multi-scale convolutions and channel recalibration modules and introduces context-aware loss to address class imbalance. However, the model's high computational complexity poses challenges for deployment on edge devices.

To improve the extraction of features from multimodal data, Moon et al. [165] proposed a wearable multimodal system that integrates

acoustic and bioelectric sensors to monitor and analyze respiratory activity in real time. The system collects lung sounds via piezoelectric transducers and acquires electrocardiogram signals through electrodes. The system extracts respiratory cycle and tidal volume features using polynomial curve fitting and feature fusion. It classifies breathing patterns using dynamic time window variance analysis and signature matrix techniques. Han et al. [166] proposed the multimodal multi-instance learning framework to address information loss in long-term ECG classification. This method treats raw ECG and Gram angular field images as multimodal inputs. Each long sequence is considered a bag, and image patches are considered instances. Features are fused via an attention mechanism to improve abnormal heartbeat detection. Sun et al. [167] addressed few-shot category incremental learning in medical time series by proposing the Meta Auto-Attention Prototype Incremter (MAPIC) framework. MAPIC extracts spatio-temporal features through an embedding encoder, prototype augmentation module, and distance classifier. It resists catastrophic forgetting via sample classification loss, prototype non-overlap loss, and knowledge distillation loss. By integrating self-attention into prototype learning, MAPIC dynamically optimizes inter-class discriminative power. He et al. [168] proposed the unsupervised domain adaptation framework MLUDAF based on graph convolutional networks to improve generalization across devices and individuals. The MLUDAF framework extracts spatio-temporal features via an Attenuated Spatial Pyramid Pooling

Residual module. It leverages GCNs to capture data structural characteristics. It designs a three-tier mechanism (domain, semantic, and structural alignment) to address data distribution shifts caused by inter-patient variability in ECG classification. However, the model is computationally complex, and its generalization capability in few-shot target domains requires improvement. To address computational resource constraints and privacy protection in cross-patient ECG classification and arrhythmia detection on edge devices, Farag [169] proposed a lightweight convolutional neural network based on matching filter theory. This approach embeds MF templates into CNN convolutional layers to extract ECG features and combines dynamically normalized RR interval features. It also optimizes the model through weight quantization and pruning.

Current ECG signal classification research is driven by deep learning, with various models achieving significant progress in tasks such as arrhythmia detection. However, existing techniques prioritize performance optimization while neglecting security and privacy protection and suffer from insufficient cross-database generalization [170]. Class imbalance, cross-center data heterogeneity, and inadequate model interpretability remain significant challenges [171, 172].

2. EEG Signal Classification

The electroencephalogram records the electrical activity of brain neurons over time. It is used to diagnose neurological disorders such as epilepsy, sleep disorders, and brain injuries. EEG signals contain different frequency components (e.g., $\delta, \theta, \alpha, \beta$ and γ waves) that reflect distinct functional states of the brain. Traditional methods primarily rely on frequency-domain analysis and statistical models to extract energy characteristics of waves at different frequencies.

Deng et al. [173] designed a reusable long short-term memory network to analyze the neural mechanisms underlying EEG responses to visual stimuli to improve the extraction of features from EEG signals. This approach extracts dual-channel features, including time-domain and frequency-domain representations. Ma et al. [174] developed a multi-channel CNN-Bi-LSTM model with an attention mechanism for epilepsy electroencephalogram classification. This model uses convolutional neural networks to extract spatial features, bidirectional long short-term memory networks to capture temporal dependencies, and attention mechanisms to select key electrode channels for dynamic channel weight optimization. To improve the classification accuracy of EEG signals, Lv et al. [175] proposed a spatiotemporal feature representation method that integrates the Gram angular field (GAF) and phase locking value (PLV). GAF captures temporal nonlinear relationships, and PLV constructs electrode topological connection matrices. Features are fused via parallel CNNs before classification. To leverage Transformers' global feature extraction capabilities, Siddharth et al. [176] explored the classification efficacy of Transformer networks on raw EEG data, addressing the limitations of manual feature extraction in traditional methods. This study utilizes the self-attention mechanism of Transformers to capture time-dependent relationships in EEG, achieving superior performance compared to traditional attention networks without requiring feature engineering. Zhao et al. [177] proposed a CTNet-based classification model that extracts spatial EEG features via CNN modules and captures global dependencies through a Transformer encoder. It enables classification through hierarchical feature extraction. Ali et al. [178] introduced ConTraNet, a hybrid model that integrates CNNs and Transformers to address EEG classification challenges with limited training data. This approach extracts local features via CNNs, while Transformers capture long-range dependencies. Attention mechanisms dynamically weight key features, yielding significant advantages in cross-subject scenarios. However, the model consumes much memory with long sequences and lacks validation in multimodal fusion settings.

Furthermore, to improve model interpretability, Cao et al. [179] designed the Interpretable Filter Bank Convolutional LSTM Network.

This approach extracts frequency-domain features using interpretable rectangular filters and combines deep separable convolutions with long short-term memory to fuse the electroencephalogram's spatiotemporal and frequency characteristics. Additionally, it integrates traditional filtering with deep learning to learn optimal frequency bands adaptively. However, it does not incorporate multimodal data fusion, and its real-time processing capability has yet to be validated. Given the immense potential of large language models, Kim et al. [180] applied LLMs to EEG classification. Using few-shot learning (only 2 % of the data) and integrating the Tree of Thought framework, they coordinated seizure detection and spike recognition tools, providing interpretable reasoning steps. The innovation lies in the LLMs' interpretability and cross-scale integration capabilities, which outperform traditional machine learning methods. However, there are risks of hallucinations and performance fluctuations during complex reasoning.

Despite new advances in EEG classification through deep learning, the field still faces core challenges, including performance degradation across individuals due to interindividual variability, insufficient robustness to high-noise signals, and the high cost of clinical annotation [181,182].

3. EMG Signal Classification

Electromyography involves recording the time series of muscle electrical activity to evaluate neuromuscular function. EMG signals exhibit high noise levels, non-stationarity, and short-term variability. Their amplitude and frequency characteristics correlate with the intensity of muscle activity. Traditional methods primarily rely on time-domain features and statistical models. For example, they extract root mean square, mean absolute value, and waveform length features.

Due to the scarcity of electromyography data resulting from ethical constraints on signal acquisition, Xiong et al. [183] proposed PatchEMG, a few-shot data augmentation method based on diffusion models. This method employs a patch-based training strategy to segment raw EMG signals into discrete pieces, generating high-quality synthetic data using conditional diffusion models and accelerated sampling algorithms. Block-wise learning captures signal context dependencies, enabling category-conditional generation without additional classifiers. For low-power EMG classification in wearable devices, Choi et al. [184] proposed a solution based on maximum overlap discrete wavelet transform and lightweight convolutional neural networks. MODWT extracts frequency-domain features and combines them with data augmentation and one-dimensional convolutional networks. Meanwhile, bit optimization and parallel computation reduce power consumption. However, the study only considered single-channel data and did not address multi-channel collaboration. To address the generalization challenge in cross-user gesture classification, Shi et al. [185] compared eight unsupervised transfer learning algorithms (e.g., CORAL). They enhanced adaptability by aligning second-order statistics between source and target domains using minimal unlabeled calibration data. However, performance degraded when traditional classifiers were combined with transfer learning. Future work may optimize network architectures to strengthen cross-modal transfer capabilities.

Deep learning has effectively reduced the reliance of EMG classification on feature engineering. Nevertheless, the field still faces challenges, including signal non-stationarity, generalization issues stemming from inter-individual variability, and the tension between wearable devices' real-time, edge deployment requirements and model complexity.

4. Respiratory Signal Classification

Respiratory signals are time-series recordings of breathing activity, including airflow, chest wall movement, and respiratory sounds. These signals are primarily used for applications such as detecting sleep apnea, diagnosing chronic obstructive pulmonary disease, and assessing asthma.

Osathitporn et al. [186] designed an end-to-end, multi-scale residual convolutional neural network called RRWaveNet to estimate respiratory rate using photoplethysmography. Multi-scale convolutions capture temporal features, and transfer learning improves performance in wearable device scenarios. Roy et al. [187] proposed a lightweight Inception network, RDLINet, for respiratory disease classification via Mel-spectrogram transformation and deep separable convolutions. However, it lacks real-time noise suppression.

Deep learning methods further advance key technologies and clinical applications. Nevertheless, signals remain susceptible to motion artifacts and interference from environmental noise. Heterogeneity across devices and populations impairs model generalization, and a lack of annotated samples hinders the training of complex models.

7.1.2. Domain-specific challenges in medical signal classification

Medical signals are time-sequenced medical measurement data characterized by temporal order, non-stationarity, noise interference, and inter-individual variability. While general time series challenges exist, medical signal classification faces unique barriers embedded in physiological data and clinical practice:

- (1) Inter-subject variability: Physiological signals exhibit significant distribution shifts across individuals due to age, gender, and pathology. A model trained on one patient cohort often fails to generalize to another, violating the independent and identically distributed assumption common in standard machine learning.
- (2) Label ambiguity and gold standard dilemma: Unlike visual object recognition, where ground truth is often objective, medical annotation requires expensive board-certified expertise. Furthermore, experts often disagree on complex signals (e.g., subtle arrhythmia or early-stage epileptic spikes), introducing inherent label noise. The lack of a unified Gold Standard makes supervised training unstable.
- (3) Data silos and privacy constraints: Due to strict regulations (e.g., HIPAA, GDPR), medical data cannot be easily centralized for large-scale training. This fragmentation creates “data silos”, making it difficult to train foundation models comparable to those in NLP or CV, necessitating federated or privacy-preserving learning paradigms.
- (4) Artifacts and clinical reliability: Medical signals are subject to specific non-stationary noise, such as muscle artifacts in EEG or baseline wander in ECG, caused by breathing. Moreover, “Black Box” deep learning models face a trust crisis in clinical settings; without explicit physiological interpretability [188], high accuracy alone is insufficient for clinical adoption.

7.2. Radar high-resolution range profile recognition

Radar Automatic Target Recognition (RATR) is a core function of modern radar systems. It is designed to automatically identify a target's category, model, or attributes by analyzing radar echo signals. As high-resolution radar technology has advanced, particularly through the widespread application of the High-Resolution Range Profile (HRRP), RATR has played an increasingly vital role in both military defense and civil security domains. HRRP represents the vector sum of target scatter point echoes projected along the radar line of sight and reflects the target's one-dimensional structural information along the radar line of sight. Due to its concise vector representation and rich structural information, HRRP is the primary focus of RATR research. The core task of radar HRRP recognition involves extracting attitude-invariant features from high-dimensional, sparse HRRP data. This task has evolved from traditional methods to machine learning approaches and, most recently, to deep learning methods. Traditional methods rely primarily on manually designed features and statistical models, whereas machine learning approaches enhance classification performance through data-

driven methods. Deep learning methods can automatically learn feature representations from raw data, significantly improving recognition accuracy. Emerging techniques such as few-shot learning have recently improved HRRP recognition performance under data-scarce conditions. This paper reviews the application of time series classification methods in HRRP recognition and provides an outlook on future development of recognition tasks.

7.2.1. Related works on the high-resolution range profile recognition

Traditional radar HRRP target recognition methods include distance-, model-, and feature-based approaches. Early methods relied on manually designed features and conventional machine learning algorithms. As the electromagnetic environment becomes more complex, deep learning has emerged as a popular topic in radar HRRP target recognition.

To improve the extraction of adaptive features from sparse data, Xiang et al. [189] proposed the multi-scale group fusion 1D convolutional neural network (MSGF-1D-CNN) to address the insufficient receptive field issue caused by small-scale convolutional kernels in standard CNNs. MSGF-1D-CNN employs Multi-Scale Group Convolution to capture features at different detail levels and combines Pointwise Convolution for feature fusion. This architecture outperforms traditional 1D-CNNs in scenarios with low signal-to-noise ratios. Zhang et al. [190] proposed a gradient-stable recurrent neural network to address gradient vanishing in RNNs during HRRP sequence processing. It enhances the estimation and recognition performance of missing samples. Gradient vanishing is mitigated through weight regularization, activation function optimization (LeakyReLU), and batch normalization. Meanwhile, sequence correlations are leveraged to estimate missing HRRP samples. Experiments demonstrate the proposed method's exceptional noise resistance. However, its capability for modeling dependency in long sequences is limited. Lin et al. [191] designed a cascaded model that combines a dual-channel CNN with a bidirectional long short-term memory network. This approach integrates spatial feature extraction from the dual-channel CNN with temporal dependency capture from the BiLSTM. Experiments demonstrate that the proposed method is more robust than single-channel models when dealing with data imbalance, though it struggles to separate target features in high sea clutter backgrounds. Wang et al. [192] proposed an attention-based ERTCN HRRP recognition model. This model extracts deep temporal features of HRRP sequences by stacking expanded causal convolutional residual blocks. It enhances generalization by replacing weight normalization with batch normalization and introduces elastic network regularization for feature selection. Additionally, it employs an attention mechanism to adaptively assign distance unit weights while suppressing local noise via average pooling. This approach dynamically captures HRRP temporal dependencies while suppressing redundant features, significantly improving noise resistance. Zhang et al. [193] utilized polarization and spatial scattering information from polarized HRRP to achieve high-precision target recognition. They constructed a fusion model integrating self-attention and convolutional long short-term memory (ConvLSTM). Self-attention focuses on discriminative distance units, and the ConvLSTM's convolutional operations and recurrent structure simultaneously capture polarization-dimensional features and spatial sequence dependencies. This approach demonstrates robust performance against noise and small-sample scenarios. However, it remains limited in its adaptability to dynamic target pose variations and does not consider higher-order feature fusion across multiple polarization channels. Li et al. [194] addressed domain shift in cross-domain, few-shot HRRP recognition by employing a masked visual transformer for self-supervised pretraining. Through masked reconstruction, they learned inter-block dependencies in HRRP and designed a Brownian Distance Covariance classifier to quantify dependencies between support class prototypes and query vectors. This approach addresses the issue of non-target region interference and target region contribution differences in HRRP, thereby enhancing recognition robustness under

target direction changes. Chen et al. [195] developed a Target Attention CNN that utilizes a 1D CNN to extract local structural features and generates attention weights via a bidirectional GRU to enhance the discriminative features of the target region adaptively. To address model failure caused by azimuth estimation errors, Chen et al. [196] leveraged imperfect side information to improve HRRP target recognition performance. This approach uses a conditional variational parameterized convolutional network that encodes azimuth information into convolution kernel parameters via variational parameterized convolution. The model uses a heteroscedastic Gaussian distribution to model estimation errors and optimizes feature extraction through a sample-dependent kernel attention module.

To improve the model's lightweight design, Zhang et al. [197] proposed a lightweight HRRPSeqNet recognition framework. The framework efficiently extracts spatio-temporal features through the Partial Inception module and jointly models HRRP structural features and sequence information through the label-guided module. The KL divergence-weighted inference algorithm optimizes open-set inference. Combining self-supervised learning with structural feature enhancement enhances the model's robustness against geometric modeling errors and improves recognition accuracy. Yue et al. [198] proposed a lightweight Transformer model to address the low efficiency of manual feature extraction and the insufficient few-shot accuracy of traditional HRRP object recognition methods. Experiments demonstrate that the proposed method significantly reduces dependence on the training sample size by embedding a recurrent neural network to extract local correlations between adjacent HRRP distance units, introducing group linear transformations to reduce the number of parameters, and combining adaptive gradient clipping to optimize training stability. To mitigate the challenges of manually designing CNN architectures for HRRP recognition, Xiang et al. [199] proposed a 1D-CNN framework that integrates Adaptive Pathway Representation (APR) blocks and the Globally Optimal Guided Artificial Bee Colony algorithm. APR blocks adaptively calibrate pathway features to enhance the extraction of structural information, and the GBL-ABC algorithm automatically prunes redundant pathways based on lottery assumptions. This approach enhances noise immunity while maintaining high accuracy. However, the pruning phase requires extensive training time, and its generalization capability for novel targets remains to be validated.

To improve adaptability in few-shot scenarios, Li et al. [200] proposed an ensemble framework that combines a prior-aided gated recurrent unit (PA-GRU) and task-shared quadratic discriminant analysis (TS-QDA). The PA-GRU learns the latent relationship between prior information and intra-class variation. Meanwhile, TS-QDA mitigates covariance matrix singularity through task-shared contraction. Experiments demonstrate that the proposed method significantly improves performance under few-shot conditions. To improve noise-resistant recognition capabilities in small-sample scenarios, Yin et al. [201] proposed a fusion of stacked denoising autoencoders (SDAE) and weighted auxiliary classification generative adversarial networks (WACGAN). The SDAE extracts noise-resistant deep features, and the WACGAN generates high-quality augmented data. Sample quality is optimized through label constraints and the optimization of weight coefficients. Significant expansion of the small sample size was achieved on a three-class military aircraft dataset. However, adaptability to complex maritime clutter backgrounds remained limited. Chen et al. [202] proposed a framework integrating dynamic graph neural networks with meta-learning. This framework dynamically generates graph representations adapted to target scattering characteristics via multi-head self-attention mechanisms. It combines these mechanisms with a layer-wise learning rate optimization meta-learning framework to rapidly adapt to new tasks. The hybrid graph mechanism effectively enhances recognition performance in few-shot scenarios by integrating physical priors with dynamic relationship learning.

Currently, deep learning methods have made significant progress in radar HRRP recognition research. However, the domain continues to

face challenges, including noise interference, severe feature variation due to target attitude sensitivity, limited global feature extraction capabilities, and data heterogeneity that restricts fusion performance. Furthermore, the scarcity of annotated samples hinders the training of complex models.

7.2.2. Domain-specific challenges in high-resolution range profile recognition

Despite the success of deep learning in RATR, radar signals differ fundamentally from optical images or generic time series. Practical application faces several inherent physical and operational barriers:

- (1) Aspect sensitivity and manifold discontinuity: The electromagnetic signature of a target is highly sensitive to the viewing aspect angle. A variation of merely a few degrees can cause the interference relationships between scattering centers to change drastically, resulting in a completely different range profile envelope. Consequently, samples of the same target class do not form a continuous cluster in the feature space but rather a highly non-linear, fragmented manifold. This forces models to require dense angular sampling during training, which is often unavailable.
- (2) Translation sensitivity and amplitude-phase coupling: The one-dimensional waveform of an HRRP is sensitive to the target's absolute position within the range window. A slight shift in distance results in a linear shift of the envelope and a rotation of the signal phase. While most existing deep learning methods rely solely on the amplitude (modulus) of the signal, the phase information contains critical details about the target's fine structure and micro-motion. Developing complex-valued neural networks that can robustly utilize both amplitude and phase while maintaining shift-invariance remains a difficult technical hurdle.
- (3) Open-set recognition in non-cooperative environments: Unlike closed-set classification in commercial applications, radar systems operate in non-cooperative military or security environments. The system frequently encounters unknown targets (e.g., new aircraft types or decoys) that were not present in the training database. Standard classifiers tend to assign these unknowns to known classes with high confidence (the overconfidence problem). Therefore, the evaluation standard cannot rely on accuracy alone; it must assess the model's ability to reject unknown samples while correctly classifying known ones.
- (4) Variable SNR and simulation-to-real gap: Radar echoes are subject to the fourth-power law of distance, leading to drastic variations in SNR. Additionally, real-world signals suffer from scintillation effects where the RCS fluctuates rapidly. Models trained on clean, high-SNR simulation data often suffer severe performance degradation when applied to measured data with low SNR and clutter, highlighting a significant domain gap that requires advanced transfer learning techniques.

7.3. Industrial fault diagnosis

The safe and stable operation of industrial equipment is essential to modern production. Equipment failures are a primary cause of production interruptions, decreased efficiency, and safety incidents. As industrial systems become more complex and automated, traditional manual inspection and reactive maintenance models can no longer meet modern industry demands for reliable and safe equipment. Vibration signal analysis is one of the most widely used and effective methods of industrial fault diagnosis. Vibration signals directly reflect operating conditions, contain rich fault information, and can be collected without damaging the equipment. The core task of vibration-based fault diagnosis is capturing subtle fault characteristics from multi-sensor, non-stationary time-series data. The rapid development of the Industrial Internet of Things (IIoT) and smart manufacturing presents new

opportunities and challenges for industrial fault diagnosis. On the one hand, the networked operation of vast industrial equipment generates massive amounts of vibration data, providing a rich foundation for data-driven diagnostic methods. However, complex and variable industrial environments, diverse equipment operating conditions, and a scarcity of fault samples impose higher demands on diagnostic approaches' adaptability, robustness, and data efficiency. This work provides a systematic review of vibration-based industrial fault diagnosis methods. Through a comparative analysis of various methods, our work aims to advance the development and application of industrial fault diagnosis technologies by providing a reference for researchers and engineers in the field.

7.3.1. Related works on the fault diagnosis of industrial vibration signals

Industrial fault diagnosis based on vibration signals relies primarily on the correlation between vibration signals and equipment status and fault characteristic frequencies, vibration transmission characteristics, and time-frequency domain analysis of vibration signals. Vibrations originating within equipment structures are transmitted to externally measurable points, and the transmission characteristics depend on the equipment's structural and material properties. Analyzing the variation characteristics of vibration signals enables the determination of equipment faults, their type, and severity. Additionally, time-domain characteristics (e.g., peak values, mean values, variance) and frequency-domain characteristics (e.g., spectral distribution, energy distribution) reflect the operational state of the equipment. Analyzing vibration transmission characteristics helps infer the location and nature of internal fault sources from externally measured signals.

To improve the low diagnostic accuracy and time-consuming nature of machine learning and deep learning for rolling bearing fault diagnosis, Zhou et al. [203] proposed a model integrating Robust Feature Ranking (RFR) and a Genetic Algorithm-optimized Broad Learning System (GA-BLS). This model enhances diagnostic efficiency and precision. After preprocessing infrared thermal images, they extracted 72-dimensional second-order statistical features using a grayscale co-occurrence matrix. These features were then ranked by robustness using RFR and introduced sequentially into a GA-optimized BLS model to select the optimal number of features and parameters. However, this feature extraction method relies on manual design, which limits its adaptability to complex composite faults. To address the complexity of data (e.g., noise, outliers, and imbalance) and data drift in diagnosing faults in aircraft engine interstage bearings, Berghout et al. [204] proposed a structured data preprocessing strategy that combines extraction, denoising, balancing, and other processes. They also used Gap statistics to evaluate the effectiveness of preprocessing. Additionally, this method uses recursive weight initialization to improve the adaptability of long short-term memory networks by optimizing weights through multiple iterative rounds. This method addresses the challenges of processing high-dimensional, complex noise signals and of insufficient feature extraction in inter-stage bearing fault diagnosis for aeroengines. It improves diagnostic accuracy and noise immunity. Wang et al. [205] proposed an MD-BiMamba-based fault diagnosis framework. This framework uses Complete Ensemble Empirical Mode Decomposition (CEEMDAN) to decompose sensor signals into intrinsic mode functions. Then, a bidirectional Mamba network extracts sequential, bidirectional, long-range correlations combined with an adaptive, attention-based fusion strategy for feature integration. Label smoothing regularization optimizes the loss function. Bidirectional Mamba enables global feature extraction with linear complexity. CEEMDAN effectively suppresses noise. The adaptive fusion strategy enhances the discriminative power of features. Experiments demonstrate that the proposed method exhibits excellent noise resistance and lower model complexity compared to similar approaches. To improve the generalization capability of rolling bearing fault diagnosis under multi-load and noisy environments, Sun et al. [206] built a parallel structure of a Gated Recurrent Unit-1D Convolutional Network that combines adaptive batch normalization to mitigate domain shift issues and reduce parameter redundancy through

1D global average pooling. This parallel architecture fuses temporal and spatial features, achieving significantly higher accuracy than traditional models in variable load and noisy scenarios, as well as notable noise resistance. However, model training remains time-consuming, and the response time to sudden faults requires optimization. Kaplan et al. [207] converted vibration signals into grayscale images and extracted texture features using Local Binary Patterns (LBP). They employed machine learning algorithms, such as K-Nearest Neighbors and Random Forest, for classification. This conversion preserves temporal texture information, and LBP features demonstrate high fault mode recognition capability. However, its capability to fuse features for complex composite faults is limited, and its real-time performance requires improvement. To improve outlier sensitivity, balance global and local information in fault diagnosis, and enhance feature separability, Zhang et al. [208] employed Structure-Balanced Discrimination Projection. This method balances local intra-class clustering and global inter-class separation through adaptive weighting, combined with Laplacian matrix optimization for determining the projection direction. Eulerian representations suppress the impact of outliers, and the structural balancing strategy enhances the discriminative power of features. To design a lightweight model, Yang et al. [209] developed a Lightweight Spatio-Temporal model that integrates a Self-Attention Mechanism. This model uses a squeeze-inflate module for fine-grained feature extraction, a bidirectional GRU to capture temporal dependencies, and self-attention to optimize feature weights. This approach enables end-to-end intelligent diagnosis.

To address the scarcity of samples in aviation engine bearing fault diagnosis and improve diagnostic accuracy with limited samples, Wang et al. [210] proposed an improved composite multi-scale phase entropy (RCMPHE) method for extracting dynamic signal features. When combined with a black chimpanzee optimization support vector machine, this approach establishes a mapping relationship between features and fault types. RCMPHE enhances entropy stability by performing segmented probability distribution calculations, thereby improving fault diagnosis accuracy. To overcome the challenges posed by limited and imbalanced samples in mechanical fault diagnosis and improve fault recognition accuracy in scenarios with few samples, Wang et al. [211] utilized digital twin technology to compensate for the lack of real-world data in bearing fault diagnosis, enabling real-time online diagnostics. They constructed a bearing lumped parameter model based on dynamic theory and used Pearson correlation coefficients for online model parameter correction. Nonlinear fault factors were introduced to generate simulated data combined with machine learning models for fault classification and parameter tuning. Integrating digital twin models with machine learning generates sparse fault data that guides practical parameter adjustments and informed maintenance decisions. This approach achieves significantly higher diagnostic accuracy than traditional methods in scenarios with imbalanced samples. Wang et al. [212] proposed a dynamic model-assisted transfer learning network. This network simulates fault scenarios through modular mathematical modeling, performs cross-domain feature transfer using convolutional neural networks, and optimizes parameters via a pre-training and fine-tuning strategy. This approach mitigates the small-sample problem by generating simulated data through dynamic, model-injected fault scenarios. However, the matching accuracy between the dynamic model and actual engines requires improvement, and its adaptability to novel fault modes is limited. Shi et al. [213] proposed a Deep Subdomain Adversarial Network (DSAN) that extracts features via one-dimensional convolutions. DSAN achieves subdomain adaptation through Local Maximum Mean Difference and employs a subdomain-level distribution alignment strategy to preserve fine-grained features effectively. Self-supervised pseudo-label generation guides target domain learning. Liu et al. [214] introduced a counterfactual augmentation contrastive learning framework. This framework extracts sparse optimal features via a feature weighting network, generates negative samples through counterfactual augmentation, and balances global-local feature

correlations with a contrastive learning module. The counterfactual augmentation strategy effectively expands the diversity of negative samples, and the adaptive weight update mechanism enhances the discriminative power of features. Experiments validate the effectiveness of the proposed method. Ping et al. [215] converted one-dimensional vibration signals into time-frequency images using a synchronous squeeze-squeeze wavelet transform. They designed a residual connection-enhanced CRR-UNet architecture and generated high-quality augmented data based on the denoising diffusion probabilistic model (DDPM). Deep residual shrinkage networks are employed for fault classification. It addresses the scarcity of diagnostic data for aircraft engine bearing faults under few-shot conditions, enhancing the model's generalization capabilities. To create a more realistic dataset, Hou et al. [216] developed a benchmark dataset for inter-shaft bearing fault diagnosis based on an aero-engine system. It advances algorithm validation and technological development. They constructed a real aero-engine test rig and induced faults artificially in both the outer and inner rings. It marks the first release of an inter-shaft bearing dataset based on an engine system, encompassing three categories: healthy, inner-ring fault, and outer-ring fault. These categories are closer to actual operating conditions.

Industrial vibration signal analysis is evolving from single-signal processing toward multimodal fusion and intelligent approaches driven by both physical and data factors [217]. However, technical bottlenecks persist in areas such as non-stationary signal modeling, edge computing optimization, and interpretability.

7.3.2. Domain-specific challenges in fault diagnosis of industrial vibration signals

Industrial environments impose harsh conditions that differentiate fault diagnosis from clean laboratory benchmarks. The core challenges stem from the physical nature of mechanical systems and production realities:

- (1) Extreme class imbalance condition: In real-world production, machinery operates in a healthy state for the vast majority of its lifecycle. Fault data is inherently rare, and data representing catastrophic failures is nearly non-existent. This results in extreme imbalance ratios (e.g., 1000:1 or higher). Standard accuracy metrics are misleading in this context (a model predicting healthy always achieves 99.9 % accuracy). Evaluation must prioritize metrics like F1-score, Recall, and the ability to detect rare anomalies without high false alarm rates.
- (2) Variable operating conditions: Rotating machinery often operates under varying loads and rotational speeds. The vibration characteristics of a fault shift linearly or non-linearly with the RPM. A model trained on data collected at a steady speed typically fails when applied to ramp-up, ramp-down phases, or different load regimes. This presents a severe Domain Shift problem, requiring models to possess strong domain generalization or unsupervised adaptation capabilities to align distributions across different working conditions.
- (3) Compound faults and signal coupling: Unlike standard classification tasks, where classes are mutually exclusive (e.g., Cat and Dog), industrial machinery often suffers from compound faults. A localized bearing defect may co-occur with gear misalignment or unbalance. The vibration signatures of these faults couple together, often non-linearly, making it difficult to disentangle and identify multiple simultaneous failure modes using single-label classification frameworks.
- (4) Strong periodicity and asynchronous noise: Industrial signals are dominated by strong periodic components related to the rotation frequency. However, factory environments are filled with asynchronous impulsive noise from adjacent heavy machinery. This background noise often has a high amplitude, burying the weak, transient impulses characteristic of early-stage faults (incipient

faults). Detecting these weak signatures under low SNR conditions remains a critical hurdle.

7.4. Network traffic detection

Cybersecurity faces increasingly complex and dynamic threats as network technology rapidly advances and digital transformation deepens. Intrusion detection technologies are among the most prevalent cybersecurity threats, making research in this field significant from both theoretical and practical standpoints. Network intrusion detection systems (NIDS) are critical technologies for ensuring network security. They primarily monitor network traffic, detect abnormal behavior or known attack patterns, and promptly issue alerts and implement protective measures. Traditional intrusion detection methods rely primarily on rule-based matching and simple statistical analysis. While these methods are effective against known attacks, they are limited due to increasingly complex network environments and evolving attack techniques. In recent years, data-driven intrusion detection methods have become a research focus. These approaches can automatically learn the characteristics of normal and abnormal traffic patterns, effectively identifying novel attacks. However, network intrusion detection faces numerous challenges, including data imbalance, high-dimensional features, real-time requirements, encrypted traffic, and small-sample learning. In practical applications, new attack types continuously emerge while labeled attack samples are scarce and difficult to obtain. Traditional learning methods that rely on large amounts of labeled data face severe challenges in this context. In this article, we review the current state of network traffic detection research, its primary challenges, and emerging trends to promote the development and application of time series classification methods in the field of network intrusion detection.

7.4.1. Related works on the network traffic intrusion detection

Network traffic detection involves analyzing temporal data of network packets to detect abnormal behavior, which requires high real-time performance.

To decrease redundant features in high-dimensional deep neural networks for intrusion detection, Thakkar et al. [218] proposed a statistical feature selection method that combines standard deviation and mean-median differences. This approach recursively screens for highly discriminative features to input into DNN classification. However, relying solely on static statistical features fails to model dynamic temporal characteristics. To address the inefficiency of traditional convolutional neural networks in Internet of Things intrusion detection due to data redundancy and excessive parameters, Yang et al. [219] proposed a lightweight LSCNN integrating a data purification algorithm (DPA) and separable convolutions. The DPA uses liquid molecular distillation principles to select representative samples and reduce data transformation redundancy. The LSCNN reconstructs the network using separable convolutions and SMLP convolutional and semi-connected layers to reduce parameters and computational complexity. Zhu et al. [220] proposed IoT-Defender, an intrusion detection framework that integrates subspace clustering and ensemble learning. This framework aims to enhance detection accuracy and robustness through multi-strategy collaboration. Feature selection utilizes mutual information, while subspace clustering algorithms, such as CLIQUE and PROCLUS, generate additional features. An ensemble model combines naive Bayes, LGBM, and XGB as base learners and logistic regression as the meta-learner. Hyperparameters are tuned via particle swarm optimization. The deep integration of subspace clustering and ensemble learning effectively captures both local data patterns and global features. The synergistic application of these three strategies enhances the model's detection capabilities against known and unknown attacks. However, relying on mutual information for feature selection may overlook nonlinear dependencies, and PSO optimization may get stuck in local optima. To address the reliance on complex feature preprocessing and

weak global feature extraction capabilities in existing IoT intrusion detection models, Wang et al. [221] proposed the TransIDS method. This approach utilizes the multi-head self-attention mechanism of Transformers to adaptively focus on key features in network traffic and incorporates label smoothing techniques to mitigate dataset imbalance. Mathina et al. [222] proposed the optimized convolutional sparse Fickian graph point trans-net. It employs a hybrid Hawk-Spider optimizer to select highly discriminative features and integrates convolutional sparse graph point networks with Fickian algorithms to extract local and global features. To address the challenges posed by high-dimensional complexity and the subtle distinctions between normal and malicious samples in IoT traffic, Wang et al. [223] introduced ImagTIDS. This intrusion detection framework integrates Gram Angular Difference Field imaging with an enhanced Transformer. The framework encodes network traffic into two-dimensional images using GADF to preserve temporal and global features. The ImagTrans model combines the advantages of convolutional neural networks and visual Transformers. It utilizes self-attention mechanisms to dynamically focus on key features and introduces Focal Loss to address class imbalance. Although the model can extract both local and global features, it suffers from high computational complexity. To address the limitations of existing methods for detecting anomalies in multi-time series within the Industrial Internet of Things, namely the insufficient capture of complex topological relationships and the high false positive rates due to reliance on local spatio-temporal representations, Zhou et al. [224] proposed a detection framework based on global-local association difference (GLAD). The framework aims to enhance the accuracy of anomaly recognition by integrating various features derived from both global and local spatio-temporal associations. The researchers designed a Gumbel-Softmax-based graph structure learning strategy that automatically captures global topological connections between sensors. Graph Attention Networks and Transformers were used to extract global and local correlations from data samples. They used symmetric Kullback-Leibler divergence to compute correlation differences, which they then combined with reconstruction error to generate anomaly scores. Maoudj et al. [225] proposed a two-step framework for minority class prediction based on one-dimensional convolutional neural networks. This hierarchical architecture utilizes a main model to combine minority classes into general labels for preliminary classification, and a submodel to classify these minority classes more specifically. The training process is optimized using class weighting techniques.

To address the challenges of dynamic environments and the scarcity of attack samples in few-shot attack detection within IoT settings, Yan et al. [226] proposed a GDE model that integrates data processing, augmentation, and variable classification. This framework balances detection accuracy and inference efficiency for few-shot intrusion detection. The model achieves feature learning and classification for few-shot attacks by transforming one-dimensional network traffic into three-dimensional RGB images through enhanced Gram Angular Fields and an improved Denoising Diffusion Probability Model for few-shot data augmentation. It also generates a 63-iteration variable network, ETNetv2, via differentiable neural architecture search based on EfficientNetV2. However, deploying the model on devices with limited resources poses challenges due to significant parameter overhead, and its ability to generalize against novel, unknown attacks remains to be validated. Ding et al. [227] proposed the TMG-GAN framework based on Generative Adversarial Networks. Its multi-generator architecture simultaneously generates different types of attack data. A classifier optimizes both generators and discriminators while filtering noisy samples. Cosine similarity is used to calculate the difference between generated and original samples, or between samples of different classes, to reduce inter-class overlap. Wang et al. [228] introduced the DIFT framework, which integrates diffusion models with transformers. These models learn feature patterns from minority class samples to generate balanced datasets. Patching techniques enhance local features in IoT traffic data and shorten sequence lengths to improve processing efficiency. A Time

Series Transformer then extracts local and global features from the balanced dataset to address data imbalance. Chen et al. [229] introduced the Synaptic Intelligent Convolutional Neural Network (SICNN). The SICNN's Synaptic Intelligence algorithm optimizes CNN synapse structures by suppressing critical parameter updates to preserve historical detection knowledge. The Double Cosine Cross-Entropy loss function mitigates gradient vanishing while focusing on hard-to-classify samples. Quantization-aware training compresses the model for devices with limited resources. To address the labeling of sparse, high-dimensional data in the Industrial Internet of Things and the limited generalization capability of traditional intrusion detection models against unknown attacks, Long et al. [230] proposed a semi-supervised ladder network model that integrates manifold regularization and dense cross-layer connections. The model aims to enhance detection accuracy by leveraging vast amounts of unlabeled data. The model preserves the low-dimensional manifold structure of high-dimensional data by introducing manifold regularization constraints in the decoder of the ladder network. Simultaneously, dense cross-layer connections between the encoder and decoder enhance feature propagation. They also propose a data fusion strategy based on random attention, which enables the collaborative use of local and global features for intrusion detection. Lu et al. [231] addressed the scarcity of IoT zero-day attack samples by proposing a few-shot detection framework based on Model-Agnostic Model Learning. This approach converts traffic data into 2D images for feature extraction using convolutional neural networks, and then utilizes MAML to optimize the initial parameters for rapid adaptation to new tasks. However, this data transformation process loses temporal dynamic information. Future work aims to integrate Transformer architectures to improve detection capabilities against time-sensitive attacks. Wu et al. [232] proposed the Multi-stage Attention Twin Network (MASiNet) based on a meta-learning framework. MASiNet employs a Transformer architecture, combined with channel and spatial attention, to extract spatio-temporal features from traffic data. Feature embeddings are optimized through feature encoding loss and contrastive loss, which enhances detection capabilities under few-shot conditions. To address the scarcity of labeled data in the Industrial Internet of Things and the lack of interpretability in anomaly detection frameworks, Abudurexiti et al. [233] proposed a local feature extraction method that integrates one-dimensional convolutional neural networks with convolutional block attention modules. They built a variational autoencoder by combining an enhanced temporal convolutional network and a Kolmogorov–Arnold network to identify long-term temporal dependencies. They computed anomaly scores using a dynamic Gaussian score function and a top-k threshold strategy and achieved model interpretability by visualizing feature contributions via SHAP values. To address the challenges of data privacy leakage and zero-day attack detection in machine learning intrusion detection systems for vehicle ad hoc networks, Cui et al. [234] proposed a lightweight hybrid intrusion detection system (LH-IDS) based on differential privacy. The LH-IDS employs a stacked ensemble model that combines one-class support vector machines and isolated forests. It uses Laplace and Gaussian mechanisms to provide differential privacy protection and detect anomaly traffic without requiring labeled data.

To address the performance bottlenecks of traditional intrusion detection systems in Internet of Things networks with resource-constrained edge devices, Saheed et al. [235] proposed the IoT-Defender framework. The framework integrates a modified genetic algorithm with a fine-tuned long short-term memory network to achieve efficient, real-time intrusion detection through edge computing. The researchers introduced a focus loss function to mitigate class imbalance issues and ultimately deployed lightweight models on edge devices, such as the Raspberry Pi 4. Zhu et al. [236] proposed the lightweight LKD-STNN model based on knowledge distillation (KD). They designed a student model with deep separable convolutions (1D-DSC) and a bidirectional long short-term memory network. This model dynamically

adjusts the knowledge transfer process via an adaptive temperature function. The student model learns under the guidance of soft labels from the teacher model (1DCNN+BiLSTM). Weight updates are optimized through a joint loss function (attention loss and cross-entropy loss). The combination of 1D-DSC and BiLSTM enables the extraction of lightweight spatio-temporal features. Meanwhile, the adaptive temperature function enhances the efficiency of knowledge distillation, resolving the conflict between the high complexity of traditional models and their challenging deployment on the edges.

7.4.2. Domain-specific challenges in network traffic intrusion detection

Unlike physical signals (e.g., ECG or vibration), which follow consistent physical laws, network traffic is generated by human behavior and automated protocols, leading to unique adversarial and evolutionary challenges:

- (1) Concept drift and rapid obsolescence: The distribution of network traffic is highly non-stationary. It changes rapidly due to legitimate updates (e.g., a new video streaming protocol) or the emergence of novel attack vectors (Zero-Day Attacks). A model trained on a dataset from six months ago often fails to detect modern attacks. This Concept Drift necessitates models with continuous learning capabilities to prevent rapid performance decay.
- (2) The “Black Box” of encrypted traffic: With the universal adoption of encryption protocols (e.g., TLS 1.3, QUIC), the packet payload, which contains the most discriminative information, is inaccessible. Deep learning models are forced to rely solely on side-channel features, such as the sequence of packet lengths and inter-arrival times. Classifying traffic based on these limited features without decrypting the data creates a significant information bottleneck compared to plaintext analysis.
- (3) Adversarial evasion and robustness: Unlike a faulty bearing, a hacker actively tries to evade detection. Network detection is a dynamic adversarial game. Attackers can inject dummy packets (padding), perturb timing characteristics, or fragment packets to alter the traffic signature without changing the malicious payload. NIDS models must, therefore, possess high adversarial robustness, not just high accuracy on static test sets.
- (4) The false positive dilemma: In cybersecurity, the cost of a False Positive (blocking legitimate traffic) can be as high as a False Negative. In high-throughput networks (Gigabits per second), even a 0.1 % false alarm rate generates thousands of alerts per hour, overwhelming security analysts (alert fatigue). Consequently, evaluation standards in NIDS must prioritize precision at extremely low false positive rates, rather than simple overall accuracy.

8. Discussions of general challenges and future works

Time Series Classification has demonstrated transformative potential across critical domains, including healthcare, defense, industrial manufacturing, and cybersecurity. However, a significant chasm remains between academic benchmarks and practical deployment, hindered by both scenario-specific bottlenecks and cross-domain systemic dilemmas. Building upon the domain-specific analysis in Section 7, this section distills the general core challenges that transcend individual applications. We then identify urgent research priorities and propose feasible technical pathways tailored to these challenges. The overarching objective is to bridge the gap between domain-specific practices and universal technological advancements, providing a roadmap for the next generation of robust TSC systems.

8.1. Synthesis of general challenges

Through synthesizing the operational constraints of medical signal

classification, radar HRRP recognition, industrial fault diagnosis, and network traffic detection, the fundamental barriers of TSC can be categorized into three interrelated dimensions: Data-centric, Model-centric, and Deployment-centric challenges.

8.1.1. Data-centric challenges: scarcity, heterogeneity, and quality degradation

The primary bottleneck across all domains is the scarcity of high-quality supervision coupled with the non-IID (Independent and Identically Distributed) nature of real-world data. All four scenarios suffer from extremely long-tailed class distributions, where critical samples, such as rare disease markers, incipient industrial faults, or zero-day network attacks, are inherently scarce. Concurrently, data collected from diverse sources (e.g., cross-center medical records, multi-sensor industrial plants) exhibit severe heterogeneity. Furthermore, environmental interference often results in a low SNR, directly compromising the fidelity of feature extraction.

Additionally, the scenario relevance of general challenges is as follows: (1) Heterogeneity: The inter-subject variability in medical signals and the multi-station differences in radar systems exemplify the non-IID challenge, where distribution shifts prevent effective transfer learning. (2) Scarcity and quality: The healthy majority problem in industrial diagnosis mirrors the label ambiguity and high expert costs in medical annotation. Both stem from the fundamental difficulty in acquiring large-scale, clean, and balanced datasets for domain-specific tasks.

8.1.2. Model-centric challenges: generalization failure and opacity

Existing Deep Learning-based TSC models exhibit limited adaptability, often suffering from catastrophic performance degradation when confronted with domain shift and concept drift. The assumption that training and testing data share identical distributions is routinely violated in dynamic environments. Furthermore, the inherent “Black-Box” nature of deep neural networks leads to a lack of interpretability. In safety-critical domains, high accuracy without physiological or physical explainability is insufficient for trust and adoption.

Additionally, the scenario relevance of general challenges is as follows: (1) Generalization: Radar models trained on high-SNR simulation data often fail in low-SNR real-world environments due to the simulation-to-real gap. Similarly, network detection models struggle to adapt to evolving attack patterns (concept drift), leading to rapid obsolescence. (2) Interpretability: In healthcare, the lack of physiological transparency hinders clinical decision-making despite high model scores. Similarly, in industrial safety, operators require causal explanations for fault predictions to schedule maintenance confidentially.

8.1.3. Deployment-centric challenges: resource constraints and trustworthiness

The practical deployment of TSC models faces a rigorous trade-off between computational complexity and real-time latency. State-of-the-art models (e.g., Transformers) impose substantial memory and computational burdens, making them ill-suited for edge devices with strict power budgets. Beyond efficiency, System Trustworthiness, encompassing data privacy and adversarial robustness, has emerged as a non-negotiable constraint.

Additionally, the scenario relevance of general challenges is as follows: (1) Real-time constraints: Wearable medical devices require ultra-low power consumption, while radar target recognition demands millisecond-level latency, conflicting with the heavy computational load of complex architecture. (2) Security and privacy: The legal mandates for patient privacy (e.g., HIPAA) create “data silos” that isolate medical datasets. In the cybersecurity domain, the vulnerability of NIDS to adversarial padding attacks highlights the critical need for robustness against malicious evasion.

8.2. Future works

Targeting the general core challenges identified above, and integrating the scenario-specific characteristics analyzed in [Section 7](#), we propose the following future works. These directions adhere to the principles of being problem-oriented, scenario-driven, and technically actionable, focusing on the most urgent bottlenecks and their corresponding technical solutions.

8.2.1. Generative augmentation and domain adaptation

The critical imperative lies in generating high-fidelity, scenario-consistent synthetic data to mitigate the scarcity of critical samples, alongside achieving robust alignment of heterogeneous data across disparate devices without compromising discriminative feature integrity.

And the feasible technical pathways include: (1) Physics-guided generative modeling: For domains with clear physical laws (Industry, Radar), future work should leverage Digital Twins to simulate the dynamic evolution of faults and targets. Integrating Denoising Diffusion Probabilistic Models or Conditional GANs with these physical simulations can generate synthetic samples that are statistically consistent with real distributions. In healthcare, generative models must incorporate physiological constraints (e.g., cardiac electrophysiology rules) to prevent the synthesis of biologically implausible artifacts. (2) Meta-learning for few-shot adaptation: To handle rapid scenario shifts, researchers should design task-agnostic meta-models (e.g., improved MAML) pre-trained on diverse multi-center datasets. This enables the model to adapt to a new hospital or machine type with only a few support samples. Furthermore, introducing Domain-Adversarial Training (DANN), inspired by subdomain adaptation strategies like DSAN, can actively align source and target feature distributions, mitigating the performance drop caused by cross-device drift. (3) Adaptive multimodal fusion: For multi-sensor environments, networks must evolve from static concatenation to Dynamic Weighting Mechanisms. Integrating temporal alignment modules is crucial to resolving the spatio-temporal misalignment inherent in asynchronous data streams. For cross-modal tasks (e.g., ECG and Clinical Notes), Cross-Attention Mechanisms should be adopted to facilitate knowledge transfer, enhancing the discriminative power of the fused representation.

8.2.2. Physics-informed learning and causal interpretability

The paramount objective is the seamless embedding of domain-specific physical mechanisms into deep learning architectures to enforce cross-scenario generalization, coupled with the transition from opaque predictions to intuitive, actionable interpretations that satisfy rigorous field-specific safety requirements.

And the feasible technical pathways include: (1) Physics-Informed Neural Networks (PINNs) for TSC: Instead of purely data-driven learning, future models should integrate domain knowledge directly into the loss function. For radar, electromagnetic scattering theories can constrain feature learning to ensure aspect-angle robustness. In industry, embedding mechanical dynamics equations (e.g., gear meshing stiffness) ensures that the learned features possess clear physical semantics, significantly improving extrapolation capabilities under variable operating conditions. (2) Multi-dimensional interpretability: Trustworthy AI requires a triad of mechanisms, visualization, and rules. In healthcare, visualization should focus on clinical landmarks (e.g., QRS complexes) rather than abstract heatmaps. In industry, deep learning models should be distilled into explicit decision rules that map to mechanical failure modes. Validating these explanations using SHAP values alongside domain expert review is essential for deployment.

(3) Drift-adaptive incremental learning: To tackle Concept Drift in network traffic and industrial aging, models must embrace Continual Learning. Techniques like Synaptic Intelligence or Knowledge Distillation can prevent catastrophic forgetting of old patterns. Crucially, integrating online drift detection modules (e.g., based on statistical

distance metrics) can trigger real-time, adaptive updates, ensuring stability in dynamic environments.

8.2.3. Lightweight architectures and privacy-preserving computing

The central challenge involves striking an optimal balance between state-of-the-art model performance and the stringent computational constraints of edge deployment, while simultaneously guaranteeing data privacy and systemic robustness within collaborative, cross-organizational ecosystems.

And the feasible technical pathways include: (1) Hardware-aware model compression: For edge deployment (wearables, IoT sensors), a prune-quantize-distill pipeline is essential. Future architectures should prioritize Depthwise Separable Convolutions combined with lightweight recurrent units, reducing parameter counts by orders of magnitude while retaining temporal receptive fields, similar to the LKD-STNN. (2) Federated learning with differential privacy: To break data silos in healthcare and cybersecurity without compromising privacy, Federated Learning is the standard path forward. Integrating Differential Privacy mechanisms (e.g., Laplace noise injection) ensures that gradient updates cannot leak raw data. Optimizing communication efficiency (e.g., gradient compression) is vital to handle the high transmission overhead of dense time series data in FL settings. (3) Adversarial robustness engineering: Recognizing the adversarial nature of radar jamming and network evasion, training pipelines must include Adversarial Training against a spectrum of attacks (e.g., perturbation, padding). For foundation models, input filtering and rigorous post-validation mechanisms are necessary to prevent "poisoning" attacks, ensuring system reliability under duress.

8.2.4. Time series foundation models

A frontier research challenge is the construction of universal time series foundation models capable of adapting to disparate modalities, thereby facilitating the effective transfer of learned representations and capabilities across distinct domains such as healthcare, industry, and defense.

And the feasible technical pathways include: (1) Universal tokenization and pre-training: The key barrier is the continuous nature of time series. Future work must focus on Patching, segmenting signals into fixed-length tokens, and developing scenario-agnostic tokenizers (e.g., Amplitude-Phase encoding). Pre-training these architectures on massive, multi-domain datasets using self-supervised objectives (e.g., Masked Reconstruction, Temporal Order Prediction) allows the model to learn universal temporal dynamics. (2) Semantic alignment and prompt tuning: To adapt a foundation model to specific tasks, Prompt Tuning and Low-Rank Adaptation (LoRA) are more efficient than full fine-tuning. Aligning the temporal token embedding space with domain-specific semantic spaces (e.g., medical ontology, fault taxonomies) significantly accelerates transfer learning. (3) Chain-of-Thought reasoning: To mitigate "hallucinations" in complex diagnostic tasks, integrating Chain-of-Thought (CoT) reasoning is a promising direction. By constraining the model to output step-by-step logical deductions (grounded in physical rules or clinical guidelines) before the final prediction, we can dramatically improve the reliability and interpretability of large-scale models.

In summary, this section establishes a rigorous logical synergy with [Section 7](#) by anchoring the methodological discussion within the concrete constraints of real-world applications. Each proposed technical pathway is distilled directly from empirical pain points, ensuring that the research directions are not only theoretically sound but also problem-oriented, feasible, and immediately actionable. As the field advances toward the convergence of physical mechanisms, edge intelligence, and foundation models, TSC technology is poised to transcend current limitations, delivering robust, interpretable, and scalable decision support for data-driven innovation across diverse sectors.

9. Conclusion

This review proposes a multidimensional review framework with three horizontal and three vertical dimensions for deep learning-driven time series classification. We systematically integrate research achievements and development trends in this domain. The three horizontal dimensions include traditional analytical, deep learning-based, and few-shot learning-based methods, which clearly illustrate the evolutionary path from manual feature engineering to data-driven modeling. The three vertical dimensions traverse data characteristics, methodological frameworks, and application scenarios, forming a data-method-application closed-loop knowledge system that addresses the shortcomings of existing review literature regarding systematic integration and scenario adaptation analysis. Methodologically, the paper thoroughly analyzes the limitations of traditional methods and focuses on four core, deep-learning-based techniques: local pattern mining, temporal dependency modeling, hybrid model architectures, and spatiotemporal fusion modeling. The paper reveals the strengths and weaknesses of each method in terms of its feature capture capabilities. Additionally, it reviews transfer learning and self-supervised learning strategies within the few-shot learning paradigm, offering technical references for scenarios with limited labels. In practice, the paper summarizes data augmentation and model architecture optimization techniques, and compiles mainstream datasets and multidimensional evaluation metrics. In terms of application, the paper focuses on four domains: medical signal classification, radar HRRP target recognition, industrial fault diagnosis, and network traffic detection. It analyzes implementation challenges and solutions to demonstrate the practical value of TSC.

In summary, the field still faces challenges, including data complexity, insufficient model generalization, efficiency bottlenecks, high annotation costs, and a lack of interpretability. Future breakthroughs should target multimodal fusion, few-shot learning, explainable AI, temporal foundation models, and lightweight edge deployment. This review offers researchers a comprehensive overview of the field and its technological trajectory, presented within a structured knowledge framework. We aim to advance TSC technology from academic exploration to industrial application, driving technological innovation in smart healthcare, Industry 4.0, and other fields while providing core support for perception and decision-making in complex, dynamic systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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